Stuff I've derived

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A Numerical Methods

A.1 Finite Difference Stencils

A.1.1 First Derivatives

Two-point forward difference stencil with error:

$$\begin{aligned} f'(x) = &\frac{f(x+\varepsilon) - f(x)}{\varepsilon} \\ &-\frac{\varepsilon}{2} f''(x) + \mathcal{O}(\varepsilon^2) \end{aligned} \tag{A.1.1}$$

Three-point central difference stencil with error:

$$\begin{aligned} f'(x) = &\frac{f(x+\varepsilon) - f(x-\varepsilon)}{2\varepsilon} \\ &-\frac{\varepsilon^2}{6} f^{(3)}(x) + \mathcal{O}(\varepsilon^3) \end{aligned} \tag{A.1.2}$$

Five-point central difference stencil with error:

$$\begin{aligned} f'(x) = &\frac{-f(x+2\varepsilon) + 8f(x+\varepsilon) - 8f(x-\varepsilon) + f(x-2\varepsilon)}{12\varepsilon} \\ &+\frac{\varepsilon^4}{30} f^{(5)}(x) + \mathcal{O}(\varepsilon^6) \end{aligned} \tag{A.1.3}$$

A.1.2 Second Derivatives

A.1.2.1 Single-Variable

Three-point stencil with error:

$$\begin{aligned} f''(x) = &\frac{f(x+\varepsilon) - 2f(x) + f(x-\varepsilon)}{\varepsilon^2} \\ &-\frac{\varepsilon^2}{12} f^{(4)}(x) + \mathcal{O}(\varepsilon^4) \end{aligned} \tag{A.1.4}$$

Five-point stencil with error:

$$\begin{aligned} f''(x) = &\frac{-f(x+2\varepsilon) + 16f(x+\varepsilon) - 30f(x) + 16f(x-\varepsilon) - f(x-2\varepsilon)}{12\varepsilon^2} \\ &-\frac{\varepsilon^4}{90} f^{(6)}(x) + \mathcal{O}(\varepsilon^6) \end{aligned} \tag{A.1.5}$$

A.1.2.2 Mixed Second Derivative

With a grid spacing of $\varepsilon_x, \varepsilon_y > 0$ in the x- and y-directions, respectively, we have

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{f_{++} + f_{--} - f_{+-} - f_{-+}}{4\varepsilon_x \varepsilon_y} - \frac{1}{6}\left(\varepsilon_x^2 \frac{\partial^4 f}{\partial x^3 \partial y} + \varepsilon_y^2 \frac{\partial^4 f}{\partial x \partial y^3}\right) + \mathcal{O}(\varepsilon^4) \tag{A.1.6}$$

Here,

$$f_{\pm_1 \pm_2} = f(x \pm_1 \varepsilon_x, y \pm_2 \varepsilon_y) \tag{A.1.7}$$

Note that the expression (A.1.6) corresponds to taking twice the central difference (A.1.2) of f, once with respect to x and once with respect to y.

A.1.3 Laplacian

27-point stencil:

$$\Delta f = \frac{1}{\varepsilon^2}((-8\lambda - 6)\cdot(\text{center}) + (4\lambda+1)\cdot(\text{faces}) - 2\lambda\cdot(\text{edges}) + \lambda\cdot(\text{corners})) \quad \text{(A.1.8)}$$

where, using $f_{+0-} = f(x+\varepsilon, y, z-\varepsilon)$ etc.,

$$\begin{aligned}
(\text{center}) &= f_{000} \\
(\text{faces}) &= f_{+00} + f_{-00} + f_{0+0} + f_{0-0} + f_{00+} + f_{00-} \\
(\text{edges}) &= f_{++0} + f_{+-0} + f_{-+0} + f_{--0} \\
&\quad\ f_{+0+} + f_{+0-} + f_{-0+} + f_{-0-} \\
&\quad\ f_{0++} + f_{0+-} + f_{0-+} + f_{0--} \\
(\text{corners}) &= f_{+++} + f_{++-} + f_{+-+} + f_{-++} \\
&\quad\ f_{+--} + f_{-+-} + f_{--+} + f_{---}.
\end{aligned} \quad \text{(A.1.9)}$$

The error is $\mathcal{O}(\varepsilon^2)$, with leading contribution given by

$$\frac{\varepsilon^2}{12}\left(\frac{\partial^4 f}{\partial x^4} + \frac{\partial^4 f}{\partial y^4} + \frac{\partial^4 f}{\partial z^4}\right). \quad \text{(A.1.10)}$$

The choice of λ doesn't change this leading error order, but it changes the anisotropy of the error. Useful choices are $\lambda = \frac{1}{22}$ or $\lambda = \frac{1}{26}$. The latter minimises the anisotropy for Fourier modes, and its full expression reads

$$\Delta f = \frac{-164\cdot(\text{center}) + 30\cdot(\text{faces}) - 2\cdot(\text{edges}) + 1\cdot(\text{corners})}{26\varepsilon^2} + \mathcal{O}(\varepsilon^2)_{\text{iso}} + \mathcal{O}(\varepsilon^4)_{\text{aniso}}. \quad \text{(A.1.11)}$$

A.2 Runge-Kutta

A.2.1 General Structure

Runge-Kutta is the name of a family of integrators for first-order differential equations of the form

$$y'(t) = F(t, y(t)). \quad \text{(A.2.1)}$$

Generally speaking, given a timestep ε, one evaluates the right-hand side of the ODE s times, at different times $t_i = \varepsilon c_i \in (t_0, t_0+\varepsilon)$ and for different guesses for y. Concretely, starting from $y(t_0) = y_0$, one evaluates

$$\begin{aligned}
k_1 &= F(t_0, y_0), \\
k_2 &= F(t_0 + \varepsilon c_2, y_0 + \varepsilon a_{21} k_1), \\
k_3 &= F(t_0 + \varepsilon c_3, y_0 + \varepsilon(a_{32}k_2 + a_{31}k_1)), \\
&\ \ \vdots \\
k_s &= F(t_0 + \varepsilon c_s, y_0 + \varepsilon\big(a_{s,s-1}k_{s-1} + ... + a_{s1}k_1\big).
\end{aligned} \quad \text{(A.2.2)}$$

These are then used to approximate $y(t+\varepsilon)$ as

$$y(t+\varepsilon) \approx y_0 + \varepsilon(b_1 k_1 + ... + b_s k_s). \quad \text{(A.2.3)}$$

An RK integrator is hence defined by picking a value of s, quadrature points $c_i \in [0,1]$, $i = 2, ..., s$, as well as the coefficients b_i, $i = 1, ..., s$ and a_{ij}, $i > j$. These coefficients are not chosen at random, but rather picked such that the error of $y(t+\varepsilon)$ is as small as possible, i.e. of as high of an order ε as can be arranged.

The coefficients can be thought of as a matrix $\mathbf{A} \in \mathbb{R}^{s\times s}$ and a pair of vectors $\mathbf{b}, \mathbf{c} \in \mathbb{R}^s$, with the coefficients arranged such that

$$\mathbf{A} = \begin{pmatrix} 0 & & & \dots & 0 \\ a_{21} & 0 & & \dots & 0 \\ a_{31} & a_{32} & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ a_{s1} & a_{s2} & \dots & a_{s,s-1} & 0 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_s \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 0 \\ c_2 \\ \vdots \\ c_s \end{pmatrix}. \tag{A.2.4}$$

Further writing $\mathbf{k} = (k_1, ..., k_s)^\top$, and denoting row evalution of a vector matrix by $[\cdot]_i$, we can write the above as follows:

$$k_i = F(t_0 + \varepsilon[\mathbf{c}]_i, y_0 + \varepsilon[\mathbf{A}\mathbf{k}]_i), \qquad y(t+\varepsilon) \approx y_0 + \varepsilon \mathbf{b}^\top \mathbf{k}. \tag{A.2.5}$$

One way of determining optimal coefficients $\mathbf{A}, \mathbf{b}, \mathbf{c}$ is to expand the approximation (A.2.3) around $\varepsilon = 0$ and comparing coefficients to the Taylor expansion of $y(t_0 + \varepsilon)$, which using $y' = F(t, y(t))$ as well as the notation

$$F = F(t_0, y_0), \quad F_t = (\partial_t F)(t_0, y_0), \quad F_y = \big(\partial_y F\big)(t_0, y_) \quad \text{etc.} \tag{A.2.6}$$

can be written explicitly as

$$\begin{aligned} y(t_0+\varepsilon) = y_0 &+ \varepsilon F + \frac{\varepsilon^2}{2}\big(F_t + F F_y\big) \\ &+ \frac{\varepsilon^3}{6}\big(F_{tt} + 2F F_{ty} + F_t F_y + F F_y^2 + F^2 F_{yy}\big) \\ &+ \frac{\varepsilon^4}{24}\big(F_{ttt} + 3F F_{tty} + 3F_t F_{ty} + 5F F_y F_{ty} + 3F^2 F_{tyy} + F_{tt} F_y \\ &\qquad + 3F F_t F_{yy} + F_t F_y^2 + F F_y^3 + 4F^2 F_y F_{yy} + F^3 F_{yyy}\big) \\ &+ \mathcal{O}(\varepsilon^5) \end{aligned} \tag{A.2.7}$$

As is evident from the number of terms quickly growing with the order in ε, the strategy of comparing coefficients quickly becomes unmanageable for higher-order RK integrators. Further, we can see that it is highly non-linear in derivatives of F; this explains why we need to use the k_i with $i < j$ in the evaluation of k_j. Without this chaining, we could not reproduce the nonlinearity in F required to match the explicit expression (A.2.7).

A.2.2 RK1 (is just Euler)

As a first illustration of how the strategy of reading off coefficients works, let us indulge in the lowest-order case of $s = 1$. Although *technically*, $s = 0$ would be the lowest-order case, and $y(t) \approx y_0$ is *technically* an approximation of the solution to $y' = F(t, y)$, it is neither a good nor interesting one, hence the decision to start at $s = 1$. We start by writing down the k's, of which there is only one—namely

$$k_1 = F(t_0, y_0) = F. \tag{A.2.8}$$

The RK method then prescribes the approximation

$$y(t_0 + \varepsilon) \approx y_0 + \varepsilon b_1 k_1 = y_0 + \varepsilon b_1 F \tag{A.2.9}$$

For this to match expansion (A.2.7) up to order $\mathcal{O}(\varepsilon)$, we need to set $b_1 = 1$. Thus, RK1 is nothing but

$$y(t_0 + \varepsilon) = y_0 + \varepsilon F(t_0, y_0) + \mathcal{O}(\varepsilon^2), \tag{A.2.10}$$

which we immediately recognise as the explicit Euler method.

A.2.3 RK2 (gets interesting)

Moving up to the $s = 2$ case, we can employ the same strategy to obtain a (hopefully) more precise integration scheme. Here, the k's, along with their expansions in ε, read

$$\begin{aligned} k_1 &= F(t_0, y_0) = F, \\ k_2 &= F(t_0 + \varepsilon c_2, y_0 + \varepsilon a_{21} k_1) = F + \varepsilon c_2 F_t + \varepsilon a_{21} k_1 F_y + \mathcal{O}(\varepsilon^2) \\ &= F + \varepsilon\big(c_2 F_t + a_{21} F F_y\big) + \mathcal{O}(\varepsilon^2). \end{aligned} \tag{A.2.11}$$

Again assembling the RK-approximation, we find

$$y(t_0 + \varepsilon) = y_0 + \varepsilon(b_1 + b_2)F + \varepsilon^2\big(b_2 c_2 F_t + b_2 a_{21} F F_y\big) + \mathcal{O}(\varepsilon^3) \tag{A.2.12}$$

Clearly, $b_1 + b_2 = 1$ and $b_2 \neq 0$ must hold for this to have a chance of matching expansion (A.2.7) up to $\mathcal{O}(\varepsilon^2)$. Further, we need $b_2 c_2 = b_2 a_{21} = \frac{1}{2}$, which upon introduction of a parameter $\alpha \in \left[\frac{1}{2}, \infty\right)$ allows us to parameterise

$$b_1 = 1 - \alpha, \quad b_2 = \alpha, \quad c_2 = \frac{1}{2\alpha}, \quad a_{21} = \frac{1}{2\alpha}. \tag{A.2.13}$$

Using this parametrisation, the RK2 integration procedure reads

$$k_1 = F(t_0, y_0), \quad k_2 = F\Big(t_0 + \frac{\varepsilon}{2\alpha}, y_0 + \frac{\varepsilon}{2\alpha} k_1\Big), \tag{A.2.14}$$

and

$$y(t_0 + \varepsilon) = y_0 + \varepsilon((1 - \alpha)k_1 + \alpha k_2) + \mathcal{O}(\varepsilon^3). \tag{A.2.15}$$

Two common choices for the free parameter α are

- $\alpha = \frac{1}{2}$ (Heun's method): This leads to the approximation scheme

$$y(t_0 + \varepsilon) = y_0 + \frac{\varepsilon}{2}(k_1 + k_2) + \mathcal{O}(\varepsilon^3) \tag{A.2.16}$$

  with

$$k_1 = F(t_0, y_0), \quad k_2 = F(t_0 + \varepsilon, y_0 + \varepsilon k_1). \tag{A.2.17}$$

- $\alpha = 1$ (Explicit Midpoint Method): In this case, we get

$$y(t_0 + \varepsilon) = y_0 + \varepsilon k_2 + \mathcal{O}(\varepsilon^3), \tag{A.2.18}$$

  where

$$k_1 = F(t_0, y_0), \quad k_2 = F\Big(t_0 + \frac{\varepsilon}{2}, y_0 + \frac{\varepsilon}{2} k_1\Big). \tag{A.2.19}$$

Although both these cases—and for that matter, any $\alpha \in \left[\frac{1}{2}, \infty\right)$ has the same *order* error, $\mathcal{O}(\varepsilon^3)$, different values for α lead to different forms of the error, which might—depending on the ODE in question—lead to a better constant factor.

A.2.4 RK3 (gets ugly)

For $s = 3$, the k's and their expansions around $\varepsilon = 0$ read

$$
\begin{aligned}
k_1 &= F(t_0, y_0) = F, \\
k_2 &= F(t_0 + \varepsilon c_2, y_0 + \varepsilon a_{21} k_1) \\
&= F + \varepsilon\big(c_2 F_t + a_{21} F F_y\big) + \frac{\varepsilon^2}{2}\big(c_2^2 F_{tt} + 2c_2 a_{21} F F_{ty} + a_{21}^2 F^2 F_{yy}\big) + \mathcal{O}(\varepsilon^3), \\
k_3 &= F(t_0 + \varepsilon c_3, y_0 + \varepsilon(a_{31} k_1 + a_{32} k_2)) \\
&= F + \varepsilon\big(c_3 F_t + (a_{31} + a_{32}) F F_y\big) + \frac{\varepsilon^2}{2}\Big(2a_{32} c_2 F_t F_y + 2a_{32} a_{21} F F_y^2 \\
&\quad + c_3^2 F_{tt} + 2c_3(a_{31} + a_{32}) F F_{ty} + (a_{31} + a_{32})^2 F^2 F_{yy}\Big) + \mathcal{O}(\varepsilon^3)
\end{aligned}
\tag{A.2.20}
$$

Next, we evaluate the RK3 integration scheme

$$
y(t_0 + \varepsilon) \approx y_0 + \varepsilon(b_1 k_1 + b_2 k_2 + b_3 k_3) \tag{A.2.21}
$$

order by order in ε, comparing to the explicit expansion (A.2.7) on the left-hand side:

$$
\text{at } \mathcal{O}(\varepsilon): \quad F \overset{!}{=} b_1 F + b_2 F + b_3 F \tag{A.2.22}
$$

which requires

$$
b_1 + b_2 + b_3 = 1 \quad \Leftrightarrow \quad b_1 = 1 - b_2 - b_3, \tag{A.2.23}
$$

for some $b_2, b_3 \neq 0$. Moving on, we get

$$
\text{at } \mathcal{O}(\varepsilon^2): \quad \frac{\varepsilon}{2}\big(F_t + F F_y\big) \overset{!}{=} \varepsilon^2\big(b_2\big(c_2 F_t + a_{21} F F_y\big) + b_3\big(c_3 F_t + (a_{31} + a_{32}) F F_y\big)\big) \tag{A.2.24}
$$

From this, we read off the requirements

$$
c_2 = a_{21}, \quad c_3 = a_{31} + a_{32}, \quad b_2 c_2 + b_3 c_3 = \frac{1}{2}. \tag{A.2.25}
$$

Note that this is not the full solution space; it does however simplify things moving forward. Concretely, it makes the right-hand side simpler; at $\mathcal{O}(\varepsilon^3)$, we find

$$
\begin{aligned}
&\frac{\varepsilon^3}{6}\big(F_{tt} + 2F F_{ty} + F_t F_y + F F_y^2 + F^2 F_{yy}\big) \\
&\overset{!}{=} \frac{\varepsilon^3}{2}\Big[b_2\big(c_2^2 F_{tt} + 2c_2 a_{21} F F_{ty} + a_{21}^2 F^2 F_{yy}\big) + b_3\Big(2a_{32} c_2 F_t F_y + 2a_{32} a_{21} F F_y^2 \\
&\quad + c_3^2 F_{tt} + 2c_3(a_{31} + a_{32}) F F_{ty} + (a_{31} + a_{32})^2 F^2 F_{yy}\Big)\Big] \\
&= \frac{\varepsilon^3}{2}\big[b_2 c_2^2\big(F_{tt} + 2F F_{ty} + F^2 F_{yy}\big) + b_3\big(2a_{32} c_2\big(F_t F_y + F F_y^2\big) + c_3^2\big(F_{tt} + 2F F_{ty} + F^2 F_{yy}\big)\big)\big]
\end{aligned}
\tag{A.2.26}
$$

This implies

$$
b_2 c_2^2 + b_3 c_3^2 = \frac{1}{3} \quad \text{and} \quad 2b_3 a_{32} c_2 = \frac{1}{3}. \tag{A.2.27}
$$

At this point, it is most convenient to introduce $c_2 = \alpha$ and $c_3 = \beta$ as freely choosable constants. In doing so, we get the system of equations

$$
b_1 = 1 - b_2 - b_3, \quad \alpha b_2 + \beta b_3 = \frac{1}{2}, \quad \alpha^2 b_2 + \beta^2 b_3 = \frac{1}{3}, \quad a_{32} = \frac{1}{6\alpha b_3}. \tag{A.2.28}
$$

This is solved by

$$
\begin{aligned}
b_1 &= -\frac{3\alpha(1-2\beta)+3\beta-2}{6\alpha\beta}, \qquad & b_2 &= \frac{2-3\beta}{6\alpha(\alpha-\beta)}, \\
b_3 &= -\frac{2-3\alpha}{6\beta(\alpha-\beta)}, & a_{32} &= -\frac{\beta(\alpha-\beta)}{\alpha(2-3\alpha)}.
\end{aligned}
\tag{A.2.29}
$$

The full RK3 integration procedure thus reads

$$
y(t_0+\varepsilon) \approx y_0 + \varepsilon\left(-\frac{3\alpha(1-2\beta)+3\beta-2}{6\alpha\beta}k_1 + \frac{2-3\beta}{6\alpha(\alpha-\beta)}k_2 - \frac{2-3\alpha}{6\beta(\alpha-\beta)}k_3\right) + \mathcal{O}(\varepsilon^4)
\tag{A.2.30}
$$

where

$$
\begin{aligned}
k_1 &= F(t_0, y_0), \\
k_2 &= F(t_0+\varepsilon\alpha, y_0+\varepsilon\alpha k_1) \\
k_3 &= F\left(t_0+\varepsilon\beta, y_0+\varepsilon\left[\left(\beta+\frac{\beta(\alpha-\beta)}{\alpha(2-3\alpha)}\right)k_1 - \frac{\beta(\alpha-\beta)}{\alpha(2-3\alpha)}k_2\right]\right).
\end{aligned}
\tag{A.2.31}
$$

A common choice for the quadrature points is $\alpha = \frac{1}{2}$, $\beta = 1$. This leads to

$$
\begin{aligned}
b_1 &= \frac{1}{6}, \quad & b_2 &= \frac{2}{3}, \quad & b_3 &= \frac{1}{6}, \\
c_1 &= 0, \quad & c_2 &= \frac{1}{2}, \quad & c_3 &= 1 \\
a_{21} &= \frac{1}{2}, \quad & a_{31} &= -1 \quad & a_{32} &= 2.
\end{aligned}
\tag{A.2.32}
$$

When inserted, this yields the scheme

$$
y(t_0+\varepsilon) = y_0 + \varepsilon\left(\frac{1}{6}k_1 + \frac{2}{3}k_2 + \frac{1}{6}k_3\right) + \mathcal{O}(\varepsilon^4)
\tag{A.2.33}
$$

with

$$
\begin{aligned}
k_1 &= F(t_0, y_0), \\
k_2 &= F\left(t_0+\frac{\varepsilon}{2}, y_0+\frac{\varepsilon}{2}k_1\right), \\
k_3 &= F(t_0+\varepsilon, y_0-\varepsilon k_1+2\varepsilon k_2)
\end{aligned}
\tag{A.2.34}
$$

A.3 Courant-Friedrichs-Lewy (CFL) Conditions for PDEs

A.3.1 Example: Parabolic Heat Equation

In this section, we derive the CFL conditions for various discretisations of the parabolic heat equation,

$$
\partial_t \phi = \alpha \Delta \phi
\tag{A.3.1}
$$

where $\alpha > 0$ is the diffusion coefficient and $\phi = \phi(t, \mathbf{x})$, $t \in \mathbb{R}$, $\mathbf{x} \in \mathbb{R}^d$ is the dynamical variable.

A.3.1.1 7-Point Laplacian Explicit Euler CFL Condition

For this first derivation, we consider the simplest discretisation of equation (A.3.1), taking the forward difference for the left-hand side, and the Laplacian stencil built from the second derivative stencil (A.1.4) with equal grid spacing Δx for all axes. In d dimensions, this amounts to

$$\frac{\phi(t+\Delta t,\mathbf{x})-\phi(t,\mathbf{x})}{\Delta t}=\alpha\sum_{i=1}^{d}\frac{\phi(t,\mathbf{x}+\mathbf{e}_i\Delta x)-2\phi(t,\mathbf{x})+\phi(t,\mathbf{x}-\mathbf{e}_i\Delta x)}{\Delta x^2}, \tag{A.3.2}$$

or when rearranged for $\phi(t+\Delta t,\mathbf{x})$,

$$\phi(t+\Delta t,\mathbf{x})=\phi(t,\mathbf{x})+C\sum_{i=1}^{d}[\phi(t,\mathbf{x}+\mathbf{e}_i\Delta x)-2\phi(t,\mathbf{x})+\phi(t,\mathbf{x}-\mathbf{e}_i\Delta x)]. \tag{A.3.3}$$

where we defined $C\coloneqq\frac{\alpha\Delta t}{\Delta x^2}$. The goal is to ensure C is such that the discrete linear operator acting on the right-hand side has no exponentially growing modes, i.e. that all its eigenvalues are non-negative. Since any discrete function $\psi(\mathbf{x})=\phi(t,\mathbf{x})$ can be decomposed into functions of the form $e^{i\mathbf{k}\cdot\mathbf{x}}$, we make the ansatz for a mode

$$\phi(t,\mathbf{x})=G^{t/\Delta x}e^{i\mathbf{k}\cdot\mathbf{x}} \tag{A.3.4}$$

where G is the growth factor per timestep, fixed by the discretised equation (A.3.3). For the evolution to be stable, we must have $|G|\leq 1$—let us examine how these modes evolve. Inserting into the left-hand side, we get

$$\phi(t,\mathbf{x})=G^{(t/\Delta x)+1}e^{i\mathbf{k}\cdot\mathbf{x}}=G\phi(t,\mathbf{x}); \tag{A.3.5}$$

hence the name *growth factor*. For the right-hand side, we expand

$$\begin{aligned}
&\phi(t,\mathbf{x})+C\sum_{i=1}^{d}[\phi(t,\mathbf{x}+\mathbf{e}_i\Delta x)-2\phi(t,\mathbf{x})+\phi(t,\mathbf{x}-\mathbf{e}_i\Delta x)]\\
&=\phi(t,\mathbf{x})+CG^{t/\Delta x}\sum_{i=1}^{d}\left[e^{i\mathbf{k}\cdot\mathbf{x}}e^{i\mathbf{k}\cdot\mathbf{e}_i\Delta x}-2e^{i\mathbf{k}\cdot\mathbf{x}}+e^{i\mathbf{k}\cdot\mathbf{x}}e^{-i\mathbf{k}\cdot\mathbf{e}_i\Delta x}\right]\\
&=\phi(t,\mathbf{x})\left(1+C\sum_{i=1}\underbrace{\left[e^{ik_i\Delta x}-2+e^{-ik_i\Delta x}\right]}_{=-4\sin^2(k_i\Delta x)}\right)\\
&=\phi(t,\mathbf{x})\left(1-4C\sum_{i=1}^{d}\sin^2(k_i\Delta x)\right).
\end{aligned} \tag{A.3.6}$$

Equating both sides and dividing by $\phi(t,\mathbf{x})\neq 0$, we establish

$$G=1-4C\sum_{i=1}^{d}\sin^2(k_i\Delta x) \tag{A.3.7}$$

The condition that $|G|\leq 1$ is satisfied if $G\leq 1$ and $G\geq -1$. The former is always satisfied since $\sin^2\geq 0$, but the latter introduces constraints on C—and by that, on the relationship between Δt, Δx and α. Concretely, the right-hand side attains a global minimum where $k_i=\frac{\pi}{2\Delta x}$, $i=1,...,d$. With this value for $\mathbf{k}$, we obtain

$$-1\overset{!}{\leq}G=1-4Cd\quad\Leftrightarrow\quad\frac{1}{2d}\geq C=\frac{\alpha\Delta t}{\Delta x^2}. \tag{A.3.8}$$

We can turn this into a condition on the timestep Δt, which leads us to the CFL condition for this discrete stepping operator,

$$\Delta t\leq\frac{\Delta x^2}{2\alpha d}. \tag{A.3.9}$$

A.3.1.2 7-Point Laplacian Implicit Euler CFL Condition

In the implicit case, one replaces the forward difference in time with the backwards difference, or equivalently, evaluates the right-hand side of the equation at $t + \Delta t$ instead of t. That is, the heat equation is discretised as

$$\frac{\phi(t+\Delta t, \mathbf{x}) - \phi(t, \mathbf{x})}{\Delta t} = \alpha \sum_{i=1}^{d} \frac{\phi(t+\Delta t, \mathbf{x} + \mathbf{e}_i \Delta x) - 2\phi(t+\Delta t, \mathbf{x}) + \phi(t+\Delta t, \mathbf{x} - \mathbf{e}_i \Delta x)}{\Delta x^2}. \tag{A.3.10}$$

Inserting the ansatz $\phi(t, \mathbf{x}) = G^{t/\Delta t} e^{i\mathbf{k}\cdot\mathbf{x}}$ and identifying terms proceeds largely identically to the explicit case, with the difference that the Laplacian stencil terms now carry an additional factor of G as they are evaluated at $t + \Delta t$ instead of t. This has as consequence that now,

$$G = 1 - 4GC \sum_{i=1}^{d} \sin^2(k_i \Delta x/2), \tag{A.3.11}$$

which rearranges to

$$G = \frac{1}{1 + 4C \sum_{i=1}^{d} \sin^2(k_i \Delta x/2)}. \tag{A.3.12}$$

The implications of this result are profound. As long as $C \geq 0$, we have $|G| \leq 1$—this means that the evolution is unconditionally stable; it allows for an arbitrarily large timestep Δt.

This, however, does not mean that one should opt to just do one's entire simulation in a single step. Although the discrete evolution is *stable* for any $\Delta t > 0$, the error of a step with the given choices of stencils is $\mathcal{O}(\Delta t + \Delta x^2)$. Increasing Δt hence also increases the error—just as increasing Δx in the explicit case (to allow for larger Δt) can destroy the evolution due to a lack of resolution. Because of this, the initially perceived superiority of implicit over explicit time-stepping becomes more subtle; if error requirements pose harsher constraints on Δt than the explicit CFL condition does, then explicit stepping is preferable over implicit as it is cheaper computationally.

A.3.1.3 27-point Isotropic Laplacian Explicit Euler CFL Condition

We now move on to derive the CFL condition for the explicit Euler case where on the right-hand side, the stencil (A.1.11) is employed. Concretely, we thus consider the stepping scheme

$$\phi(t+\Delta t, \mathbf{x}) = \phi(t, \mathbf{x}) + C(-164 \cdot (\text{center}) + 30 \cdot (\text{faces}) - 2 \cdot (\text{edges}) + 1 \cdot (\text{corners})), \tag{A.3.13}$$

where $C = \frac{\alpha \Delta t}{26 \Delta x^2}$. Similar steps to the previous two derivations, together with the fact that the stencil applied to a constant yields 0, leads to

$$G(\mathbf{k}) = 1 - 4C \Bigg(30 \sum_{i=1}^{3} \sin^2(k_i \Delta x/2) - 2 \sum_{i<j} \big[\sin^2\big((k_i + k_j)\Delta x/2\big) + \sin^2\big((k_i - k_j)\Delta x/2\big) \big] + \sum_{\sigma_y, \sigma_z \in \{\pm 1\}} \sin^2\big((k_x + \sigma_y k_y + \sigma_z k_z)\Delta x/2\big) \Bigg) \tag{A.3.14}$$

Denoting the large parentheses by $S(\mathbf{k})$, we obtain the stability condition

$$-1 \leq 1 - 4CS(\mathbf{k}) \quad \Leftrightarrow \quad S(\mathbf{k}) \leq \frac{1}{2C}. \tag{A.3.15}$$

We are thus left to find the maximum of $S(\mathbf{k})$. Introducing the auxiliary variables

$$\xi_i = \frac{k_i \Delta x}{2}, \quad u_i = \sin^2(\xi_i), \tag{A.3.16}$$

we can rewrite the sum over the faces as

$$\sum_{i=1}^{3} \sin^2(k_i \Delta x/2) = \sum_i u_i. \tag{A.3.17}$$

For the sum over the edges, we make use of

$$\begin{aligned} \sin^2(\xi_i + \xi_j) + \sin^2(\xi_i - \xi_j) &= 2\big(\sin^2\xi_i \cos^2\xi_j + \cos^2\xi_i \sin^2\xi_j\big) \\ &= 2\big(u_i(1-u_j) + (1-u_i)u_j\big) \\ &= 2\big(u_i + u_j - 2u_i u_j\big) \end{aligned} \tag{A.3.18}$$

so that

$$\begin{aligned} \sum_{i<j} \big[\sin^2\big((k_i + k_j)\Delta x/2\big) + \sin^2\big((k_i - k_j)\Delta x/2\big)\big] &= 2\sum_{i<j}\big[u_i + u_j - 2u_i u_j\big] \\ &= 4\sum_i u_i - 4\sum_{i<j} u_i u_j. \end{aligned} \tag{A.3.19}$$

For the corner terms, we consider

$$\begin{aligned} &\sin\big(\xi_x + \sigma_y\xi_y + \sigma_z\xi_z\big) = \sin\big(\xi_x + \sigma_y\xi_y\big)\cos(\xi_z) + \sigma_z \cos\big(\xi_x + \sigma_y\xi_y\big)\sin(\xi_z) \\ &= \sin(\xi_x)\cos\big(\xi_y\big)\cos(\xi_z) + \sigma_y\cos(\xi_x)\sin\big(\xi_y\big)\cos(\xi_z) \\ &\quad + \sigma_z\cos(\xi_x)\cos\big(\xi_y\big)\sin(\xi_z) - \sigma_y\sigma_z\sin(\xi_x)\sin\big(\xi_y\big)\sin(\xi_z) \end{aligned} \tag{A.3.20}$$

After squaring, this turns into a horrible mess, but since we are summing over all values of $\sigma_y, \sigma_z \in \{\pm 1\}$, any term linear in one of the σ will cancel in the sum. That is, only the squares of the individual summands appear in the final result, so that

$$\begin{aligned} &\sum_{\sigma_y,\sigma_z\in\{\pm1\}} \sin^2\big((k_x + \sigma_y k_y + \sigma_z k_z)\Delta x/2\big) \\ &\quad = 4\big(\sin^2\xi_x\cos^2\xi_y\cos^2\xi_z + \cos^2\xi_x\sin^2\xi_y\cos^2\xi_z \\ &\qquad + \cos^2\xi_x\cos^2\xi_y\sin^2\xi_z - \sin^2\xi_x\sin^2\xi_y\sin^2\xi_z\big) \\ &\quad = 4\big(u_x(1-u_y)(1-u_z) + (1-u_x)u_y(1-u_z) \\ &\qquad + (1-u_x)(1-u_y)u_z + u_x u_y u_z\big) \\ &\quad = 4\left(\sum_i u_i - 2\sum_{i<j} u_i u_j + 4u_x u_y u_z\right) \end{aligned} \tag{A.3.21}$$

Hence, $S(\mathbf{k})$, now as a function of $\mathbf{u} = (u_x, u_y, u_z)$ reads

$$\begin{aligned} S(\mathbf{u}) &= 30\sum_i u_i - 2\left(4\sum_i u_i - 4\sum_{i<j} u_i u_j\right) + 4\left(\sum_i u_i - 2\sum_{i<j} u_i u_j + 4u_x u_y u_z\right) \\ &= 26\big(u_x + u_y + u_z\big) + 16u_x u_y u_z. \end{aligned} \tag{A.3.22}$$

Since $u_i \in [0,1]$, S clearly takes its maximum where $u_x = u_y = u_z = 1$, yielding

$$\max_{[0,1]^3} S(\mathbf{u}) = 94. \tag{A.3.23}$$

Thus, the condition (A.3.15) is turned into

$$94 \leq \frac{1}{2C} \quad \Leftrightarrow \quad \frac{\alpha \Delta t}{26 \Delta x^2} = C \leq \frac{1}{188} \tag{A.3.24}$$

Rearranging for Δt yields the final condition

$$\Delta t \leq \frac{13}{94} \frac{\Delta x^2}{\alpha}. \tag{A.3.25}$$

Note that

$$\frac{13}{94} \approx 0.13892..., \tag{A.3.26}$$

making the 27-point stencil CFL condition slightly more restrictive than for the 7-point case, where the factor in front of $\Delta x^2/\alpha$ is

$$\frac{1}{2d} = \frac{1}{6} \approx 0.166\overline{6}. \tag{A.3.27}$$

This trade-off is typically worth it, given that the isotropy of the error is much higher.

A.3.1.4 27-point Isotropic Laplacian Implicit Euler CFL Condition

In the implicit case of the preceding section, where the right-hand side of equation (A.3.13) is evaluated at $t + \Delta t$ instead, we obtain

$$G = 1 - 4GCS(\mathbf{k}). \tag{A.3.28}$$

Solving for G, we find

$$G = \frac{1}{1 + 4CS(\mathbf{k})}. \tag{A.3.29}$$

As we have shown before,

$$S(\mathbf{k}(\mathbf{u})) = 26\big(u_x + u_y + u_z\big) + 16 u_x u_y u_z \in [0, 94], \tag{A.3.30}$$

so that $|G| \leq 1$ is always satisfied and hence, the implicit Euler procedure remains unconditionally stable even with the isotropic stencil for the Laplacian.

A.3.2 Example: Hyperbolic Wave Equation

The derivation of CFL conditions for the wave equation,

$$\Box \phi = 0 \quad \Leftrightarrow \quad \partial_t^2 \phi = c^2 \Delta \phi, \tag{A.3.31}$$

where c is the propagation velocity. The simplest discretisation for this equation (and the only we will consider here) is

$$\frac{\phi(t + \Delta t, \mathbf{x}) - 2\phi(t, \mathbf{x}) + \phi(t - \Delta t, \mathbf{x})}{\Delta t^2} = c^2 \sum_i \frac{\phi(t, \mathbf{x} + \mathbf{e}_i \Delta x) - 2\phi(t, \mathbf{x}) + \phi(t, \mathbf{x} - \mathbf{e}_i \Delta x)}{\Delta x^2} \tag{A.3.32}$$

This form, as is, is unsuited for a time-stepping procedure, but we can derive stability conditions from it. Again employing the ansatz $\phi(t, \mathbf{x}) = G^{t/\Delta t} e^{i\mathbf{k}\cdot\mathbf{x}}$ for an amplification number G, we can rewrite the above as

$$
\begin{aligned}
(G - 2 + G^{-1})\phi(t, \mathbf{x}) &= C^2 \sum_i \big(e^{ik_i \Delta x} - 2 + e^{-ik_i \Delta x}\big)\phi(t, \mathbf{x}) \\
&= -4C^2 \sum_i \sin^2(k_i \Delta x/2)\phi(t, \mathbf{x}) \\
&=: -C^2 S(\mathbf{k})\phi(t, \mathbf{x}).
\end{aligned} \tag{A.3.33}
$$

Rearranging and multiplying by G leads to

$$
G^2 - 2\Big(1 - \frac{1}{2}C^2 S(\mathbf{k})\Big)G + 1 = 0 \tag{A.3.34}
$$

The allowed amplification factors are the two roots $G_\pm$ of this polynomial in G. Since the constant term is 1, they must satisfy $G_+ G_- = 1$, whence we must have $|G_\pm| = 1$ to satisfy the no-growth condition $|G| \leq 1$.

For a general polynomial of this form,

$$
G^2 - 2\beta G + 1 = 0, \quad \beta \in \mathbb{R}, \tag{A.3.35}
$$

the roots read

$$
G_\pm = \beta \pm \sqrt{\beta^2 - 1}. \tag{A.3.36}
$$

If $\beta^2 > 1$, we have strictly real and distinct roots, of which at least one has a magnitude greater than 1. Thus, we must have $\beta^2 \leq 1$, or equivalently, $-1 \leq \beta \leq 1$. Inserting the β from equation (A.3.34), this turns into

$$
-1 \leq 1 - \frac{1}{2}C^2 S(\mathbf{k}) \leq 1. \tag{A.3.37}
$$

The right-hand bound holds trivially; the left-hand bound requires

$$
C^2 S(\mathbf{k}) \leq 4 \quad \Rightarrow \quad C^2 S_{\max} \leq 4 \tag{A.3.38}
$$

Since $S_{\max} = 4d$, this yields the final CFL condition,

$$
\frac{c^2 \Delta t^2}{\Delta x^2} = C^2 \leq \frac{1}{d} \quad \Leftrightarrow \quad \Delta t \leq \frac{\Delta x}{c\sqrt{d}}. \tag{A.3.39}
$$

This has a very nice physical interpretation: the distance travelled by a wave within one timestep, $c\Delta t$, must not exceed a value proportional to the grid spacing, Δx. Since the proportionality factor of $1/\sqrt{d}$ is less than 1, this means that the numerical domain of dependence is contained in the physical domain of dependence.

A.3.3 Stability Regions of RK Schemes

In this section, we derive and study the stability regions of different RK schemes applied to a PDE of the form

$$
\partial_t \phi = F(\phi). \tag{A.3.40}
$$

In a sense, this will be a generalisation of the analysis we did in the preceding sections, where we derived the CFL conditions for the heat and wave equations. The main advantage to the approach in this section is that we will derive stability conditions for the *time integrator* separately (at least in the linear case), where the only input from the right-hand side are the eigenvalues (or state-dependent growth factors when linearising) of the right-hand side. This separation is useful because frequently in numerical solvers for PDEs, one wants to replace the time integrator or RHS implementation while leaving the other untouched, whence it is more convenient to know

the stability conditions of both components individually to then be able to bring them together into a final stability condition.

A.3.3.1 Linear Equations

We first consider the case where F is linear,

$$\partial_t \phi = F\phi. \tag{A.3.41}$$

Assuming F admits a complete set of spatial eigenmodes ψ_n with corresponding eigenvalues λ_n —as is typical for spatial discretisations of linear operators—we can expand any solution as

$$\phi(t, \mathbf{x}) = \sum_n c_n(t)\psi_n(\mathbf{x}). \tag{A.3.42}$$

Substituting this expression into the linear PDE yields decoupled ODEs for each mode coefficient:

$$\dot{c}_n(t) = \lambda_n c_n(t) \tag{A.3.43}$$

Because the modes evolve independently, the stability of the full system reduces to studying the linear test equation

$$\partial_t \phi = \lambda\phi \tag{A.3.44}$$

where $\lambda \in \mathbb{C}$ represents an eigenvalue of the spatial operator F. A Runge-Kutta scheme applied to a PDE will be stable only if it produces a non-amplifying update for every eigenvalue λ in the spectrum of F.

Due to the simplicity of this test equation, we can evaluate the full update performed by an RK step explicitly. Let us begin by looking at RK1/Forward Euler. Written in full, RK1 reads

$$\begin{aligned} k_1 &= F(\phi(t_0)), \\ \phi(t_0 + \Delta t) &= \phi(t_0) + \Delta t \cdot k_1. \end{aligned} \tag{A.3.45}$$

Inserting k_1 into the definition, and making use of $F\phi = \lambda\phi$, we get

$$\phi(t_0 + \Delta t) = (1 + \lambda\Delta t)\phi(t_0). \tag{A.3.46}$$

Introducing the complex variable $z = \lambda\Delta t$, we can write this as

$$\phi(t_0 + \Delta t) = p_1(z)\phi(t_0), \quad p_1(z) = 1 + z. \tag{A.3.47}$$

The no-growth condition now simply demands that $|p_1(z)| \leq 1$, which defines a region in the complex plane for z where the RK1 integration scheme is stable. Because of the relationship $z = \lambda\Delta t$, stability hence depends on the spectrum of F through λ, and on the size of the time step through the factor Δt; a smaller Δt generally improves stability. We will look at the *stability region* $\{z \in \mathbb{C} : |p_1(z)| \leq 1\}$ later on, when we have derived the polynomials of higher-order schemes to compare it to.

Moving on to RK2, taking e.g. the Heun scheme, the update expands to

$$\begin{aligned} k_1 &= F(\phi(t_0)) = \lambda\phi(t_0), \\ k_2 &= F(\phi(t_0) + \Delta t k_1) = \lambda\phi(t_0) + \Delta t\lambda^2\phi(t_0), \\ \phi(t_0 + \Delta t) &= \phi(t_0) + \frac{\Delta t}{2}(k_1 + k_2) \\ &= \phi(t_0) + \frac{\Delta t}{2}(\lambda + \lambda + \Delta t\lambda^2)\phi(t_0) \\ &= \left(1 + \lambda\Delta t + \frac{1}{2}(\lambda\Delta t)^2\right)\phi(t_0). \end{aligned} \tag{A.3.48}$$

This implies the RK2 stability condition

$$|p_2(z)| \leq 1, \quad p_2(z) \coloneqq 1 + z + \frac{1}{2}z^2. \tag{A.3.49}$$

Though already now, one might start to see a pattern, let us briefly also derive $p_3(z)$, the growth factor polynomial for RK3. Expanding the RK3 update yields

$$\begin{aligned}
k_1 &= F(\phi(t_0)) = \lambda\phi(t_0), \\
k_2 &= F\left(\phi(t_0) + \frac{\Delta t}{2}k_1\right) = \lambda\phi(t_0) + \frac{\Delta t}{2}\lambda^2\phi(t_0), \\
&= \left(\lambda + \frac{\Delta t}{2}\lambda^2\right)\phi(t_0), \\
k_3 &= F(\phi(t_0) - \Delta t k_1 + 2\Delta t k_2) \\
&= \lambda\phi(t_0) - \Delta t\lambda^2\phi(t_0) + 2\Delta t\lambda^2\phi(t_0) + (\Delta t)^2\lambda^3\phi(t_0) \\
&= (\lambda + \Delta t\lambda^2 + (\Delta t)^2\lambda^3)\phi(t_0), \\
\phi(t_0 + \Delta t) &= \phi(t_0) + \frac{\Delta t}{6}(k_1 + 4k_2 + k_3) \\
&= \left(1 + \frac{\Delta t}{6}\left(\lambda + 4\left(\lambda + \frac{\Delta t}{2}\lambda^2\right)\right) + (\lambda + \Delta t\lambda^2 + (\Delta t)^2\lambda^3)\right)\phi(t_0) \\
&= \left(1 + \lambda\Delta t + \frac{1}{2}(\lambda\Delta t)^2 + \frac{1}{6}(\lambda\Delta t)^3\right)\phi(t_0).
\end{aligned} \tag{A.3.50}$$

Hence, at third order, we get the stability condition

$$|p_3(z)| \leq 1, \quad p_3(z) = 1 + z + \frac{1}{2}z^2 + \frac{1}{6}z^3. \tag{A.3.51}$$

At this point, the pattern is unmistakable: for an n-th order RK method with n stages, the stability polynomial $p_n(z)$ is preciely the n-th degree Taylor polynomial of the exponential function,

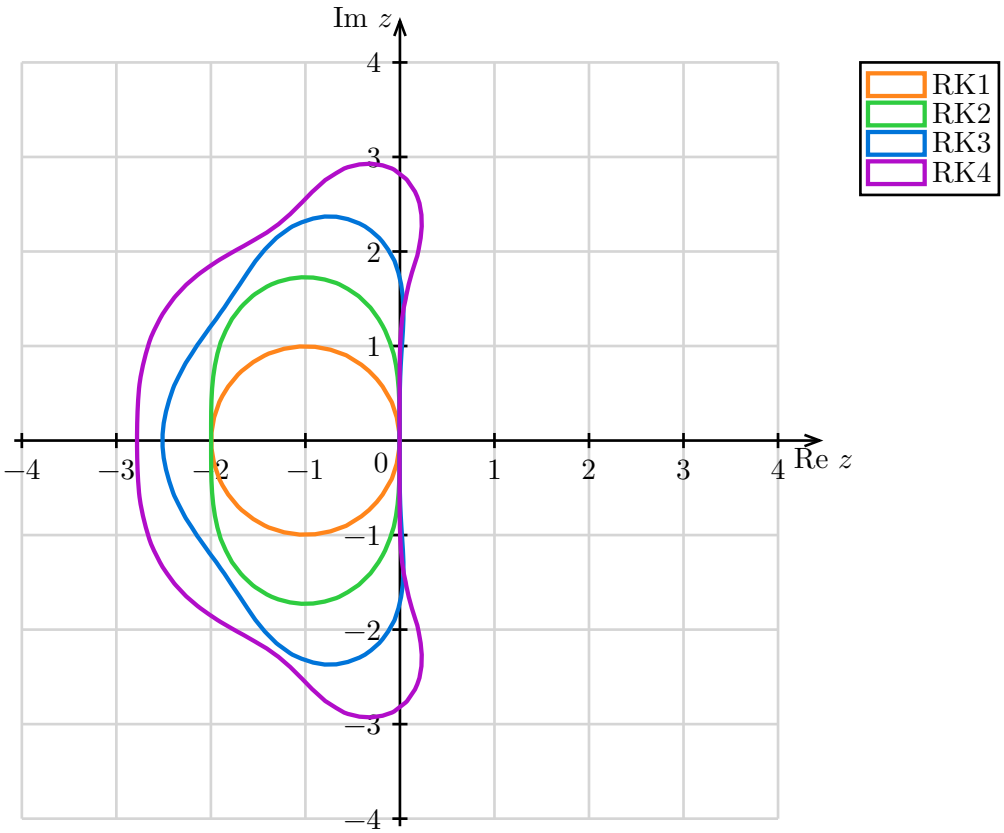
$$p_n(z) = \sum_{k=0}^{n} \frac{z^k}{k!}. \tag{A.3.52}$$

This is not a coincidence. The exact solution to the linear test equation over one time step is

$$\phi(t_0 + \Delta t) = e^{\lambda\Delta t}\phi(t_0) = e^z\phi(t_0). \tag{A.3.53}$$

By definition, an n-th order numerical scheme must match the Taylor expansion of the exact evolution operator e^z up to $\mathcal{O}(z^n)$. Furthermore, an m-stage explicit RK scheme performs m nested evaluations of F, so its amplification factor $p(z)$ is necessarily a polynomial in z of degree $\leq m$.

When $m = n$ (which Butcher's barriers permit for $n \leq 4$), the polynomial degree bound matches the order-matching condition, which fixes $p_n(z)$ uniquely to the partial sum of the exponential series. For $n \geq 5$, where maintaining order n requires $m > n$ stages, the polynomial $p(z)$ contains higher-degree terms $\mathcal{O}(z^{n+1})$ that depend on the specific choice of Butcher tableau, though the first n terms always remain $1 + z + ... + z^n/n!$.



(A.3.54)

In the plot above, the stability regions are the interiors of the lines above. We can see that the higher the order of the method, the larger the region generally becomes. Many important linear PDEs have negative real or purely imaginary eigenvalues, so it makes sense to know the largest possible value of z for which the scheme is still stable. For reference, we provide a table of these values below:

Method	**Min. z along $\mathbb{R}_{\leq 0}$, $z_{\text{min}}^{\text{Re}}$**	**Max. $\lvert z\rvert$ along $i\mathbb{R}$, $z_{\text{max}}^{\text{Im}}$**
RK1/Euler	-2	0
RK2	-2	0
RK3	≈ -2.5127	$\sqrt{3} \approx 1.732$
RK4	≈ -2.7853	$2\sqrt{2} \approx 2.8284$

(A.3.55)

There is an important observation to be made about the shape of the stability regions of the higher-order schemes RK3, RK4 and beyond. Recalling that the growth factor of the exact polynomial is e^z, it is clear that we can only have stability if $\operatorname{Re} z \leq 0$. However, both RK3 and RK4 incorporate regions with $\operatorname{Re} z$ in their stability regions, producing stable evolution for modes which physically speaking would be unstable. Though this is often irrelevant as the RHS operators of interest typically do not have unstable eigenmodes, it is worth keeping in mind.

To see how these affect integration schemes for different operators, let us reconsider the heat and wave equations. The main input into the RK stability considerations are the eigenvalues of the right-hand side operator when put into first-order form. The heat equation,

$$\partial_t \phi = \Delta \phi, \tag{A.3.56}$$

is already in first-order form; the wave equation needs to be reexpressed slightly. By introducing the additional variable $\Pi = \partial_t \phi$, we can write it as

$$\partial_t \begin{pmatrix} \phi \\ \Pi \end{pmatrix} = \begin{pmatrix} \Pi \\ \Delta\phi \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \Delta & 0 \end{pmatrix} \begin{pmatrix} \phi \\ \Pi \end{pmatrix}, \tag{A.3.57}$$

so that the RHS operator is the matrix operator $\left(\begin{smallmatrix} 0 & 1 \\ \Delta & 0 \end{smallmatrix}\right)$.

We first consider RK stability by only discretising time, i.e. considering Δ a continuum operator. Starting with the heat equation, we recall that on $\mathbb{R}^3$, we can decompose any function as a (continuous) linear combination of the mode functions $\phi(\mathbf{x}) = e^{i\mathbf{k}\cdot\mathbf{x}}$. These are eigenfunctions of the Laplacian, since

$$\Delta\phi(\mathbf{x}) = -\mathbf{k}^2 e^{i\mathbf{k}\cdot\mathbf{x}} = -\mathbf{k}^2\phi(\mathbf{x}). \tag{A.3.58}$$

Hence, the eigenvalues of the heat equation's RHS are of the form $\lambda_{\mathbf{k}}^{\Delta} = -\mathbf{k}^2 \in \mathbb{R}_{\leq 0}$, so that the timestep must satisfy

$$z_{\min}^{\mathrm{Re}} \leq \lambda_{\mathbf{k}}^{\Delta}\Delta t \quad \Leftrightarrow \quad \Delta t \leq \frac{z_{\min}^{\mathrm{Re}}}{\lambda_{\mathbf{k}}^{\Delta}} = -\frac{z_{\min}^{\mathrm{Re}}}{\mathbf{k}^2} > 0. \tag{A.3.59}$$

Given an initial condition containing mode contributions of arbitrarily large $|\mathbf{k}|$, this requires $\Delta t = 0$. If however, we limit our initial conditions to have contributions only up to some fixed maximum $|\mathbf{k}|$, a finite non-zero value of Δt can provide stability across all of them. In essence, due to $z = \lambda\Delta t$, we can use Δt to rescale the relevant portion of the spectrum to lie inside the stability region.

We now move on to the spatial-continuous wave equation, whose eigenvalues are derived as follows:

$$0 = \left(\begin{pmatrix} 0 & 1 \\ \Delta & 0 \end{pmatrix} - \lambda I \right) \begin{pmatrix} \phi \\ \Pi \end{pmatrix} = \begin{pmatrix} -\lambda & 1 \\ \Delta & -\lambda \end{pmatrix} \begin{pmatrix} \phi \\ \Pi \end{pmatrix} = \begin{pmatrix} \Pi - \lambda\phi \\ \Delta\phi - \lambda\Pi \end{pmatrix}. \tag{A.3.60}$$

Combining these two linear equations, we obtain

$$\Delta\phi = \lambda^2\phi, \quad \Pi = \lambda\phi, \tag{A.3.61}$$

implying that the eigenvalues for the wave equation RHS are the square roots of those of the Laplacian, meaning that

$$\lambda_{\mathbf{k}}^{\mathrm{wave}} = \pm\sqrt{\lambda_{\mathbf{k}}^{\Delta}} \pm i|\mathbf{k}|, \quad \mathbf{k} \in \mathbb{R}^3, \tag{A.3.62}$$

with corresponding eigenfunction

$$\phi(\mathbf{x}) = e^{i\mathbf{k}\cdot\mathbf{x}}, \quad \Pi(\mathbf{x}) = \pm i|\mathbf{k}|e^{i\mathbf{k}\cdot\mathbf{x}}. \tag{A.3.63}$$

In particular, this means that the eigenvalues of Since RK1 and RK2 do not contain any nonzero part of the imaginary axis, these schemes are unconditionally unstable. Both RK3 and RK4 contain parts of the imaginary axis, and limit Δt by

$$\Delta t \leq \frac{|z_{\max}^{\mathrm{Im}}|}{|\mathbf{k}|} = \begin{cases} \frac{\sqrt{3}}{|\mathbf{k}|} & \text{for RK3,} \\ \frac{2\sqrt{2}}{|\mathbf{k}|} & \text{for RK4.} \end{cases} \tag{A.3.64}$$

Again, an initial condition containing contributions at arbitrarily large $|\mathbf{k}|$ would require $\Delta t = 0$. For frequency-limited initial data, however, finite non-zero values for Δt exist for which RK$\geq$3 is stable.

Recalling the derivations of the CFL conditions, we note that there, the concept of a "maximum allowed wavenumber" $|\mathbf{k}|$ to make it possible for Δt to be non-zero. On one hand, this has to do with the fact that there, we were working with a discretised grid, which automatically sets an upper limit for the wave number—any mode above the Nyquist frequency aliases onto a mode below it. However, there is another effect at play as well: discretising an operator changes its spectrum.

Hence, we are led to compute the spectrum of discretised versions of the Laplacian, so that we can compare to the continuum case.

(i) The most basic discretisation of the Laplacian is

$$\Delta_1^{\text{disc}} = \frac{1}{\Delta x^2} \sum_{i=1}^{d} (E_i^{\Delta x} - 2I + E_i^{-\Delta x}), \tag{A.3.65}$$

where $E_i^a f(\mathbf{x}) = f(\mathbf{x} + a\mathbf{e}_i)$ and $\mathbf{e}_i$ is the unit vector in the i-th direction. Clearly, $e^{i\mathbf{k}\cdot\mathbf{x}}$ are eigenfunctions of E_i^a and I, and hence also of Δ_1^{disc}. Since any function on $\mathbb{R}^d$ can be written as a (continuous) linear combination of these eigenmodes, they form a complete basis. Concretely, the eigenvalues appear as

$$\begin{aligned} \lambda_{\mathbf{k}}^{\Delta,1} e^{i\mathbf{k}\cdot\mathbf{x}} = \Delta_1^{\text{disc}} e^{i\mathbf{k}\cdot\mathbf{x}} &= \frac{1}{\Delta x^2} \sum_{i=1}^{d} (e^{ik_i\Delta x} - 2 + e^{-ik_i\Delta x}) e^{i\mathbf{k}\cdot\mathbf{x}} \\ &= -\frac{4}{\Delta x^2} \left(\sum_{i=1}^{d} \sin^2\left(\frac{k_i \Delta x}{2}\right) \right) e^{i\mathbf{k}\cdot\mathbf{x}}, \end{aligned} \tag{A.3.66}$$

whence

$$\lambda_{\mathbf{k}}^{\Delta,1} = -\frac{4}{\Delta x^2} \sum_{i=1}^{d} \sin^2\left(\frac{k_i \Delta x}{2}\right) \tag{A.3.67}$$

For small $k_i \Delta x$, we can approximate $\sin(k_i \Delta x/2) \approx k_i \Delta x/2$, whence

$$\lambda_{\mathbf{k}}^{\Delta,1} \approx -\frac{4}{\Delta x^2} \sum_{i=1} \left(\frac{k_i \Delta x}{2}\right)^2 = -\mathbf{k}^2, \tag{A.3.68}$$

reproducing the true spectrum. However, the closer the period of the spatial oscillation gets to the grid spacing, the more the spectral points of the continuous and discretised operators diverge. Moreover, unlike $-\mathbf{k}^2$, which grows more and more negative for larger wave vectors, the discretised spectrum is bounded and periodic in the components of $\mathbf{k}$. This is simply due to aliasing of higher onto lower frequency modes; the relevant portion of spatial frequency space is within the Brillouin zone $\left[-\frac{\pi}{\Delta x}, \frac{\pi}{\Delta x}\right]^d$.

The largest magnitude that $\lambda_{\mathbf{k}}^{\Delta,1}$ can attain is $\frac{4d}{\Delta x^2}$. Putting this together with e.g. RK1 reproduces the heat equation's CFL condition

$$\Delta t \leq \min_{\mathbf{k}} \left| \frac{z_{\min}^{\text{Re}}}{\lambda_{\mathbf{k}}^{\Delta,1}} \right| = \frac{2}{\frac{4d}{\Delta x^2}} = \frac{\Delta x^2}{2d} \tag{A.3.69}$$

that we have derived before. However, this more general approach of deriving stability conditions—split into determining stability regions of the time integrator and the calculating the eigenvalues of the right-hand side operator—is a lot more versatile and insightful; we could easily replace the time integrator by RK2 or higher, and all that would change is an already computed value of $z_{\min}^{\text{Re}}$.

(ii) Alternative to changing the time integrator, we can also pick a different discretisation of the right-hand side operator. Let us take a look at a more precise stencil, the Laplacian built from the higher-order stencil equation (A.1.3):

$$\Delta_2^{\text{disc}} = \frac{1}{12\Delta x^2}\sum_{i=1}^{d}(-E_i^{2\Delta x} + 16E_i^{\Delta x} - 30I + 16E_i^{-\Delta x} - E_i^{-2\Delta x}), \tag{A.3.70}$$

Inserting the eigenmodes $e^{i\mathbf{k}\cdot\mathbf{x}}$, we get

$$\begin{aligned}\lambda_{\mathbf{k}}^{\Delta,2} e^{i\mathbf{k}\cdot\mathbf{x}} = \Delta_2^{\text{disc}} e^{i\mathbf{k}\cdot\mathbf{x}} &= \frac{1}{12\Delta x^2}\sum_{i=1}^{d}(-e^{i2k_i\Delta x} + 16e^{ik_i\Delta x} - 30 + 16e^{-ik_i\Delta x} - e^{-i2k_i\Delta x})e^{i\mathbf{k}\cdot\mathbf{x}} \\ &= -\frac{1}{3\Delta x^2}\sum_{i=1}^{d}\left(16\sin^2\left(\frac{k_i\Delta x}{2}\right) - \sin^2(k_i\Delta x)\right),\end{aligned} \tag{A.3.71}$$

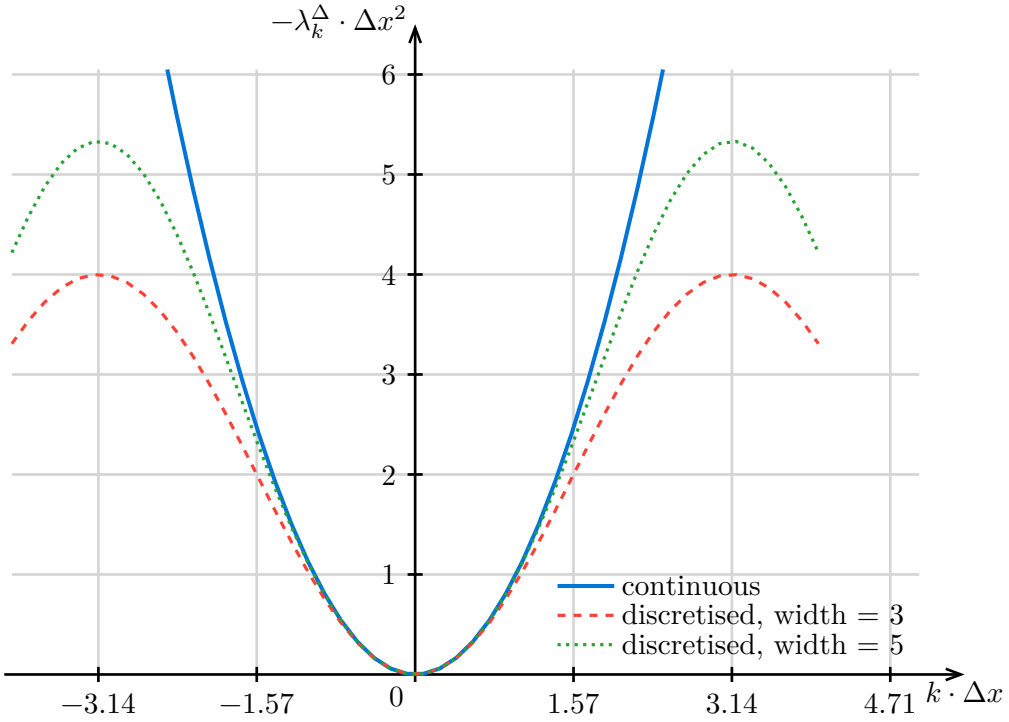
so that

$$\lambda_{\mathbf{k}}^{\Delta,2} = -\frac{1}{3\Delta x^2}\sum_{i=1}^{d}\left(16\sin^2\left(\frac{k_i\Delta x}{2}\right) - \sin^2(k_i\Delta x)\right) \tag{A.3.72}$$

Again, assuming $k_i\Delta x$ to be small, we can approximate

$$\lambda_{\mathbf{k}}^{\Delta,2} \approx -\frac{1}{3\Delta x^2}\sum_{i=1}^{d}\left(16\left(\frac{k_i\Delta x}{2}\right)^2 - (k_i\Delta x)^2\right) = -\frac{1}{\Delta x^2}\sum_{i=1}^{d}(k_i\Delta x)^2 = -\mathbf{k}^2. \tag{A.3.73}$$

This confirms that we again reproduce the continuum spectrum for small enough $|\mathbf{k}|$, though now with higher accuracy (as we will see in a plot below).



(A.3.74)

In this figure, the one-dimensional comparison between the continuous eigenvalues and the two discretised spectra is shown. We can see that around $k = 0$, both discretised spectra match the graph of the continuum spectrum well, but diverge moving away from $k = 0$. The divergence is much faster with the lower-order stencil; the 5-wide stencil resolves higher-frequency modes better. This, however, comes at the cost of having a larger maximum value for $\lambda_{\mathbf{k}}^{\Delta,2}$ than for $\lambda_{\mathbf{k}}^{\Delta,1}$, which reduces the maximum admissible timestep due to equation (A.3.69).

A.3.3.2 Non-Linear Equations

While an in-depth analysis of the numerical stability of non-linear equations of the form

$$\partial_t \phi = F(\phi), \quad \phi : \mathbb{R}^{1,d} \to \mathbb{R}^N, \tag{A.3.75}$$

should be done on a case-by-case basis, we can make some preliminary observations of this general case. Since a general functional F cannot be assigned a spectrum, the eigenvalue-based approach for the stability analysis of the linear case breaks down. We can, however, linearise the right-hand side around some background state $\phi_0 : \mathbb{R}^d \to \mathbb{R}^N$, as

$$F(\phi) = F(\phi_0) + J(\phi_0)(\phi - \phi_0) + \mathcal{O}(\|\phi - \phi_0\|^2), \tag{A.3.76}$$

with

$$J_{ij}(\phi) = \frac{\delta F_i}{\delta \phi_j}|_\phi \tag{A.3.77}$$

This allows us to approximate the fully non-linear equation around ϕ_0 into the linear equation

$$\begin{aligned} \partial_t \phi &= F(\phi_0) + J(\phi_0)(\phi - \phi_0) + \mathcal{O}(|\phi - \phi_0|^2) \\ &= J(\phi_0)\phi + S_0 + \mathcal{O}(\|\phi - \phi_0\|^2) \end{aligned} \tag{A.3.78}$$

where $S_0 = F(\phi_0) - J(\phi_0)\phi_0$ is the source term encoding the non-linear dynamics near the background ϕ_0. A general solution to the linearised equation

$$\partial_t \phi = J(\phi_0)\phi + S_0 \tag{A.3.79}$$

Has the form

$$\phi(t, x) = \phi_h(t, x) + \phi_s(x), \tag{A.3.80}$$

where ϕ_h solves the source-free equation

$$\partial_t \phi_h = J(\phi_0)\phi_h, \tag{A.3.81}$$

and ϕ_s solves the elliptic problem

$$J(\phi_0)\phi_s + S_0 = 0. \tag{A.3.82}$$

Since ϕ only gains time-dependence through ϕ_h, any stability considerations (near ϕ_0) only concern ϕ_h. Concretely, the relevant eigenvalues of $J(\phi_0)$ need to lie inside the stability region of the RK scheme being used. To be as accurate as possible, for a time-iteration scheme

$$\phi^{n+1} = \phi^n + \sum_{i=1}^{n} b_i k_i, \tag{A.3.83}$$

one typically takes as background the initial state, that is, $\phi_0 = \phi^n$. In this case, the linearised approximation typically remains valid, since ϕ^{n+1} remains close to ϕ^n—their difference is $\mathcal{O}(\Delta t)$.

Let us consider a few concrete PDEs to see how they linearise, and how the spectrum of $J(\phi_0)$ behaves.

(i) As a first example, we examine the non-linear heat equation

$$\partial_t \phi = \Delta \phi + \gamma \phi |\phi|^p \tag{A.3.84}$$

for some constants $\gamma \in \mathbb{R}$ and $p > 1$. We pick an arbitrary field configuration ϕ_0 to linearise around, writing $\phi = \phi_0 + \delta\phi \Leftrightarrow \delta\phi = \phi - \phi_0$. The simplest way to find the relevant quantities associated with

$$F(\phi) = \Delta\phi + \gamma\phi|\phi|^p \tag{A.3.85}$$

is to insert and expand directly:

$$\begin{aligned} F(\phi) = F(\phi_0 + \delta\phi) &= \Delta\phi_0 + \Delta\delta\phi + \gamma(\phi_0 + \delta\phi)|\phi_0 + \delta\phi|^p \\ &= \Delta\phi_0 + \gamma\phi_0|\phi_0|^p + \Delta\delta\phi + \gamma(p+1)|\phi_0|^p\delta\phi + \mathcal{O}(\|\delta\phi\|^2). \end{aligned} \tag{A.3.86}$$

Inserting back into the original equation and dropping higher-order terms in $\delta\phi$, we obtain

$$\begin{aligned} \partial_t\phi = \partial_t\delta\phi &= (\Delta + \gamma(p+1)|\phi_0|^p)\delta\phi + \Delta\phi_0 + \gamma\phi_0|\phi_0|^p \\ &= (\Delta + \gamma(p+1)|\phi_0|^p)\phi + S_0 \end{aligned} \tag{A.3.87}$$

where

$$S_0 = \Delta\phi_0 + \gamma\phi_0|\phi_0|^p - \Delta\phi_0 - \gamma(p+1)\phi_0|\phi_0|^p = -\gamma p\phi_0|\phi_0|^p \tag{A.3.88}$$

The relevant operator for stability analysis is thus

$$J(\phi_0) = \Delta + \gamma(p+1)|\phi_0|^p, \tag{A.3.89}$$

where the second term is to be understood as acting by pointwise multiplication of functions.

For an arbitrary background state ϕ_0, it is generally not possible to determine the full spectrum of $J(\phi_0)$. Only in certain cases—e.g. when ϕ_0 is constant, or when it has certain symmetries—we can determine $\sigma(J(\phi_0))$ analytically. Nonetheless, we can make some qualitative observations:

- If the second term is small, the spectrum of $J(\phi_0)$ is approximated reasonably well by that of the Laplacian Δ.
- The dependence of the operator on γ is continuous, meaning that the eigenvalues of $J(\phi_0)$ follow a continuous path from the Laplacian spectrum at $\gamma = 0$ to their value for $\gamma \neq 0$.
- The multiplication operator $\gamma(p+1)|\phi_0|^p$ is a relatively compact perturbation of the Laplacian if ϕ_0 vanishes as $|x| \to \infty$. By Weyl's theorem, compact perturbations do not change the essential spectrum; hence we retain the negative real axis as the essential spectrum. However, discrete eigenvalues can cross into the right half-plane; this is necessary, since depending on the sign of γ, we have finite-time blowup, meaning that we have *physical* instabilities (that any RK implementation should respect).

(ii) Moving to second-order in time equations, we may consider the non-linear wave equation

$$\partial_t^2\phi = \Delta\phi + \gamma|\nabla\phi|^2\phi. \tag{A.3.90}$$

To be able to address it using RK, we have to cast it into the first-order form

$$\begin{aligned} \partial_t\phi &= \pi, \\ \partial_t\pi &= \Delta\phi + \gamma|\nabla\phi|^2\phi \end{aligned} \tag{A.3.91}$$

To linearise this around some fixed background state (ϕ_0, π_0), we write $\phi = \phi_0 + \delta\phi$, $\pi = \pi_0 + \delta\pi$, so that the linearised equation reads

$$\begin{aligned} \partial_t\pi &= \Delta(\phi_0 + \delta\phi) + \gamma|\nabla\phi_0 + \nabla\delta\phi|^2(\phi_0 + \delta\phi) \\ &= \Delta\phi_0 + \Delta\delta\phi + \gamma|\nabla\phi_0|^2\delta\phi + 2\gamma\phi_0(\nabla\phi_0)\cdot\nabla\delta\phi + \gamma|\nabla\phi_0|^2\phi_0 \\ &= \Delta\phi + \gamma|\nabla\phi_0|^2\phi + 2\gamma\phi_0\nabla\phi_0\cdot\nabla\phi - 2\gamma|\nabla\phi_0|^2\phi_0. \end{aligned} \tag{A.3.92}$$

Since the equation $\partial_t\phi = \pi$ is linear already, we do not need to expand it further. Hence, the full equation system linearises as

$$\partial_t \begin{pmatrix} \phi \\ \pi \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \Delta + \gamma(|\nabla\phi_0|^2 + 2\phi_0\nabla\phi_0 \cdot \nabla) & 0 \end{pmatrix} \begin{pmatrix} \phi \\ \pi \end{pmatrix} + S_0, \tag{A.3.93}$$

where

$$S_0 = \begin{pmatrix} 0 \\ -2\gamma|\nabla\phi_0|^2\phi_0 \end{pmatrix}. \tag{A.3.94}$$

In this case, the linearised operator contains—besides the second-order Laplacian and the zero-th order pointwise-multiplication terms that we have seen before—a first-order derivative term. We recall that for an operator of the form

$$T = \begin{pmatrix} 0 & 1 \\ D & 0 \end{pmatrix}, \tag{A.3.95}$$

the spectra are related by $\sigma(T) = \sqrt{\sigma(D)} \cup -\sqrt{\sigma(D)}$. In the case above where

$$D = \Delta + \gamma(|\nabla\phi_0|^2 + 2\phi_0\nabla\phi_0 \cdot \nabla), \tag{A.3.96}$$

we again have a continuous dependence of the operator on γ; hence, for γ small enough, we have a perturbed Laplacian spectrum for D, which turns into the standard perturbed $i\mathbb{R}$ spectrum for T. Notably, due to the gradient term, D is no longer necessarily self-adjoint, and it hence has eigenvalues with nonzero imaginary part. Since for $\lambda^2 \in \sigma(D)$, both $\pm\lambda \in \sigma(T)$, this implies that we must have unstable modes, i.e. modes with $\operatorname{Re}\lambda > 0$. Because the magnitude of the gradient term grows with the frequency of the mode, this instability predominantly affects high-frequency modes.

There are, however, backgrounds ϕ_0 (and hence, initial conditions for the evolution equation), around which the evolution is stable. For this to be the case, D must be negative definite, that is—for all test functions ψ—we must have

$$\begin{aligned} 0 \geq \langle\psi, D\psi\rangle &= \int_{\mathbb{R}^d} \left[\bar{\psi}\Delta\psi + \gamma\big(|\nabla\phi_0|^2|\psi|^2 + 2\phi_0\nabla\phi_0 \cdot (\nabla\psi)\bar{\psi}\big)\right] \\ &= \int_{\mathbb{R}^d} [-|\nabla\psi|^2 + \gamma(|\nabla\phi_0|^2 - \nabla\cdot(\phi_0\nabla\phi_0))|\psi|^2]. \end{aligned} \tag{A.3.97}$$

For this to hold for any test function ψ, we must have

$$\begin{aligned} 0 &\geq \gamma(|\nabla\phi_0|^2 - \nabla\cdot(\phi_0\nabla\phi_0)) \\ &= -\gamma\phi_0\Delta\phi_0. \end{aligned} \tag{A.3.98}$$

A.4 Boundaries and Grid Stability

A.4.1 Sommerfeld Radiation Boundaries

In this section, we derive *Sommerfeld radiation boundary conditions* for the wave equation on flat Minkowski space,

$$\partial_t^2\phi = c^2\Delta\phi. \tag{A.4.1}$$

Although these boundary conditions cannot be applied directly to the more complex BSSN system, it serves as a useful entry point for more advanced radiation boundaries that "absorb" all outgoing radiation, and produce no incoming radiation.

To distinguish what is "outgoing" and "incoming", we assume our source/region of interest to be located near the origin, and the boundaries to be far enough away from the source that we can reasonably approximate the wavefront of any radiation to be a sphere centered around the

origin. Whenever origin-centered spheres come up, a reasonable choice is spherical coordinates (t, r, θ, φ). Since we assume the wavefronts to be spherical for large enough r, the field ϕ becomes independent of θ and φ, so

$$\phi = \phi(t, r). \tag{A.4.2}$$

This turns the wave equation into

$$\partial_t^2 \phi = \frac{c^2}{r^2} \partial_r (r^2 \partial_r \phi). \tag{A.4.3}$$

We rewrite this using $\phi = u/r$, yielding

$$\frac{1}{r} \partial_t^2 u = \frac{c^2}{r^2} \partial_r (r \partial_r u - u) = \frac{c^2}{r} \partial_r^2 u, \tag{A.4.4}$$

or equivalently,

$$\partial_t^2 u = c^2 \partial_r^2 u, \tag{A.4.5}$$

meaning that $u(t, r)$ satisfies a one-dimensional wave equation. Its general solution is a superposition of in- and outgoing waves,

$$u(t, r) = f_+(r - ct) + f_-(r + ct), \tag{A.4.6}$$

for two functions $f_\pm : \mathbb{R} \to \mathbb{R}$. Factoring the one-dimensional wave operator as

$$\Box = \partial_t^2 - c^2 \partial_r^2 = (\partial_t - c\partial_r)(\partial_t + c\partial_r) =: X_- X_+, \tag{A.4.7}$$

we can see that $X_+ = \partial_t + c\partial_r$ annihilates the outgoing wave f_+, while $X_- = \partial_t - c\partial_r$ annihilates the incoming wave f_-.

Let us consider how X_+ acts on the incoming wave f_-. We get

$$X_+ f_-(r + ct) = c f'(r + ct) + c f'(r + ct) = 2c f'(r + ct), \tag{A.4.8}$$

whence

$$X_+ u = \underbrace{X_+ f_+}_{=0} + X_+ f_- = 2c f_-'(r + ct). \tag{A.4.9}$$

This means that if we require $X_+ u = 0$, the only "incoming" component we have is an irrelevant constant, $f_- = \text{const.}$. As we have just seen, the wave operator factors into $\Box = X_- X_+$, meaning that if $X_+ u = 0$, then also $X_- X_+ u = 0$. Thus, requiring $X_+ u = 0$ achieves our two goals; u is both a solution to the wave equation *and* it has no incoming components.

We can hence, at our boundary away from the origin, require u to satisfy as boundary condition the equation

$$X_+ u = 0 \quad \Leftrightarrow \quad \partial_t u + c\partial_r u = 0. \tag{A.4.10}$$

By construction, this is compatible with the wave equation, and further ensures that no incoming waves enter the domain through the boundary—exactly what we want. We are left to translate this into a boundary condition for our original field $\phi = u/r$. This is done simply by inserting $u = r\phi$ into the above, yielding

$$\begin{aligned} X_+(r\phi) = 0 \quad &\Leftrightarrow \quad \partial_t(r\phi) + c\partial_r(r\phi) = 0 \\ &\Leftrightarrow \quad r\partial_t \phi + cr\partial_r \phi + c\phi = 0. \end{aligned} \tag{A.4.11}$$

After dividing both sides by r, we finally arrive at the *Sommerfeld radiation boundary condition*

$$\partial_t \phi + c\partial_r \phi + \frac{c}{r}\phi = 0. \tag{A.4.12}$$

Having now derived this boundary condition, let us consider some more practical aspects of how to implement it in a numerical simulation. In most numerical simulations, one breaks down the second-order in time wave equation into two coupled first-order in time equations for ϕ and its momentum $\pi = \partial_t \phi$, which explicitly read

$$\begin{aligned} \partial_t \phi &= \pi, \\ \partial_t \pi &= c^2 \Delta \phi. \end{aligned} \tag{A.4.13}$$

We may turn the Sommerfeld radiation boundary condition (A.4.12) into

$$\pi + c\partial_r \phi + \frac{c}{r}\phi = 0 \quad \Leftrightarrow \quad \partial_r \phi = -\frac{\pi}{c} - \frac{\phi}{r}. \tag{A.4.14}$$

Taking the first time derivative, we can obtain an analogous equation for π:

$$\partial_r \pi = \partial_t \partial_r \phi = -\frac{1}{c}\partial_t \pi - \frac{1}{r}\partial_t \phi = -c\Delta\phi - \frac{\pi}{r} \tag{A.4.15}$$

To summarise, Sommerfeld boundary conditions prescribe

$$\begin{aligned} \partial_r \phi &= -\frac{\pi}{c} - \frac{\phi}{r}, \\ \partial_r \pi &= -c\Delta\phi - \frac{\pi}{r}. \end{aligned} \tag{A.4.16}$$

These two equations specify a first spatial derviative of both ϕ and π which—in some domains—can be used to isolate the normal derivative of ϕ needed to fill exterior boundary ghost cells.

The "some" qualifier in front of the term "domains" above is unfortunately rather important. If one is running the simulation with a spherical boundary and works in spherial coordinates, then ∂_r is the normal direction to the boundary, and everything works out fine. However, most simulations are carried out on a Cartesian grid, where the boundary is a box. This has two consequences: computing ∂_r is not as straightforward, and the normal derivative of a boundary is one of the coordinate derviatives ∂_x, ∂_y or ∂_z which we need to solve for. As we will see below, solving this system can become a problem once the boundary conditions are discretised, and can lead one to be forced to take certain compromises.

So, let us proceed with an attempt to discretise the boundary conditions for a cartesian grid, and see where it fails. Then, we will find ways to remedy the failure, at the expense of losing some of its properties. Firstly, we note that the radial basis vector ∂_r is related to the cartesian ones by

$$\partial_r = \frac{x}{r}\partial_x + \frac{y}{r}\partial_y + \frac{z}{r}\partial_z, \tag{A.4.17}$$

where of course, $r(x, y, z) = \sqrt{x^2 + y^2 + z^2}$. Without loss of generality, let us consider a face of the box-shaped domain's exterior boundary whose normal is $\pm\partial_x$—that is, a portion of the boundary parallel to the yz-plane. To assert its boundary conditions, we need to solve for the normal derivative $\partial_x \phi$. Inserting the expanded expression for ∂_r into equation (A.4.14) allows us to do this as

$$\begin{aligned} &\frac{x}{r}\partial_x \phi + \frac{y}{r}\partial_y \phi + \frac{z}{r}\partial_z \phi = -\frac{\pi}{c} - \frac{\phi}{r} \\ \Leftrightarrow \qquad &\partial_x \phi = -\frac{y}{x}\partial_y \phi - \frac{z}{x}\partial_z \phi - \frac{r}{cx}\pi - \frac{1}{x}\phi. \end{aligned} \tag{A.4.18}$$

The corresponding equation for π is

$$\begin{aligned}\partial_x\phi &= -\frac{y}{x}\partial_y\phi - \frac{z}{x}\partial_z\phi - \frac{r}{cx}\pi - \frac{1}{x}\phi,\\ \partial_x\pi &= -\frac{y}{x}\partial_y\pi - \frac{z}{x}\partial_z\pi - \frac{cr}{x}\Delta\phi - \frac{1}{x}\pi.\end{aligned} \tag{A.4.19}$$

In this form, we can compute the normal derivatives at the boundary in the continuum formulation. However, it is worth noting that when replacing the boundary-tangent derivatives ∂_y and ∂_z with finite difference stencils, some numerical error is introduced, which causes the boundary conditions to become satisfied less precisely the larger the angle between ∂_r and ∂_x.

As an additional remark, we should note that the above form assumes $x \neq 0$ at the boundary (as well as $y \neq 0$ and $z \neq 0$ at the respective other boundaries). In practice, this is no restriction, since we have built up this entire discussion on the assumption that the boundaries are far enough from the origin that outgoing waves can be approximated as being emanated from a point source at the origin.

Let us consider how we would discretise this to $\mathcal{O}(\Delta x^2)$ for a cell-centered grid with a halo region of width 1, where the ghost cell index is -1, the first interior cell is at 0, and the boundary is located at $-\frac{1}{2}$. The left-hand sides are straightforwardly discretised to $\mathcal{O}(\Delta x^2)$, since the finite difference between -1 and 0 is naturally located at $-\frac{1}{2}$ to order $\mathcal{O}(\Delta x^2)$:

$$\begin{aligned}\partial_x\phi|_{-\frac{1}{2}} &= \varepsilon\frac{\phi_0 - \phi_{-1}}{\Delta x} + \mathcal{O}(\Delta x^2),\\ \partial_x\pi|_{-\frac{1}{2}} &= \varepsilon\frac{\pi_0 - \pi_{-1}}{\Delta x} + \mathcal{O}(\Delta x^2).\end{aligned} \tag{A.4.20}$$

Here, the sign ε is -1 for a lower boundary and $+1$ for an upper boundary, since for a lower boundary we need to correct for the fact that ϕ_{-1} lies to the left of ϕ_0. On the right-hand sides, we have an issue: we only know $\phi|_0$ and $\pi|_0$, which approximate $\phi|_{-\frac{1}{2}}$ and $\pi|_{-\frac{1}{2}}$ to order $\mathcal{O}(\Delta x)$. Though solving for (φ_{-1}, π_{-1}) multiplies this by Δx to achieve the desired $\mathcal{O}(\Delta x^2)$ error order, we can do better. Higher accuracy is achieved by using interpolation,

$$\begin{aligned}\phi_{-\frac{1}{2}} &= \frac{\phi_{-1} + \phi_0}{2} + \mathcal{O}(\Delta x^2),\\ \pi_{-\frac{1}{2}} &= \frac{\pi_{-1} + \pi_0}{2} + \mathcal{O}(\Delta x^2).\end{aligned} \tag{A.4.21}$$

This turns the discretised boundary conditions into an implicit (but linear) system in (ϕ_{-1}, π_{-1}), which we will have to solve later.

Another term that we have not yet adressed is the Laplacian. Since it consists of second derivatives, it is much harder to colocate on the boundary at $-\frac{1}{2}$—luckily, we do not have to. This is because, when multiplying through by the Δx factor from the first derivative of the left-hand side, it obtains an additional factor of Δx. This increases the error order by one, so it is sufficient to have

$$\Delta\phi|_{-\frac{1}{2}} = \Delta\phi|_0 + \mathcal{O}(\Delta x). \tag{A.4.22}$$

The $\Delta\phi|_0$ term, we separate into a normal and a transverse part,

$$\Delta\phi|_0 = \frac{\phi_{-1} - 2\phi_0 + \phi_1}{\Delta x^2} + \Delta_\perp\phi|_0 + \mathcal{O}(\Delta x^2), \tag{A.4.23}$$

with $\Delta_\perp\phi = \partial_y^2\phi + \partial_z^2\phi$ the transverse Laplacian that we can evaluate entirely from interior cells, to order $\mathcal{O}(\Delta x^2)$. This makes it so that

$$\Delta\phi|_{-\frac{1}{2}} = \frac{\phi_{-1} - 2\phi_0 + \phi_1}{\Delta x^2} + \Delta_\perp \phi|_0 + \mathcal{O}(\Delta x). \tag{A.4.24}$$

The same is true for the transverse first derivative terms: e.g.

$$\partial_y \phi|_{-\frac{1}{2}} = \partial_y \phi|_0 + \mathcal{O}(\Delta x), \tag{A.4.25}$$

so that we can simply calculate the derivative using a stencil at 0.

Let us now insert everything back into the continuous equation while keeping track of error orders:

$$\begin{aligned}
\varepsilon\frac{\phi_0 - \phi_{-1}}{\Delta x} &= -\frac{y}{x}\partial_y\phi|_0 - \frac{z}{x}\partial_z\phi|_0 - \frac{r}{x}\frac{\pi_{-1} + \pi_0}{2c} - \frac{\phi_{-1} + \phi_0}{2x} + \mathcal{O}(\Delta x^2) \\
\varepsilon\frac{\pi_0 - \pi_{-1}}{\Delta x} &= -\frac{y}{x}\partial_y\pi|_0 - \frac{z}{x}\partial_z\pi|_0 - c\frac{r}{x}\frac{\phi_{-1} - 2\phi_0 + \phi_1}{\Delta x^2} - c\frac{r}{x}\Delta_\perp\phi|_0 - \frac{\pi_{-1} + \pi_0}{2x} + \mathcal{O}(\Delta x).
\end{aligned} \tag{A.4.26}$$

Multiplying through by $-\varepsilon\Delta x$, we get

$$\begin{aligned}
\phi_{-1} - \phi_0 &= \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\phi + \frac{z}{x}\partial_z\phi\Big)|_0 + \varepsilon\frac{r}{x}\frac{\Delta x}{2c}(\pi_{-1} + \pi_0) + \varepsilon\frac{\Delta x}{2x}(\phi_{-1} + \phi_0) + \mathcal{O}(\Delta x^3) \\
\pi_{-1} - \pi_0 &= \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\pi + \frac{z}{x}\partial_z\pi\Big)|_0 + \varepsilon\frac{r}{x}\frac{c}{\Delta x}(\phi_{-1} - 2\phi_0 + \phi_1) + \varepsilon\frac{r}{x}c\Delta x\Delta_\perp\phi|_0 + \varepsilon\frac{\Delta x}{2x}(\pi_{-1} + \pi_0) + \mathcal{O}(\Delta x^2).
\end{aligned} \tag{A.4.27}$$

As stated previously, this is a linear system in the unknowns (ϕ_{-1}, π_{-1}). To solve it, we first have to separate evaluations at -1 from those at 0 and 1, yielding

$$\begin{aligned}
&\Big(1 - \varepsilon\frac{\Delta x}{2x}\Big)\phi_{-1} - \varepsilon\frac{r}{x}\frac{\Delta x}{2c}\pi_{-1} \\
&= \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\phi + \frac{z}{x}\partial_z\phi\Big)|_0 + \varepsilon\frac{r}{x}\frac{\Delta x}{2c}\pi_0 + \Big(1 + \varepsilon\frac{\Delta x}{2x}\Big)\phi_0
\end{aligned} \tag{A.4.28}$$

$$\begin{aligned}
&\Big(1 - \varepsilon\frac{\Delta x}{2x}\Big)\pi_{-1} - \varepsilon\frac{r}{x}\frac{c}{\Delta x}\phi_{-1} \\
&= \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\pi + \frac{z}{x}\partial_z\pi\Big)|_0 - \varepsilon\frac{r}{x}\frac{2c}{\Delta x}\Big(\phi_0 - \frac{1}{2}\phi_1\Big) + \varepsilon\frac{r}{x}c\Delta x\Delta_\perp\phi|_0 + \Big(1 + \varepsilon\frac{\Delta x}{2x}\Big)\pi_0
\end{aligned}$$

To have a less cumbersome time in the following, we introduce the substitutions

$$\alpha = \frac{\Delta x}{2x}, \quad \beta = \frac{\Delta x}{2c}, \quad \gamma = \frac{r}{x}, \tag{A.4.29}$$

which collapses the above down to

$$\begin{aligned}
&(1 - \varepsilon\alpha)\phi_{-1} - \varepsilon\beta\gamma\pi_{-1} \\
&\quad = (1 + \varepsilon\alpha)\phi_0 + \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\phi + \frac{z}{x}\partial_z\phi\Big)|_0 + \varepsilon\beta\gamma\pi_0,
\end{aligned} \tag{A.4.30}$$

$$\begin{aligned}
&-\frac{\varepsilon}{2}\beta^{-1}\gamma\phi_{-1} + (1 - \varepsilon\alpha)\pi_{-1} \\
&\quad = (1 + \varepsilon\alpha)\pi_0 + \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\pi + \frac{z}{x}\partial_z\pi\Big)|_0 - \varepsilon\beta^{-1}\gamma\Big(\phi_0 - \frac{1}{2}\phi_1\Big) + \frac{\varepsilon}{2}\beta^{-1}\gamma(\Delta x)^2\Delta_\perp\phi|_0
\end{aligned}$$

Summarising the right-hand sides into the vector $\mathbf{b} = \big(b_\phi, b_\pi\big)$ with

$$b_\phi = (1+\varepsilon\alpha)\phi_0 + \varepsilon\Delta x\left(\frac{y}{x}\partial_y\phi + \frac{z}{x}\partial_z\phi\right)|_0 + \varepsilon\beta\gamma\pi_0, \tag{A.4.31}$$
$$b_\pi = (1+\varepsilon\alpha)\pi_0 + \varepsilon\Delta x\left(\frac{y}{x}\partial_y\pi + \frac{z}{x}\partial_z\pi\right)|_0 - \varepsilon\beta^{-1}\gamma\left(\phi_0 - \frac{1}{2}\phi_1\right) + \frac{\varepsilon}{2}\beta^{-1}\gamma(\Delta x)^2\Delta_\perp\phi|_0$$

as well as introducing the matrix

$$\mathbf{A} = \begin{pmatrix} 1-\varepsilon\alpha & -\varepsilon\beta\gamma \\ -\frac{\varepsilon}{2}\beta^{-1}\gamma & 1-\varepsilon\alpha \end{pmatrix}, \tag{A.4.32}$$

we can write the system (A.4.28) as

$$\mathbf{A}\mathbf{x} = \mathbf{b} + \mathcal{O}(\Delta x^2) \quad \Leftrightarrow \quad \mathbf{x} = \mathbf{A}^{-1}\mathbf{b} + \mathcal{O}(\Delta x^2), \tag{A.4.33}$$

with $\mathbf{x} = (\phi_{-1}, \pi_{-1})$ and

$$\mathbf{A}^{-1} = \frac{1}{(1-\varepsilon\alpha)^2 - \frac{1}{2}\gamma^2}\begin{pmatrix} 1-\varepsilon\alpha & \varepsilon\beta\gamma \\ \frac{\varepsilon}{2}\beta^{-1}\gamma & 1-\varepsilon\alpha \end{pmatrix}. \tag{A.4.34}$$

We have now completed the discretisation of the Sommerfeld boundary conditions. However, there is a subtle issue with this implementation; at points where $y^2 + z^2 = x^2$, we have

$$\gamma = \frac{r}{x} = \frac{\sqrt{x^2+y^2+z^2}}{x} = \frac{\sqrt{2x^2}}{x} = \sqrt{2}, \tag{A.4.35}$$

so that

$$\det\mathbf{A} = (1-\varepsilon\alpha)^2 - \frac{1}{2}\gamma^2 = (1-\varepsilon\alpha)^2 - 1 = -2\varepsilon\alpha + \alpha^2 = \mathcal{O}(\Delta x) \tag{A.4.36}$$

Even though this is nonzero, Δx is small, so that the inversion of $\mathbf{A}$ produces large factors in the update $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. This makes it unstable near points where $x^2 = y^2 + z^2$, that is, at points where we have an angle of 45° between the normal and ∂_r. Since such angles are unavoidable in a box, there will always be an instability.

Although this is somewhat disheartening to learn, there are two ways in which we can fix this. Let us go over both of them.

(i) The first approach requires us to analyse our steps in deriving the previous, failed result. The matrix $\mathbf{A}$ which we need to invert is shaped largely by the fact that we have introduced an additional dependency on ϕ_{-1} and π_{-1} in the right-hand side of the original discretised equations (A.4.26). This came about because we decided to use interpolation, which is more accurate than approximating $\phi_{-\frac{1}{2}}$ by ϕ_0. However, since $\phi_{-\frac{1}{2}} = \phi_0 + \mathcal{O}(\Delta x)$, the subsequent multiplication by Δx from the divisor of the left-hand side finite difference is sufficient to push the error order to Δx^2. We can hence drop the interpolation, which leads us to the discretisation

$$\varepsilon\frac{\phi_0 - \phi_{-1}}{\Delta x} = -\frac{y}{x}\partial_y\phi|_0 - \frac{z}{x}\partial_z\phi|_0 - \frac{r}{x}\frac{1}{c}\pi_0 - \frac{1}{x}\phi_0 + \mathcal{O}(\Delta x) \tag{A.4.37}$$
$$\varepsilon\frac{\pi_0 - \pi_{-1}}{\Delta x} = -\frac{y}{x}\partial_y\pi|_0 - \frac{z}{x}\partial_z\pi|_0 - c\frac{r}{x}\frac{\phi_{-1} - 2\phi_0 + \phi_1}{\Delta x^2} - c\frac{r}{x}\Delta_\perp\phi|_0 - \frac{1}{x}\pi_0 + \mathcal{O}(\Delta x).$$

Multiplying through by $-\varepsilon\Delta x$ and dropping error orders, we arrive at

$$
\begin{aligned}
&\phi_{-1} - \phi_0 \\
&= \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\phi + \frac{z}{x}\partial_z\phi\Big)|_0 + 2\varepsilon\frac{r}{x}\frac{\Delta x}{2c}\pi_0 + 2\varepsilon\frac{\Delta x}{2x}\phi_0 + \mathcal{O}(\Delta x^2) \\
&= \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\phi + \frac{z}{x}\partial_z\phi\Big)|_0 + 2\varepsilon\gamma\beta\pi_0 + 2\varepsilon\alpha\phi_0
\end{aligned}
\tag{A.4.38}
$$

$$
\begin{aligned}
&\pi_{-1} - \pi_0 \\
&= \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\pi + \frac{z}{x}\partial_z\pi\Big)|_0 + \varepsilon\frac{r}{x}\frac{c}{\Delta x}\phi_{-1} - \varepsilon\frac{r}{x}\frac{2c}{\Delta x}\Big(\phi_0 - \frac{1}{2}\phi_1\Big) + \varepsilon\frac{r}{x}\frac{c}{\Delta x}(\Delta x^2)\Delta_\perp\phi|_0 + 2\varepsilon\frac{\Delta x}{2x}\pi_0 \\
&= \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\pi + \frac{z}{x}\partial_z\pi\Big)|_0 + \frac{\varepsilon}{2}\gamma\beta^{-1}\phi_{-1} - \varepsilon\gamma\beta^{-1}\Big(\phi_0 - \frac{1}{2}\phi_1\Big) + \frac{\varepsilon}{2}\gamma\beta^{-1}(\Delta x^2)\Delta_\perp\phi|_0 + 2\varepsilon\alpha\pi_0
\end{aligned}
$$

Again bringing all evaluations at -1 to the left-hand side, we get

$$
\begin{aligned}
\phi_{-1} &= (1+2\varepsilon\alpha)\phi_0 + \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\phi + \frac{z}{x}\partial_z\phi\Big)|_0 + 2\varepsilon\gamma\beta\pi_0, \qquad (A.4.39)\\
\pi_{-1} - \frac{\varepsilon}{2}\gamma\beta^{-1}\phi_{-1} &= (1+2\varepsilon\alpha)\pi_0 + \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\pi + \frac{z}{x}\partial_z\pi\Big)|_0 - \varepsilon\gamma\beta^{-1}\Big(\phi_0 - \frac{1}{2}\phi_1\Big) + \frac{\varepsilon}{2}\gamma\beta^{-1}(\Delta x^2)\Delta_\perp\phi|_0
\end{aligned}
$$

Although this is again a system of the form $\mathbf{Ax} = \mathbf{b}$, here,

$$
\mathbf{A} = \begin{pmatrix} 1 & 0 \\ -\frac{\varepsilon}{2}\gamma\beta^{-1} & 1 \end{pmatrix} \tag{A.4.40}
$$

has determinant 1 and can stably be inverted. Further, since the wave equation only contains spatial derivatives of ϕ and none of π, it is usually even sufficient to only solve the first, which is in explicit form already—this even enables us to compute ϕ_{-1} first, and then evaluate

$$
\begin{aligned}
\pi_{-1} = \frac{\varepsilon}{2}\gamma\beta^{-1}\phi_{-1} + (1+2\varepsilon\alpha)\pi_0 + \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\pi + \frac{z}{x}\partial_z\pi\Big)|_0 \\
-\varepsilon\gamma\beta^{-1}\Big(\phi_0 - \frac{1}{2}\phi_1\Big) + \frac{\varepsilon}{2}\gamma\beta^{-1}(\Delta x^2)\Delta_\perp\phi|_0
\end{aligned}
\tag{A.4.41}
$$

directly. Because we now have the full second normal derivative on the right hand side again, we can compress it to

$$
\begin{aligned}
\pi_{-1} = (1+2\varepsilon\alpha)\pi_0 + \varepsilon\Delta x\Big(\frac{y}{x}\partial_y\pi + \frac{z}{x}\partial_z\pi\Big)|_0 \\
+\frac{\varepsilon}{2}\gamma\beta^{-1}(\phi_{-1} - 2\phi_0 + \phi_1) + \frac{\varepsilon}{2}\gamma\beta^{-1}(\Delta x^2)\Delta_\perp\phi|_0.
\end{aligned}
\tag{A.4.42}
$$

(ii) The second option to remedy the unstable inversion of $\mathbf{A}$ is to fundamentally simplify the continuum boundary conditions. Instead of expecting radially-incident waves at each point on the boundary, we can assume that the incidence is purely normal, so that along each coordinate direction, the problem simplifies to a one-dimensional one. In one dimension—where $r = x$—the Sommerfeld boundary conditions simplify to

$$
\partial_x\phi = -\frac{\pi}{c} - \frac{\phi}{x}. \tag{A.4.43}
$$

Taking the time derivative to establish an analogous equation for π, we again introduce the three-dimensional operator $\Delta\phi$, so the approach is not truly one-dimensional, but it is consistent with the idea of absorbing normally incident waves everywhere on the boundary. This equation reads

$$\partial_x \pi = -c\Delta\phi - \frac{\pi}{x}. \tag{A.4.44}$$

In discretising this pair of equations to get a prescription for (ϕ_{-1}, π_{-1}), we could opt in or opt out of using interpolation for the field evaluations on the boundary. The error order is bounded to $\mathcal{O}(\Delta x^2)$ by the Laplacian term anyways, so not much is gained from interpolating —so let us simply proceed without it. Again accepting the non-interpolation error

$$\phi_{-\frac{1}{2}} = \phi_0 + \mathcal{O}(\Delta x), \tag{A.4.45}$$

and similar for π, these boundary conditions discretise simply as

$$\begin{aligned} \varepsilon\frac{\phi_0 - \phi_{-1}}{\Delta x} &= -\frac{1}{c}\pi_0 - \frac{1}{x}\phi_0, \\ \varepsilon\frac{\pi_0 - \pi_{-1}}{\Delta x} &= -\frac{c}{(\Delta x)^2}(\phi_{-1} - 2\phi_0 + \phi_1) - c\Delta_\perp\phi|_0 - \frac{1}{x}\pi_0, \end{aligned} \tag{A.4.46}$$

which can be solved for ϕ_{-1} and π_{-1} by first multiplying with $-\varepsilon\Delta x$, and adding the known ϕ_0 and π_0, respectively:

$$\begin{aligned} \phi_{-1} &= \left(1 + \varepsilon\frac{\Delta x}{x}\right)\phi_0 + \varepsilon\frac{\Delta x}{c}\pi_0, \\ \pi_{-1} &= \left(1 + \varepsilon\frac{\Delta x}{x}\right)\pi_0 + \varepsilon\frac{c}{\Delta x}(\phi_{-1} - 2\phi_0 + \phi_1 + (\Delta x)^2\Delta_\perp\phi|_0). \end{aligned} \tag{A.4.47}$$

These are significantly simpler to implement.

A.4.2 Absorbing Boundary Layers

As we have seen in the previous section, implementing boundary conditions that make a finitely sized domain behave like an infinite one is rather involved, and forces unwanted compromises to work at all. Sommerfeld boundary conditions, in the continuum, are fully absorbing only for a certain direction of incidence at the boundary. Once discretised, the above radial formulation (A.4.39) achieves an amplitude absorption coefficient of about 83.5%, which certainly constitutes the majority, but far from all of the field amplitude. It takes waves multiple bounces for their reflected amplitude to decay below machine precision. For a linear wave equation, one can argue that even if present, due to the superposition principle, the reflected waves do not affect the dynamics of the physical field components in the bulk of the domain, so that depending on the application, they are acceptable. However, for non-linear applications, the reflected waves will interact with the bulk and change their dynamics, which is a much more severe effect. Moreover, the discretisation of the Sommerfeld boundary condition we derived in the above only works up to order $\mathcal{O}(\Delta x^2)$, so moving the bulk PDE to a wider and more accurate stencil becomes impossible as we are only able to fill a single layer of ghost cells.

While there certainly are more sophisticated boundary conditions one can use, discretising and implementing them in code becomes correspondingly harder. Because of this, in this section, we entertain an alternative approach that can be used to make a simulation domain appear as if its extent was infinite. Namely, instead of imposing and having to solve complicated equations *at* the boundary, we modify the PDE in a layer *near* the boundary, to incorporate terms that force a decay of the field towards the configuration it should have at the boundary. This makes it so that the boundary conditions can be simple Dirichlet boundaries, with this "absorbing layer" taking care of matching the dynamical solution smoothly onto the imposed boundary values, in such a way that as little as possible is reflected back into the bulk.

We assume that we have a PDE in the first-order form

$$\partial_t \Phi = F(t, \Phi), \tag{A.4.48}$$

where F is a potentially non-linear functional depending on t and Φ. Here, $\Phi : \mathbb{R}^{1,d} \to \mathbb{R}^N$ is the N-component field configuration; in the example of the wave equation,

$$\Phi = \begin{pmatrix} \phi \\ \pi \end{pmatrix}, \tag{A.4.49}$$

or for BSSN (without gauge evolution),

$$\Phi = \begin{pmatrix} \phi \\ \tilde{\gamma}_{ij} \\ K \\ \tilde{A}_{ij} \\ \tilde{\Gamma}^i \end{pmatrix}. \tag{A.4.50}$$

For now, let us try to solve the more general problem uf having some open and connected subset $U \subseteq \mathbb{R}^d$ containing the origin, in which we want to have dynamics prescribed entirely by the bulk PDE (A.4.48). Outside of U, that is, within the complement $U^c = \mathbb{R}^d \setminus U$, we want Φ to quickly approach some prescribed constant configuration $\Phi_0 \in \mathbb{R}^d$, in such a way that the dynamics of the bulk are not affected by reflections or unwanted artifacts. For the wave equation, we could have e.g. $\Phi_0 = 0$, and for BSSN, where we want asymptotically flat spacetimes, e.g.

$$\Phi_0 = \begin{pmatrix} 1 \\ \delta_{ij} \\ 0 \\ 0 \\ 0 \end{pmatrix}, \tag{A.4.51}$$

or some more sophisticated choice matching/damping onto a Kerr exterior[1].

To this end, we modify the PDE (A.4.48) by a term that drives exponential decay of Φ towards Φ_0, scaled by a function $\sigma = \sigma(x)$ which controls said decay:

$$\partial_t \Phi = F(t, \Phi) - \sigma(\Phi - \Phi_0). \tag{A.4.52}$$

Since we want the PDE to remain unchanged in the bulk, we require $\sigma|_U = 0$. Within U^c, σ should be positive, so that the damping becomes active and drives Φ towards Φ_0. The naive approach would be to just set $\sigma|_{U^c}$ to some large constant value and be done with; however, intuition already tells us that that would lead to reflections in hyperbolic propagation. "Easing into" the decay by applying a gradual increase in σ starting at 0 on ∂U and smoothly growing as one moves away from the bulk.

To turn this intuition into a more rigorous strategy, we can study how the introduction of the locally scaled exponential damping terms affect the linear wave equation in $1 + 1$ dimensions. In first order form we can implement damping terms straightforwardly as

$$\begin{aligned} \partial_t \phi &= \pi - \sigma\phi, \\ \partial_t \pi &= \partial_x^2 \phi - \sigma\pi. \end{aligned} \tag{A.4.53}$$

Solving the first equation for π,

$$\pi = \partial_t \phi + \sigma\phi, \tag{A.4.54}$$

[1]For which Φ_0 would no longer be constant

allows us to bring the modified PDE back into second-order form,

$$\begin{aligned} \partial_t(\partial_t\phi + \sigma\phi) &= \partial_x^2\phi - \sigma(\partial_t\phi + \sigma\phi) \\ \Leftrightarrow \qquad \partial_t^2\phi &= \partial_x^2\phi - 2\sigma\partial_t\phi - \sigma^2\phi \end{aligned} \tag{A.4.55}$$

To analyse reflection coefficients based on the choice of σ, we introduce the mode ansatz $\phi(t,x) = e^{-i\omega t}\hat{\phi}(x)$. This turns the second-order form into

$$\begin{aligned} \partial_x^2\hat{\phi} + \omega^2\hat{\phi} + 2i\omega\sigma\hat{\phi} - \sigma^2\hat{\phi} &= 0 \\ \Leftrightarrow \qquad \partial_x^2\hat{\phi} + (\omega + i\sigma(x))^2\hat{\phi} &= 0. \end{aligned} \tag{A.4.56}$$

For simplicity, we will assume $U = \mathbb{R}_{\geq 0}$, so $U^c = \mathbb{R}_{<0}$. Let us consider different ways of "switching on" σ from its zero value for $x > 0$ as the wave passes the origin.

(i) The simplest choice for σ would be

$$\sigma(x) = \begin{cases} 0, & x \geq 0, \\ \sigma_0, & x < 0. \end{cases} \tag{A.4.57}$$

For this choice, the equation is solved by plane waves on both sides of the domain. Assuming a left-moving incident, right-moving reflected and left-moving transmitted wave, we arrive at

$$\hat{\phi}(x) = \begin{cases} e^{i\omega x} + Re^{-i\omega x}, & x \geq 0, \\ Te^{-\sigma_0 x}e^{i\omega x}, & x < 0, \end{cases} \tag{A.4.58}$$

where T and R are the reflection and transmission/absorption coefficients. For this solution to be C^1, we must match the field values and first derivatives at $x = 0$. This yields

$$\begin{aligned} \hat{\phi}(0^-) \overset{!}{=} \hat{\phi}(0^+) \quad &\Rightarrow \qquad T = 1 + R \\ \hat{\phi}'(0^-) \overset{!}{=} \hat{\phi}'(0^-) \quad &\Rightarrow \quad \omega(1-R) = (\omega + i\sigma_0)T = (\omega + i\sigma_0)(1+R) \end{aligned} \tag{A.4.59}$$

These equations combine to yield

$$R = -\frac{i\sigma_0}{2\omega + i\sigma_0}. \tag{A.4.60}$$

Consequently, the reflected energy is

$$|R|^2 = \frac{\sigma_0^2}{4\omega^2 + \sigma_0^2}. \tag{A.4.61}$$

In particular, this means that the reflection coefficient grows with larger jumps $0 \to \sigma_0$, with higher frequencies being reflected less strongly. Although one could simply opt to pick σ_0 small enough for the reflection coefficient to satisfy whatever requirements one has for it, this causes the decay to be very slow, and the field to take a significant amount of time (and hence distance) to decay to the target configuration Φ_0. In numerical implementations, this distance is to be kept minimal, as it takes up part of the simulation domain without providing physical insight—it is "wasted" but necessary computational effort.

(ii) Building on this insight that smaller jumps cause less reflections, we can view a gradually increasing envelope for σ, interpolating between a constant value of 0 in U and a constant value of σ_0 within $(-\infty, L)$—over the interval $[-L, 0]$—as a series of small jumps. Intuitively, we'd expect this to have the advantages of both a low reflection cofficient *as well as* the potential of enabling larger final values of σ_0, so that the thickness of the boundary layer that needs to be simulated is as low as possible.

Concretely, to study this, we assume that

$$\sigma(x) = \begin{cases} 0, & x \geq 0, \\ \sigma_0 \ell(-\frac{x}{L}), & -L \leq x < 0, \\ \sigma_0, & x \leq L \end{cases} \tag{A.4.62}$$

with $\ell : [0,1] \to \mathbb{R}_{>0}$ a smooth interpolation function with $\ell(0) = 0$, $\ell(1) = 1$.

Calling the regions I, II and III from top to bottom, this implies that a solution with incoming component, reflected component and transmitted component is of the form

$$\hat{\phi}(x) = \begin{cases} e^{i\omega x} + Re^{-i\omega x}, & \text{in region I,} \\ \hat{\psi}(x), & \text{in region II,} \\ Te^{-\sigma_0(x+L)}e^{i\omega x}, & \text{in region III.} \end{cases} \tag{A.4.63}$$

When applying the WKB approximation to the function $\hat{\psi}(x)$, one can derive that R is approximated by

$$R \approx \frac{1}{2\omega} \int_{-L}^{0} \frac{\sigma'(x)}{\omega + i\sigma(x)} \exp\left(2i\omega x - 2 \int_{x}^{0} \sigma(s)\, ds \right). \tag{A.4.64}$$

Firstly, we note that R scales with σ' and hence ℓ'. This means that a slower, more gradual change in ℓ, implemented by a smooth choice of ℓ as well as a wider L, reduces the reflection coefficient. Further, the rapidly oscillating $e^{2i\omega x}$ term interacts with the integral to average the integrand to smaller values; this effect gets stronger at larger values for ω, and for smoother profiles of ℓ.

Having now motivated what kinds of functions $\sigma(x)$ are effective at reducing reflections while retaining sufficient damping, a few examples follow below:

- $\ell(x) = x^k$: Provides a C^{k-1}-smooth transition at the interface between the bulk and the boundary layer. Although larger values for k improve the smoothness at this interface, they also produce larger derivatives within the boundary layer since $\ell'(x) = kx^{k-1}$. Common choices are $k = 2$ and $k = 3$.
- $\ell(x) = \frac{1}{2}(1 - \cos(\pi x))$: If C^1-smoothness is required at both interfaces, this is a suitable choice. The derivative is bounded by $\frac{\pi}{2}$.
- Smoothstep polynomials can achieve C^k-smoothness for arbitrary k at both interfaces. They are generally given by (cf. [Wikipedia: Smoothstep])

$$S_k(x) = \sum_{i=1}^{k} (-1)^i \binom{k+i}{i} \binom{2k+1}{k-i} x^{k+i+1}, \tag{A.4.65}$$

with the first few given explicitly by

$$\begin{aligned} S_0(x) &= x, \\ S_1(x) &= -2x^3 + 3x^2, \\ S_2(x) &= 6x^5 - 15x^4 + 10x^3, \\ S_3(x) &= -20x^7 + 70x^6 - 84x^5 + 35x^4. \end{aligned} \tag{A.4.66}$$

Their first derivative is given by

$$\frac{d}{dx} S_k(x) = (2k+1) \binom{2k}{k} (x - x^2)^k, \tag{A.4.67}$$

which has its maximum at $x = \frac{1}{2}$, where

$$\frac{d}{dx} S_k\left(\frac{1}{2}\right) = \binom{2k}{k} \frac{2k+1}{2^k}. \tag{A.4.68}$$

This grows rapidly with increasing k.

Besides reflections, there is a second consideration to be made regarding the allowed magnitude of σ_0 when employing boundary damping in a discrete time integration scheme. The artifical exponential decay terms in the modified PDE (A.4.52) introduce additional stiffness, and change the spectrum of the right-hand side operator. Hence, the maximum allowed Δt becomes limited by σ_0 as well. However, this is generally far less strict than the time step considerations to be made for the bulk dynamics; the boundary layer's decay must only be stable, not match the continuum decay exactly—it is an unphysical effect anyways.

A.4.3 Kreiss-Oliger Dissipation

When evolving non-linear PDEs, the non-linear terms can introduce higher-frequency components through the mechanism we discussed in Remark A.1. Although there, we were considering spectral methods—where the mechanism presents itself most clearly—it is irrespective of the field representation used; in particular, it is also present when working with discretised field values. In that specific case, the high-frequency modes produced by non-linear terms may exceed the spatial grid cutoff set by the Nyquist limit. Such modes alias back onto lower-frequency modes, introducing unphysical "energy" that can cause a simulation to become unstable and diverge.

Clearly, one should—as a first step—choose the resolution of the grid fine enough so that all physical modes one expects to be present can be resolved, i.e. do not fall below the grid's Nyquist wavelength of $2\Delta x$. However, numerical and discretisation error can introduce unwanted, unphysical high-frequency modes. Non-linearities then transform these contributions beyond the grid's spatial cutoff, which alias into lower frequencies and can cause the simulation to diverge. To resolve this, we need to dampen or *dissipate* unphysical high-frequency modes, while leaving the physical modes unchanged, and without harming the integration accuracy of the solver.

So, let us analyse how to do this. We assume that we have a $1+1$-dimensional PDE that is in a first-order in time formulation,

$$\partial_t \phi = F(t, \phi). \tag{A.4.69}$$

We want to add a term that dissipates unphysical high-frequency modes while leaving low-frequency modes untouched. A first guess might be to add a diffusion term, turning it into

$$\partial_t \phi = F(t, \phi) + \partial_x^2 \phi. \tag{A.4.70}$$

We can analyse what this does to different frequency components by switching to a spatial frequency representation, $\phi(t, x) \to \tilde{\phi}(t, k)$, where the PDE turns into

$$\partial_t \tilde{\phi} = \tilde{F}\big(t, \tilde{\phi}\big) - k^2 \tilde{\phi}. \tag{A.4.71}$$

For modes where $k^2 \tilde{\phi} \gg \tilde{F}\big(t, \tilde{\phi}\big)$, the diffusion term dominates, and the mode behaves as

$$\tilde{\phi} \sim e^{-k^2 t}. \tag{A.4.72}$$

That is, the mode is suppressed exponentially, with falloff k^2; the higher the frequency, the stronger the dissipation. However, this modification has two issues:

(i) Even though the dissipation is weaker for low spatial frequencies k, it still modifies the behaviour of all modes; we would like to differentiate between low- and high-frequency modes more strongly.

(ii) Since the dissipation term is not present in the physical PDE, it essentially acts as an error term. If introduced as above, without appropriate normalisation, it is an $\mathcal{O}(1)$ error term—completely dwarfing any numerical or discretisation errors, and causing the numerical solution to deviate strongly from the physical continuum solution.

Luckily, these issues are straightforward to resolve. To address the first, we can increase the power of k in the exponential decay factor. Moving from k^2 to k^{2r} gives us a "knob" to adjust in the form of the integer r, which turns the falloff function from a Gaussian in the frequency domain into a steeper step-like threshold around $k = 0$, leaving modes close to 0 virtually unaffected while rapidly decaying those at larger $|k|$.

Since obtaining a $-k^{2r}$ factor in the frequency domain requires an operator proportional to ∂_x^{2r} in the spatial domain (recalling that $\partial_x \to -ik$ and thus $\partial_x^2 \to -k^2$), we must track the sign,

$$\partial_x^{2r} \to (-1)^r k^{2r}. \tag{A.4.73}$$

Thus, the continuum dissipation term takes the form

$$\partial_t \phi = F(t, \phi) + (-1)^{r+1} \partial_x^{2r} \phi. \tag{A.4.74}$$

To resolve issue (ii) and prevent the dissipation from degrading the accuract of a $(2r-2)$-th order spatial discretisation scheme, we multiply the operator by $\sigma \Delta x^{2r-1}$, where $\sigma > 0$ is a dimensionless, tunable parameter. This yields the modified continuous PDE

$$\partial_t \phi = F(t, \phi) + (-1)^{r+1} \sigma \Delta x^{2r-1} \partial_x^{2r} \phi. \tag{A.4.75}$$

For resolved physical modes, where $k \ll k_{\text{Nyq}}$, the damping term is of order $\mathcal{O}(\Delta x^{2r-1})$, acting purely as a high-order truncation error that vanishes rapidly in the continuum limit $\Delta x \to 0$. However, near the grid cutoff, where $k \approx k_{\text{Nyq}} = \pi/\Delta x$, the k^{2r} factor yields a dissipation rate scaling as $\mathcal{O}(\frac{1}{\Delta} x)$, which rapidly suppresses unphysical grid-scale modes.

The final step is to discretise this operator on the spatial grid. Because the dissipation operator is already multiplied explicitly by Δx^{2r-1}, any discretisation error introduced by the operator itself is pushed to even higher powers of Δx. We are therefore free to use the simplest centered finite-difference stencil. Defining the forward and backward difference operators as

$$D_+ f(x) = \frac{f(x + \Delta x) - f(x)}{\Delta x}, \quad D_- f(x) = \frac{f(x) - f(x - \Delta x)}{\Delta x}, \tag{A.4.76}$$

we discretise the evolution equations as

$$\partial_t \phi = F(t, \phi) + (-1)^{r+1} \Delta x^{2r-1} (D_+ D_-)^r \phi. \tag{A.4.77}$$

The term

$$\mathcal{D}_{\text{KO}} \phi = (-1)^{r+1} \sigma \Delta x^{2r-1} (D_+ D_-)^r \phi \tag{A.4.78}$$

is the *Kreiss-Oliger dissipation operator of order* $2r - 1$. For reasonable choices of $\sigma \ll 1$, the $\mathcal{D}_{\text{KO}}$ term does not alter the CFL condition limiting the timestep for the dissipation-free PDE.

To get an explicit expression for the stencil $(D_+ D_-)^r$, it is useful to introduce the shift operator

$$E^k f(x) \coloneqq f(x + k\Delta x), \quad k \in \mathbb{R}. \tag{A.4.79}$$

We can write

$$D_+ D_- f(x) = f(x + \Delta x) - 2f(x) + f(x - \Delta x) = \left(E^{1/2} - E^{-1/2}\right)^2 f(x), \tag{A.4.80}$$

so that according to the binomial theorem,

$$
\begin{aligned}
(D_+D_-)^r = \left(E^{1/2} - E^{-1/2}\right)^{2r} &= \sum_{k=0}^{2r} \binom{2r}{k} \left(E^{1/2}\right)^k \left(-E^{-1/2}\right)^{2r-k} \\
&= \sum_{k=0}^{2r} (-1)^{2r-k} \binom{2r}{k} E^{k-r} = \sum_{k=-r}^{r} (-1)^{r-k} \binom{2r}{r+k} E^k
\end{aligned}
\tag{A.4.81}
$$

Thus,

$$
\mathcal{D}_{\mathrm{KO}}\phi(x) = -\sigma \Delta x^{2r-1} \sum_{k=-r}^{r} (-1)^k \binom{2r}{r+k} \phi(x + k\Delta x). \tag{A.4.82}
$$

A.5 Implicit ODE and PDE Solvers

In this section, we consider numerical integration schemes for differential equations of the form

$$
\partial_t \phi = F(t, \phi), \tag{A.5.1}
$$

where F is either a function (for ODEs) or an operator/functional (for PDEs) acting on ϕ.

Formally, integrating equation (A.5.1) over a time interval $[t, t + \Delta t]$ yields the exact integral equation

$$
\phi(t + \Delta t) = \phi(t) + \int_t^{t+\Delta t} F(t', \phi(t'))\, dt'. \tag{A.5.2}
$$

Numerical time-stepping algorithms replace this integral with a discrete quadrature rule. Let $\phi^n \approx \phi(n\Delta t)$ denote the discrete numerical solution at step n. A general single-step time-integration scheme approximates the integral via a function G,

$$
\phi^{n+1} \approx \phi^n + \Delta t G(t_n, \Delta t, \phi^n, \phi^{n+1}), \tag{A.5.3}
$$

up to some local truncation error $\mathcal{O}\big(\Delta t^{k+1}\big)$, yielding a scheme of order k.

We classify the integration scheme based on how G depends on the state vector:

(i) *Explicit*: G depends solely on known state values from the current or previous time steps $(\phi^n, \phi^{n-1}, ...)$. The update step for ϕ^{n+1} is a direct assignment operation.

(ii) *Implicit*: G depends on the unknown state ϕ^{n+1} at the new time level (or intermediate stage evaluations requiring ϕ^{n+1}).

Consequently, for an implicit scheme, the update formula cannot be evaluated directly; instead, ϕ^{n+1} must be obtained by solving a system of algebraic equations (linear or non-linear) at every time step.

As the name would suggest, explicit/forward Euler,

$$
\phi(t + \Delta t) = \phi(t) + \Delta t F(t, \phi(t)) + \mathcal{O}(\Delta t) \tag{A.5.4}
$$

is an example of an explicit integration scheme. Further examples are the Runge-Kutta integrators we considered in Section A.2. Another class of examples, which incorporates past values $\phi(t - \Delta t), \phi(t - 2\Delta t), ...$ exist, and are known as *Adams-Bashforth schemes*.

A.5.1 Example: Backwards Euler

In this section, we introduce the most basic of implicit time integration schemes, the *backwards/ implicit Euler method*. Although it has a rather large error, it will serve well in introducing a number of concepts, and provide a reference to compare more elaborate methods against.

The arguably simplest way to approximate the integral

$$\int_t^{t+\Delta t} F(t', \phi(t'))\, dt' \tag{A.5.5}$$

is to take the value of ϕ at one of the boundary times, t, $t + \Delta t$, and multiply it by the width Δt of the integration interval. Since we are interested in an implicit scheme, we take the boundary value at $t + \Delta t$, that is,

$$\int_t^{t+\Delta t} F(t', \phi(t'))\, dt' = \Delta t\, F(t + \Delta t, \phi(t + \Delta t)) + \mathcal{O}(\Delta t^2) \tag{A.5.6}$$

Neglecting to explicitly write the error term, this leads to the time-stepping scheme

$$\phi(t + \Delta t) = \phi(t) + \Delta t\, F(t + \Delta t, \phi(t + \Delta t)). \tag{A.5.7}$$

Clearly, this is an equation to be solved for $\phi(t + \Delta t)$. In the case of F being a function, it is purely algebraic; if F however depends on derivatives of ϕ, it is a differential equation.

Let us go over some examples below.

(i) **Exponential Decay:** For $\phi : \mathbb{R} \to \mathbb{R}$, we consider the ODE

$$\partial_t \phi(t) = -\phi(t). \tag{A.5.8}$$

Clearly, the solution space is given by $\phi(t) = \phi_0 e^{-t}$, $\phi_0 \in \mathbb{R}$. The implicit time-stepping scheme prescribes the update equation

$$\phi(t + \Delta t) = \phi(t) + \Delta t\, (-\phi(t + \Delta t)). \tag{A.5.9}$$

In this case—and many other ODE cases—it is possible to explicitly solve this for $\phi(t + \Delta t)$:

$$\phi(t + \Delta t) = \frac{\phi(t)}{1 + \Delta t}. \tag{A.5.10}$$

This is a recursion, which can be made closed-form as

$$\phi(n\Delta t) = \frac{\phi_0}{(1 + \Delta t)^n} \tag{A.5.11}$$

As expected from the analytical solution $\phi(t) = \phi_0 e^{-t}$, this is an exponential decay; however, each step truncates the exponential series to linear order. One timestep of the analytical solution scales it by a factor

$$e^{-\Delta t} = \frac{1}{e^{\Delta t}} = \frac{1}{1 + \Delta t + \frac{1}{2}\Delta t^2 + ...}. \tag{A.5.12}$$

The implicit Euler scheme hence neglects the $\mathcal{O}(\Delta t^2)$ terms in the denominator.

Lastly, we note that the stepping scheme is unconditionally stable—for any positive Δt, equation (A.5.11) defines a stable exponential decay. The explicit/forward Euler scheme, which prescribes

$$\phi(t + \Delta t) = \phi(t) + \Delta t(-\phi(t)) = (1 - \Delta t)\phi(t), \tag{A.5.13}$$

has the closed-form solution

$$\phi(n\Delta t) = (1 - \Delta t)^n \phi_0, \tag{A.5.14}$$

which for $\Delta t > 1$ becomes oscillatory, and for $\Delta t > 2$ starts growing exponentially in magnitude. Further, it is the truncation of the implicit solution:

$$\frac{\phi_0}{(1+\Delta t)^n} = (1 - \Delta t + \Delta t^2 + ...)^n \phi_0 \approx (1 - \Delta t)^n \phi_0 \tag{A.5.15}$$

What we can take away from this example is the following: implicit Euler handles exponential decay much more gracefully than explicit Euler.

(ii) **1d Heat Equation, Discretised**: We now proceed to the one-dimensional heat equation,

$$\partial_t \phi = \partial_x^2 \phi. \tag{A.5.16}$$

Applying the implicit Euler method to this PDE leads to the time-step

$$\phi(t + \Delta t, x) = \phi(t, x) + \Delta t (\partial_x^2 \phi)(t + \Delta t, x). \tag{A.5.17}$$

Here, separating quantities evaluated at $t + \Delta t$ to the left-hand side, we obtain the equation

$$(\phi - \Delta t \partial_x^2 \phi)(t + \Delta t, x) = \phi(t, x). \tag{A.5.18}$$

This is a second-order linear ODE with constant coefficients, where the previous field state $\phi(t, x)$ acts as a source. Although technically, depending on the initial field configuration $\phi_0(x)$, we can sometimes integrate this equation for each step, let us entertain a different (and more versatile) approach.

Concretely, we discretise ϕ not just in time but also in space, with a grid spacing of Δx. Since otherwise, notation will get heavy, let us introduce the shorthand

$$\phi_i^n = \phi(n\Delta t, i\Delta x). \tag{A.5.19}$$

Employing the stencil (A.1.4) for the second derivative, the discretised step equation reads

$$\phi_i^{n+1} - C\big(\phi_{i-1}^{n+1} - 2\phi_i^{n+1} + \phi_{i-1}^{n+1}\big) = \phi_i^n \tag{A.5.20}$$

where $C = \Delta t / \Delta x^2$. This is a linear system of equations—let us write it in matrix-vector form. Denoting $\boldsymbol{\phi}^n = (\phi_i^n)_{i=1}^N$ for some grid size N, and additionally imposing Dirichlet boundary conditions, $\phi_{-1} = \phi_{N+1} = 0$, we can turn the step equation into

$$\mathbf{M}\boldsymbol{\phi}^{n+1} = \boldsymbol{\phi}^n \tag{A.5.21}$$

with

$$\mathbf{M} = \begin{pmatrix} 1-2C & -C & & & & \\ -C & 1-2C & -C & & & \\ & -C & 1-2C & \ddots & & \\ & & \ddots & \ddots & -C & \\ & & & -C & 1-2C \end{pmatrix} \tag{A.5.22}$$

The first instinct might be to make the step equation explicit by writing

$$\boldsymbol{\phi}^{n+1} = \mathbf{M}^{-1}\boldsymbol{\phi}^n, \tag{A.5.23}$$

and precomputing the inverse $\mathbf{M}^{-1}$ once at simulation start. Although this works, this has a significant computational drawback: the matrix $\mathbf{M}$ is sparse, but $\mathbf{M}^{-1}$ is not. this makes the multiplication between $\mathbf{M}^{-1}$ and the state vector $\boldsymbol{\phi}^n$ computationally expensive, especially for larger grids. Because of this, one usually opts to solve the linear system using other methods, such as an LDU decomposition or Gauss-Seidel.

(iii) **Heat Equation, Spectral**: We now approach the heat equation in d dimensions,

$$\partial_t \phi = \Delta\phi, \quad \Delta = \sum_{i=1}^{d} \partial_i^2, \tag{A.5.24}$$

on the domain $\Omega = [0, \pi]^d$ with Dirichlet boundary conditions, $\phi|_{\partial\Omega} = 0$. Its implicit Euler step equation is clearly

$$\phi(t + \Delta t, \mathbf{x}) = \phi(t, \mathbf{x}) + \Delta t(\Delta\phi)(t + \Delta t, \mathbf{x}) \tag{A.5.25}$$

This rearranges into an elliptic problem at each step,

$$(\phi - \Delta t\Delta\phi)(t + \Delta t, \mathbf{x}) = \phi(t, \mathbf{x}). \tag{A.5.26}$$

Just as in the $d = 1$ case above, we could now discretise the Laplacian operator and proceed solving the linear system using Gauss-Seidel or any other linaer solver. Writing this down explicitly is more tedious, since the index gymnastics are far more involved, and there is not much gained conceptually from doing so.

Instead, we entertain another approach of approximating the solution ϕ and by that the action of Δ on it—by means of a spectral expansion, in the special case where $d = 3$. Concretely, we define the functions

$$\psi_{ijk}(\mathbf{x}) = \sin(ix)\sin(jy)\sin(kz), \quad i, j, k \in \mathbb{N}^3. \tag{A.5.27}$$

Clearly, these functions form an orthogonal basis with respect to the standard L^2-inner product on the space of L^2-functions on Ω satisfying the boundary conditions. We can hence express $\phi(t, \mathbf{x})$ as a sum with time-variable coefficients $c_{ijk}(t)$,

$$\phi(t, \mathbf{x}) = \sum_{i,j,k\in\mathbb{N}} c_{ijk}(t)\psi_{ijk}(\mathbf{x}). \tag{A.5.28}$$

Since computers cannot deal with infinities, we must truncate the sum at some highest-frequency mode (N, N, N), turning it into

$$\phi(t, \mathbf{x}) = \sum_{i,j,k=1}^{N} c_{ijk}(t)\psi_{ijk}(\mathbf{x}). \tag{A.5.29}$$

One might argue that this loses information just as discretising ϕ does. However, the kind of information that is lost is different; While discretising the ϕ leaves the function values at the grid points exact but makes derivatives introduce errors, truncating the basis expansion at some finite point (typically) allows derivatives to remain exact at the cost of introducing some error in raw function values. Let us examine this. The action of the Laplacian on a basis function ψ_{ijk} is

$$\Delta\psi_{ijk} = -(i^2 + j^2 + k^2)\psi_{ijk}. \tag{A.5.30}$$

Since the right-hand side involves no modes with larger indices, it can still be resolved fully by a series truncated at the mode (N, N, N). Thus, the Laplacian applied to the truncated basis expansion of $\phi(t, \mathbf{x})$ is exact, reading

$$\Delta\phi = -\sum_{i,j,k=1}^{N} (i^2 + j^2 + k^2)c_{ijk}\psi_{ijk}. \tag{A.5.31}$$

We can insert this expansion into equation (A.5.26) to obtain

$$\sum_{i,j,k=1}^{N} \left(c_{ijk}^{n+1} + \Delta t(i^2 + j^2 + k^2)c_{ijk}^{n+1}\right)\psi_{ijk} = \sum_{i,j,k=1}^{N} c_{ijk}^{n}\psi_{ijk}. \tag{A.5.32}$$

Since basis expansions are unique, identification of coefficients implies the update equation

$$c_{ijk}^{n+1} = \frac{c_{ijk}^{n}}{1 + \Delta t(i^2 + j^2 + k^2)}. \tag{A.5.33}$$

This results in a geometric decay, with higher-frequency modes being suppressed more rapidly —just as expected for the heat equation.

Remark A.1 The reason we can invert equation (A.5.32) nicely for the explicit iteration step above is that the operator acting on the basis coefficient vector on the left-hand side is diagonal. In other words, the chosen basis diagonalises the Laplacian. Unfortunately, we cannot always do this; given a differential operator L, it is not always possible to find an explicit basis that diagonalises it so that the associated system

$$\partial_t \phi = L\phi \tag{A.5.34}$$

can be time-integrated spectrally as cleanly as above. However, there are certain valuable properties of a basis $\{\psi_n\}$ (with n some generalised index) that typically can be achieved, which we now outline.

- The arguably most important advantage that spectral methods have over finite differences is that derivatives can be exact. For this to be the case, however, the truncation subpace $V_N = \text{span}\{\psi_1, ..., \psi_N\}$ must be closed under differentiation. Just as we can adjust the resolution of a discretisation grid, we want to be able to adjust N freely (up to stability and error considerations), which makes closure under differentiation require

$$\partial\psi_n \in \text{span}\{\psi_1, ..., \psi_n\} \quad \forall n \in \mathbb{N}. \tag{A.5.35}$$

  Here, ∂ represents any of the relevant first derivatives.

- Further, we should choose our basis such that L becomes lower triangular with respect to it on V_N. This makes solving the equivalent of equation (A.5.26) more straightforward to solve by employing a single Gauss elimination. Lower triangularity however is not a strict necessity, there are other forms where similarly efficient linear solvers exist

  Polynomial bases such as Legendre, Laguerre or Chebyshev polynomials work well for this application, since differentiation reduces their degree and hence ensures closure. For periodic domains, plane waves ($e^{i\mathbf{k}\cdot\mathbf{x}}$) are highly versatile, and for infinite domains, one can either map to a finite one with a conformal transformation and use a polynomial basis or employ Hermite functions if features are highly localised around the origin.

  For linear equations with constant coefficients, the exact spatial differentiation of spectral methods is a clear gain over discrete finite differences. The only errors emerge from the approximation of initial conditions when projecting onto the basis and truncating, as well as the error introduced by the time integrator. However, as soon as terms which are non-linear in the basis functions emerge, we start to introduce additional truncation error. To illustrate this, consider terms of the form

$$f\phi \quad \text{or} \quad \phi^2. \tag{A.5.36}$$

  When inserting basis expansions for the field ϕ and auxiliary functions such as f, products $\psi_n\psi_m$ of basis functions emerge. Although these can be expressed as a linear combination of ψ_n's again, this may introduce contributions from modes that are above the truncation cutoff N. Concretely, consider for example the real Fourier basis,

$$\psi_n(x) = \cos(nx), \quad \chi_n(x) = \sin(nx). \tag{A.5.37}$$

  The product of two such basis functions yields linear combinations like

$$\begin{aligned}
\psi_n(x)\psi_m(x) = \cos(nx)\cos(mx) &= \frac{1}{2}\cos((n-m)x) + \frac{1}{2}\cos((n+m)x) \\
&= \frac{1}{2}\psi_{n-m}(x) + \frac{1}{2}\psi_{n+m}(x), \\
\chi_n(x)\psi_m(x) = \sin(nx)\cos(mx) &= \frac{1}{2}\sin((n-m)x) + \frac{1}{2}\sin((n+m)x) \\
&= \frac{1}{2}\chi_{n-m}(x) + \frac{1}{2}\chi_{n+m}(x), \\
\chi_n(x)\chi_m(x) = \sin(nx)\sin(mx) &= \frac{1}{2}\cos((n-m)x) - \frac{1}{2}\cos((n+m)x) \\
&= \frac{1}{2}\psi_{n-m}(x) - \frac{1}{2}\psi_{n+m}(x).
\end{aligned} \tag{A.5.38}$$

  This means that the product of two modes will alias as a low-frequency mode ψ_{n-m} or χ_{n-m} *as well as* a high-frequency mode ψ_{n+m} or χ_{n+m}. If $n + m > N$, the high-frequency alias is truncated, introducing an error. For this error not to be devastating, N has to be chosen large enough so that modes close to it are low in amplitude.

A.5.2 Example: Crank-Nicolson

A better way of approximating an integral than by taking one of its endpoint values multiplied by the interval width is to approximate the integrand as the linear polynomial passing through both endpoints. This is the so-called *trapezoidal* integration rule,

$$\int_a^b f(t)\,dt = \frac{\Delta t}{2}(f(b) + f(a)) + \mathcal{O}(\Delta t^3), \tag{A.5.39}$$

which, in particular, improves the error from $\mathcal{O}(\Delta t^2)$ to $\mathcal{O}(\Delta t^3)$.

Applying this approximation to the exact step equation (A.5.2) leads to

$$\phi(t + \Delta t) = \phi(t) + \frac{\Delta t}{2}(F(t, \phi(t)) + F(t + \Delta t, \phi(t + \Delta t))) + \mathcal{O}(\Delta t^3). \tag{A.5.40}$$

This still leaves us with an equation to solve at each step, but improves the per-step error from $\mathcal{O}(\Delta t^2)$ to $\mathcal{O}(\Delta t^3)$.

Even though this is still implicit, the example of exponential decay is no longer unconditionally stable in any useful way. Let us consider this more precisely, starting from the differential equation

$$\partial_t \phi(t) = -\phi(t). \tag{A.5.41}$$

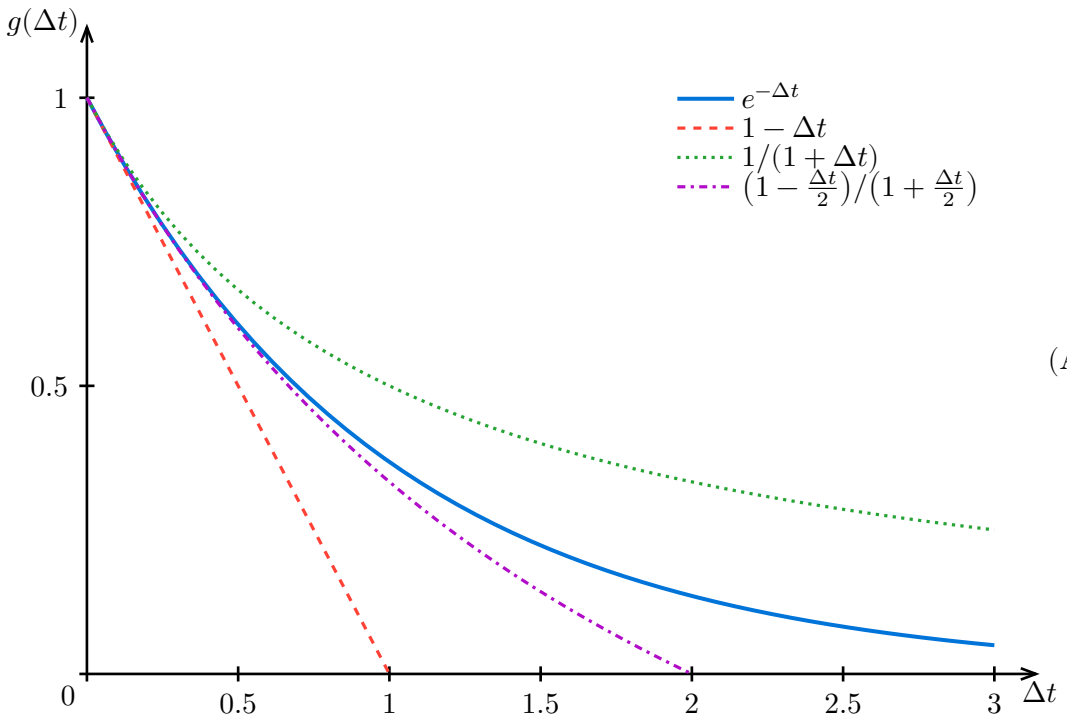
The step equation (dropping error terms) then becomes

$$\phi(t + \Delta t) = \phi(t) - \frac{\Delta t}{2}\phi(t) - \frac{\Delta t}{2}\phi(t + \Delta t), \tag{A.5.42}$$

which is easily solved for $\phi(t + \Delta t)$ as

$$\phi(t + \Delta t) = \frac{1 - \frac{\Delta t}{2}}{1 + \frac{\Delta t}{2}}\phi(t) \tag{A.5.43}$$

Just as with the forward and backward Euler approaches for this equation, ϕ is multiplied by an approximation of $e^{-\Delta t}$ in each timestep.



(A.5.44)

As can be seen in the plot above, the factor

$$g(\Delta t) = \frac{1 - \frac{\Delta t}{2}}{1 + \frac{\Delta t}{2}} \tag{A.5.45}$$

is much closer to the exact solution's step factor of $e^{-\Delta t}$ than the explicit Euler (red) and implicit Euler (green) factors. However, even though $|g(\Delta t)| \leq 1$ for all $\Delta t \geq 0$—implying unconditional stability—the factor becomes negative for $\Delta t > 2$, producing unwanted oscillatory behaviour. However, the Crank-Nicolson procedure is still superior to explicit Euler, since the timestep may have a larger value while retaining the same integration accuracy.

A.5.3 Example: IMEX Schemes (Implicit-Explicit)

What we have seen so far is that, roughly speaking, implicit Euler is more or even unconditionally stable than explicit Euler—allowing for a timestep size Δt limited only by truncation error—but explicit Euler steps are much less costly computationally speaking, as they do not require solving algebraic or differential equations at each step. Hence, there is always a tradeoff between the two; stiff systems, like the heat equation

$$\partial_t \phi = \Delta \phi, \tag{A.5.46}$$

are better approached using implicit Euler. This works well since one trades CFL-limited timesteps for having to solve a linear system of equations at each step—something for which plenty of efficient algorithms exist. However, if we introduce a nonlinear term, and consider e.g.

$$\partial_t \phi = \Delta \phi - \phi^3, \tag{A.5.47}$$

the implicit timesteps become much more costly—the ϕ^3 term destroys the linearity of the system of equations.

It would be nice if we could keep the advantage of large timesteps for the $\Delta\phi$ smoothing term, while not having to invert the cubic term. It turns out, such methods exist—they fall under the category of *IMEX* or *Implicit-Explicit* integrators. Their starting point is a differential equation

$$\partial_t \phi = F(t, \phi) + G(t, \phi), \tag{A.5.48}$$

where on the right-hand side F is a function or functional containing all stiff terms for which we want implicit-like timestepping, and $G(t, \phi)$ contains all other terms like nonlinear contributions, for which inversion is too computationally expensive. In the example (A.5.47), we would pick

$$F(t, \phi) = \Delta \phi, \quad G(t, \phi) = -\phi^3. \tag{A.5.49}$$

Clearly, the integral equation associated to equation (A.5.48) can be rearranged into the step equation

$$\phi(t + \Delta t) = \phi(t) + \int_t^{t+\Delta t} F(t', \phi(t'))\, dt' + \int_t^{t+\Delta t} G(t', \phi(t'))\, dt' \tag{A.5.50}$$

where we have used the linearity of the integral to split it up into two separate integrations. The advantage of this is that we are free to pick approximation schemes for both of the terms.

The simplest IMEX scheme consists of using the implicit Euler approximation

$$\int_t^{t+\Delta t} F(t', \phi(t'))\, dt' = F(t + \Delta t, \phi(t + \Delta t))\Delta t + \mathcal{O}(\Delta t^2) \tag{A.5.51}$$

for the stiff term involving F and the explicit Euler approximation

$$\int_t^{t+\Delta t} G(t', \phi(t'))\, dt' = G(t, \phi(t))\Delta t + \mathcal{O}(\Delta t^2) \tag{A.5.52}$$

for the remainder involving G. This produces a step equation (neglecting error) reading

$$\phi(t + \Delta t) = \phi(t) + \Delta t(F(t + \Delta t, \phi(t + \Delta t)) + G(t, \phi(t))). \tag{A.5.53}$$

Abusing notation slightly, we may rearrange this into

$$(I - \Delta t F(t + \Delta t))(\phi(t + \Delta t)) = \phi(t) + \Delta t G(t, \phi(t)). \tag{A.5.54}$$

In this form all explicit evaluations at t remain on the right-hand side, while the unknown value of $\phi(t + \Delta t)$ is moved to the left-hand side. This leaves us to invert only the (typically but not necessarily linear) operator $I - \Delta t F$, alleviating the need to solve non-linear contributions from G.

Applying this 1st-order IMEX scheme to our non-linear heat equation example equation (A.5.47) with spatial finite differences yields:

$$\mathbf{M}\boldsymbol{\phi}^{n+1} = \boldsymbol{\phi}^n - \Delta t(\boldsymbol{\phi}^n)^3, \tag{A.5.55}$$

where $\mathbf{M}$ is the same linear system matrix derived in equation (A.5.21), and the power $(\boldsymbol{\phi}^n)^3$ is evaluated component-wise. The non-linear term is computed purely as an explicit source update on the right-hand side, leaving $\mathbf{M}$ untouched.

A.6 Discretisation- and Code Validation

A.6.1 Known Exact Solutions

The simplest way to validate the consistency of a discretisation as well as its implementation in code is to compare its result against a known solution of the continuum equation. Denoting the numerical solution by $\phi_{\text{num}}(t, x)$ and the analytical solution by $\phi(t, x)$, one can then study the time evolution of e.g. the (logarithmic) difference of field values,

$$e(t, x) = \log_{10}|\phi(t, x) - \phi_{\text{num}}(t, x)|. \tag{A.6.1}$$

Since this yields a spatial distribution of the error, one can use it to get an overview of which features of the solution accrue the most error when approximated numerically.

In certain simulations, such as binary black hole mergers, over time, a phase drift can occur between the analytical and numerical solutions. Though we will get into this in more detail later on, one can account for this empirically by introducing a drift function $s(t)$ and modifying the error distribution as

$$e(t, x) = \log_{10}|\phi(t, x) - \phi_{\text{num}}(t + s(t), x)|. \tag{A.6.2}$$

By adjusting the function $s(t)$, one can attempt to reduce the error and hence quantify the accumulated phase drift.

One downside of an error distribution $e(t, x)$ is that its interpretation requires visual inspection of a full $d + 1$-dimensional dataset. This is simplified by considering certain norms of the difference instead, such as e.g.

$$\mathscr{e}_p(t) = \log_{10}\left(\int_\Omega |\phi(t, x) - \phi_{\text{num}}|^p \; d^d x\right)^{1/p} \tag{A.6.3}$$

(including a factor of $\sqrt{g}$ if appropriate)or its limit as $p \to \infty$,

$$e_\infty(t) = \log_{10}\left(\sup_{x\in\Omega}|\phi(t,x) - \phi_{\text{num}}(t,x)|\right). \tag{A.6.4}$$

Though this removes the spatial information of *where* the errors are largest, these norms (potentially incorporating a drift function as well), the resulting error-vs-time graph can typically also provide sufficient insight into whether a numerical scheme is sufficiently accurate. Naturally, with all the examples given so far, it is always possible to consider relative errors instead of absolute ones.

If one is working with a constrained system, such as electrodynamics ($\nabla \cdot \mathbf{E} = 0$) or general relativity ($\mathcal{H} = \mathcal{M}_i = 0$) the constraints themselves are additional variables to derive insights from. Concretely, even if an analytical solution for the given initial conditions is not known, on physical grounds, one knows that the constraints still have to be satisfied. We are hence motivated to study the spatial distribution or norms of the constraints over time.

The likely largest drawback of comparing against known analytical solutions is that one needs to know said analytical solutions in the first place. For certain linear equations, this is achievable, though it might lead us to question why we are solving the equation numerically while having an analytical equation on hand. If the PDE cannot be solved generally, one can test the numerical implementations on the analytical solutions one does know; e.g. Schwarzschild and Kerr for GR. Though this is a good starting point, it does leave a lot to be desired. Since we do not know any analytical two-body solutions (beyond dual extremal Reissner-Nordström BHs or similarly exotic solutions), we cannot really verify dynamical configurations, and by that lack the ability to e.g. quantify phase drift in BBH inspirals. We thus clearly need to develop more verification tools—ones that do not rely on having an analytical solution of the PDE at hand.

A.6.2 Grid and Temporal Convergence Tests

A.6.3 Time-Reversal Tests

A.6.4 Conservation Laws and Constraint Monitoring

A.6.5 Method of Manufactured Solutions

A.7 Newton-Raphson for Elliptic Equations

A.7.1 Recap of N-R for Root Finding in $d = 1$

In this section, we discuss the Newton-Raphson method of finding a root of a function $f : \mathbb{R} \to \mathbb{R}$, in preparation of the more general, higher dimensional case where we study maps $\mathbb{R}^m \to \mathbb{R}^n$.

The starting point is an initial guess $x_0 \in \mathbb{R}$ for where a root of f might be. To refine this guess, we approximate f to linear order around x_0,

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0). \tag{A.7.1}$$

This linear approximation is sure to have a root, which we use as the refined guess x_1 of the root of f itself. That is, we set

$$0 = f(x_0) + f'(x_0)(x_1 - x_0). \tag{A.7.2}$$

Rearranging for x_1, we obtain

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}. \tag{A.7.3}$$

Repeating this step yields the *Newton-Raphson iteration*

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}. \tag{A.7.4}$$

Clearly, roots of f are fixed points of this iteration, since if for x_* such that $f(x_*) = 0$,

$$x_* = x_* - \frac{f(x_*)}{f'(x_*)}. \tag{A.7.5}$$

However, for this iteration to actually converge requires $f'(x_*) \neq 0$, and that the fixed point be an attractor. That is, there must be an interval $I = (x_* - \varepsilon, x_* + \varepsilon) \subseteq \mathbb{R}$ such that the iteration map

$$g(x) = x - \frac{f(x)}{f'(x)} \tag{A.7.6}$$

is a contraction on I. That is, we require $|g'(x_*)| < 1$. Evaluating this, we get

$$1 > \left| 1 - \underbrace{\frac{f'(x_*)}{f'(x_*)}}_{=1} + \frac{\overbrace{f(x_*)}^{=0} f''(x_*)}{(f'(x_*))^2} \right| = 0, \tag{A.7.7}$$

which is unconditionally satisfied. Hence, such a contraction neighbourhood I of the root x_* exists, and by the Banach fixed point theorem, for any $x_0 \in I$, the Newton-Raphson iteration (A.7.4) converges to x_*.

Let us examine the nature of this convergence, provided that x_* is a simple root ($f'(x_*) \neq 0$). To this end, we define the error e_n as

$$e_n = |x_* - x_n|, \tag{A.7.8}$$

the distance of x_n to the true value of the root. We recall that $g(x_*) = x_*$, $g'(x_*) = 0$, so that

$$\begin{aligned} x_{n+1} = g(x_n) &= \underbrace{g(x_*)}_{=x_*} + \underbrace{g'(x_*)}_{=0}(x_n - x_*) + \frac{1}{2}g''(x_*)(x_n - x_*)^2 + \mathcal{O}\Big((x_n - x_*)^3\Big) \\ &= x_* + \frac{1}{2}g''(x_*)(x_n - x_*)^2 + \mathcal{O}\Big((x_n - x_*)^3\Big). \end{aligned} \tag{A.7.9}$$

Subtracting x_* from both sides and taking the modulus, we obtain

$$e_{n+1} = |x_{n+1} - x_*| \leq \frac{1}{2}\,|g''(x_*)|e_n^2 + \mathcal{O}(e_n^3) \tag{A.7.10}$$

Since $e_n \to 0$ as $n \to \infty$, this implies

$$\lim_{n\to\infty} \frac{e_{n+1}}{e_n^2} \leq \frac{1}{2}|g''(x_*)| = \frac{1}{2}\left|\frac{f''(x_*)}{f'(x_*)}\right| \tag{A.7.11}$$

showing that the convergence order is quadratic if $f''(x_*) \neq 0$, and at least cubic if $f''(x_*) = 0$.

A.7.2 Higher-Dimensional Generalisation

We now move to the multi-dimensional setting, where $f : \mathbb{R}^m \to \mathbb{R}^n$. To derive a Newton-Raphson iteration for finding roots of f, we pick an initial guess $x_0 \in \mathbb{R}^m$, and approximate f to linear order around x_0, which yields

$$f(x) \approx f(x_0) + J(f)(x_0)(x - x_0), \tag{A.7.12}$$

where $[J(f)]_{ij} = \partial_j f_i$ is the Jacobian matrix of f. As our refined guess $x_1 \in \mathbb{R}^m$, we pick a root of g, that is,

$$J(f)(x_0)(x_1 - x_0) = -f(x_0). \tag{A.7.13}$$

This is now a linear system. If $n > m$, this generally has no solutions; if $n < m$, the system is underdetermined an x_1 is non-unique—we thus focus on the case where $n = m$. In that case, the system can be solved iff $\det J(f)(x_0) \neq 0$, which is equivalent to the condition that $f'(x_0) \neq 0$ in the one-dimensional case.

These considerations lead us to define the *Newton-Raphson iteration* for $f : \mathbb{R}^n \to \mathbb{R}^n$ as

$$x_{n+1} = x_n - J(f)(x_n)^{-1} f(x_n). \tag{A.7.14}$$

Though this is nice to work with analytically, inverting the Jacobian explicitly is computationally expensive, whence one typically opts to solve

$$J(f)(x_n)\Delta x = -f(x_n) \tag{A.7.15}$$

and to then update $x_{n+1} = x_n + \Delta x$.

A.8 Adaptive Mesh Refinement

A.8.1 Refinement Conditions

The goal of adaptive mesh refinement (AMR) is to increase the resolution of a simulation wherever there are features in the field configuration which cannot be resolved adequately at the current resolution. Hence, we need a predicate to decide whether a grid cell should be refined or not; for this, we need to be able to detect features.

A first intuition for how to detect features might be something gradient-based—something like

$$\text{refine if} \quad |\nabla\phi| > C \tag{A.8.1}$$

for some threshold value C. Unfortunately, this has multiple issues:

(i) It is entirely decoupled from the current grid resolution. If large enough, the same gradient value will always lead to refinement, regardless of what resolution we already have.

(ii) Further, the gradient is scale-dependent. A rescaling $\phi \to \lambda\phi$ for some real constant $\lambda > 0$ does not affect the accuracy of a finite-difference stencil, since it simply gets scaled by the same factor. However, since $|\nabla\phi| \to \lambda|\nabla\phi|$, the condition depends upon the scale of ϕ.

(iii) For affine linear functions $\phi(\mathbf{x}) = \mathbf{a} \cdot \mathbf{x} + \mathbf{b}$, finite difference approximations of its derivatives are exact. However, since $\nabla\phi = \mathbf{a}$, so we refine if $|\mathbf{a}| > C$. This is bad: we do not need to increase the resolution for field configurations where the error of finite difference approximations vanishes already.

Let us address these issues in order. The first is rather simple to fix: we simply multiply by the grid spacing Δx, and turn the refinement condition into

$$\Delta x|\nabla\phi| > C. \tag{A.8.2}$$

Effectively, this removes the division by the grid spacing in the finite difference approximation. Concretely, in the one-dimensional case, we have

$$C < \Delta x|\nabla\phi| \approx \Delta x \frac{\phi(x + \Delta x) - \phi(x - \Delta x)}{2\Delta x} = \frac{1}{2}(\phi(x + \Delta x) - \phi(x - \Delta x)). \tag{A.8.3}$$

We hence respect the grid spacing now; we refine if the difference between neighbouring cells' values becomes too large. This is already a much more sensible condition—we no longer refine indefinitely, but rather, until the change from cell to cell is small enough.

To address the second issue, (ii), we might think to rescale the gradient by $|\phi|$, turning it into

$$\frac{\Delta x|\nabla\phi|}{|\phi|} > C. \tag{A.8.4}$$

Although this is now invariant under $\phi \to \lambda\phi$, we have introduced two new issues: a potential division by zero, and a sensitivity to shift, $\phi \to \phi + \alpha$. The division by zero can be remedied by adding a typical scale/noise floor ϕ_0 of the field ϕ to the denominator, turning it into

$$\frac{\Delta x|\nabla\phi|}{|\phi| + \phi_0} > C. \tag{A.8.5}$$

This is resolution- and scale-invariant, so we have solved (i) and (ii), but (iii) is still unresolved—it got even worse—and we are now sensitive to shifts, $\phi \to \phi + \alpha$.

So, let us try to address (iii), together with this new issue. The shift-covariance issue is simple enough to deal with; any derivative of ϕ is shift-invariant, so we only use its derivatives in our condition. We are left to address (iii). Since we would like our condition's left-hand side to ideally evaluate to zero for affine-linear functions, we are bound to make it proportional to *second* derivatives of ϕ. The likely most infamous scalar second derivative of a function is its Laplacian, $\Delta\phi$. Being a second derivative, it maps affine-linear functions $\mathbf{a}\cdot\mathbf{x} + \mathbf{b}$ to zero, but unfortunately, it also annihilates the entire class of harmonic functions as well. For example, the function $\phi(x, y) = e^x \sin y$ is harmonic, but its finite difference approximations of derivatives are not trivially exact. Thus, using $\Delta\phi$ is not viable.

The most general second derivative of ϕ is its Hessian, which we denote by

$$H = \left[\frac{\partial^2\phi}{\partial x^i \partial x^j}\right]_{i,j=1}^{d}. \tag{A.8.6}$$

We would like to build a scalar from it that is sensitive to any second-order features, and which is isotropic. Under an orthogonal transformation $\mathbf{x} \to R\mathbf{x}$, with $R^\top R = I$, the Hessian transforms as $H \to RHR^\top$. Unfortunately, we have already ruled out $\operatorname{tr} H = \Delta\phi$ as an option, so the next-best isotropic scalar is the Frobenius norm

$$\|H\|_F^2 = \operatorname{tr}(H^\top H). \tag{A.8.7}$$

It is indeed isotropic, since

$$\operatorname{tr}(H^\top H) \to \operatorname{tr}\Big(\big(RHR^\top\big)^\top RHR^\top\Big) = \operatorname{tr}(RH^\top R^\top RHR^\top) = \operatorname{tr}(H^\top H). \tag{A.8.8}$$

In e.g. $d = 3$, we can write it out explicitly as

$$\|H\|_F^2 = \phi_{xx}^2 + \phi_{yy}^2 + \phi_{zz}^2 + 2\phi_{xy}^2 + 2\phi_{yz}^2 + 2\phi_{zx}^2. \tag{A.8.9}$$

Since all terms are non-negative, any non-zero second-order derivative will be picked up by $\|H\|_F$. Moreover, due to isotropy, it is agnostic to the feature's orientation—we have thus found an ingredient for a refinement condition solving (iii).

Taking into account everything we have established in the above, a good refinement condition is

$$\text{refine if} \quad \frac{\Delta x^4 \|H\|_F^2}{\Delta x^2 |\nabla \phi|^2 + \phi_0^2} > C. \tag{A.8.10}$$

This is a generalisation of the Löhner error estimate. At this point it is worth considering its behaviour in different regimes. In a region of small gradients—where $\Delta x^2 |\nabla \phi|^2 \ll \phi_0^2$, such as at the top of a bell curve—the condition collapses to

$$\frac{\Delta x^4}{\phi_0^2} \|H\|_F^2 > C. \tag{A.8.11}$$

This means that refinement is proportional to the curvature $\|H\|_F^2$. In the other limiting case—$\Delta x^2 |\nabla \phi|^2 \gg \phi_0^2$, such as near shocks or wavefronts—the condition turns into

$$\Delta x^2 \frac{\|H\|_F^2}{|\nabla \phi|^2} > C. \tag{A.8.12}$$

This is a measure for the relative rate of change of the gradient, loosely to be interpreted as $\nabla \log \nabla \phi$.

A.8.2 Subcycling

A.9 Discontinuous Galerkin Methods

B Numerical Relativity

B.1 Conventions

In the following, the spacetime metric g always has signature $(-+++)$. The induced metric γ and conformal metric $\tilde{\gamma}$ are Riemannian, that is, of signature $(+++)$. The Riemann tensor has the sign convention

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z. \tag{B.1.1}$$

We use the Einstein equations in the form

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}, \tag{B.1.2}$$

that is, with $c = G = 1$.

B.2 The $3+1$-Formalism

B.2.1 Foliations and Projectors

On a Lorentzian 3-manifold $(\mathcal{M}, g)$, given a function $t : \mathcal{M} \to \mathbb{R}$ such that $g(\,dt, dt) \leq 0$ everywhere, we call the covering of $\mathcal{M}$ by the sets

$$\Sigma_{t_0} = \{p \in \mathcal{M} \mid t(p) = t_0\} \tag{B.2.1}$$

a *spacelike foliation* of $(\mathcal{M}, g)$. If $(\mathcal{M}, g)$ is time-orientable and dt future-oriented, then the foliation $\{\Sigma_t\}$ defines a unique timelike unit normal vector field n, perpendicular to the foliation everywhere, by

$$n = -\alpha\, dt^\sharp, \tag{B.2.2}$$

where $\alpha > 0$ is a function fixed by the normalisation condition

$$-1 = g(n,n) = \alpha^2 g^{tt} \quad \Leftrightarrow \quad g^{tt} = -\frac{1}{\alpha^2}. \tag{B.2.3}$$

In components, we have

$$n_\mu = -\alpha \delta^t_\mu, \quad n^\mu = -\alpha g^{\mu t}. \tag{B.2.4}$$

The vector n can be used to define an induced metric γ / a projection operator P,

$$\gamma = g + n^\flat \otimes n^\flat \quad \Leftrightarrow \quad P = \delta + n \otimes n^\flat \tag{B.2.5}$$

or equivalently in components,

$$\gamma_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu \quad \Leftrightarrow \quad P^\mu{}_\nu = \delta^\mu{}_\nu + n^\mu n_\nu. \tag{B.2.6}$$

Note: although related by metric-induced isomorphism, the fully contra- and covariant object is denoted by $\gamma_{\mu\nu}$ and $\gamma^{\mu\nu}$, respectively, wherease the (1,1)-tensor is denoted P. This is because in its fully contra- and covariant forms, it is best interpreted as an induced metric, whereas the (1,1)-form is better interpreted as a projector.

This projector has the following properties:

(i) $P^\mu{}_\nu n^\nu = 0$;

(ii) If $g(X,n) = 0$ ($\Leftrightarrow X \in \Gamma(T\Sigma)$) then $P^\mu{}_\nu X^\nu = X^\mu$;

(iii) $P^\mu{}_\nu P^\nu{}_\lambda = P^\mu{}_\lambda$;

(iv) $\gamma_{\mu\nu} = \gamma_{\nu\mu}$;

(v) $P^{\mu}{}_{\mu} = 3$.

(vi) For $X, Y \in \Gamma(T\Sigma)$, it holds that $g(X,Y) = \gamma(X,Y)$.

These properties establish that pointwise, $P : T_p\ \mathcal{M} \to T_p\Sigma_{t(p)}$ is a surjective orthogonal projection. In particular, $P|_{T\Sigma} = \mathrm{id}_{T\Sigma}$, and $P|_{N\Sigma} = 0$. The last property in the above shows that γ is indeed the induced metric on Σ when restricting arguments to $T\Sigma$.

At this point, we introduce some additional notation. Given a tensor $T \in T^{(r,s)}_p\ \mathcal{M}$, its projection $PT \in T^{(r,s)}_p\Sigma$ is defined by

$$(PT)^{\mu\ldots}{}_{\nu\ldots} = P^{\mu}{}_{\lambda}\ldots P^{\rho}{}_{\nu}\ldots T^{\lambda\ldots}{}_{\rho\ldots}, \tag{B.2.7}$$

that is, PT has as components those of T contracted with a projector on each index. In this notation we have, for example,

$$Pn = 0 \quad \text{and} \quad Pg = \gamma. \tag{B.2.8}$$

B.2.2 Extrinsic Curvature

In the codimension 1 case, the extrinsic curvature tensor is defined by

$$K(X,Y) = g(n, \nabla_{PX} PY) \quad \Leftrightarrow \quad K_{\mu\nu} X^{\mu} Y^{\nu} = n_{\mu} (PX)^{\nu} \nabla_{\nu} (PY)^{\mu} \tag{B.2.9}$$

for $X, Y \in \Gamma(T\ \mathcal{M})$ and ∇ the Levi-Civita connection on $(\mathcal{M}, g)$. For $X, Y \in \Gamma(T\Sigma)$, this reduces to

$$K(X,Y) = g(n, \nabla_X Y) = -g(\nabla_X n, Y), \tag{B.2.10}$$

which hence measures the failure of parallel transport of Y along X to remain tangent. The second expression shows that equivalently, it measures the rate of change of the normal vector n along X, as projected onto Y.

Defining the acceleration a as

$$a = \nabla_n n \quad \Leftrightarrow \quad a^{\mu} = n^{\nu} \nabla_{\nu} n^{\mu}, \tag{B.2.11}$$

we have the following properties:

(i) $K_{\mu\nu} = -P^{\lambda}{}_{\mu} \nabla_{\lambda} n_{\nu} = -\nabla_{\mu} n_{\nu} - n_{\mu} a_{\nu}$;

(ii) $K_{\mu\nu} = K_{\nu\mu}$ as a consequence of vanishing torsion;

(iii) $n^{\mu} K_{\mu\nu} = K_{\mu\nu} n^{\nu} = 0$.

These follow straightforwardly either from the definition or the alternative characterisation of the extrinsic curvature above.

We denote the trace of the extrinsic curvature by K, and note that

$$K = g^{\mu\nu} K_{\mu\nu} = \gamma^{\mu\nu} K_{\mu\nu} \tag{B.2.12}$$

due to the transverse property (iii).

Finally, we can relate $K_{\mu\nu}$ to the Lie derivative of γ along n as follows:

$$K_{\mu\nu} = -\frac{1}{2} (\mathcal{L}_n \gamma)_{\mu\nu}. \tag{B.2.13}$$

This follows by expanding the right-hand side and making use of

$$\mathcal{L}_n n^{\flat} = a^{\flat}. \tag{B.2.14}$$

Note that the Lie derivative of the *vector* n would be zero, $\mathcal{L}_n n = [n, n] = 0$—in the identity above, we take the Lie derivative of the *covector* $n^\flat$.

B.2.3 Induced Connection and Intrinsic Curvature

B.2.3.1 Induced Connection

Given the Levi-Civita connection ∇ on $(\mathcal{M}, g)$, we define the *induced/three-dimensional/spatial covariant derivative* $\bar{\nabla}$ on the foliation $\Sigma = \{\Sigma_t\}$ as

$$\bar{\nabla} T = P(\nabla T) \quad \Leftrightarrow \quad \big(\bar{\nabla}_\lambda T\big)^{\mu\ldots}{}_{\nu\ldots} = P^\alpha{}_\lambda P^\mu{}_\beta \ldots P^\gamma{}_\nu \ldots \nabla_\alpha T^{\beta\ldots}{}_{\gamma\ldots} \tag{B.2.15}$$

for any $T \in \Gamma\big(T^{(r,s)}\Sigma\big)$. Note that T must be tangent to Σ—i.e. have no normal components—for this to define a connection on $T\Sigma$. The direction X of the derivative does not necessarily have to be in $T\Sigma$, but if it is, it holds that

$$\bar{\nabla}_X T = P(\nabla_X T), \quad \text{if } X \in \Gamma(T\Sigma). \tag{B.2.16}$$

Otherwise, for general $X \in \Gamma(T\mathcal{M}))$, we have

$$\bar{\nabla}_X T = P(\nabla_{PX} T). \tag{B.2.17}$$

The fact that $\bar{\nabla}$ defines a connection on $T\Sigma$ is clear from the fact that ∇ is a connection, and that P is $C^\infty(\mathcal{M})$-linear. In particular, we have the Leibniz rule

$$\begin{aligned} \bar{\nabla}_X(T \otimes S) &= P(\nabla_{PX}(T \otimes S)) = P(\nabla_{PX} T) \otimes PS + PT \otimes P(\nabla_{PX} S) \\ &= \big(\bar{\nabla}_X T\big) \otimes S + T \otimes \big(\bar{\nabla}_X S\big). \end{aligned} \tag{B.2.18}$$

Hence, for example, on a 1-form $\eta \in \Gamma(T^*\Sigma)$,

$$\big(\bar{\nabla}_X \eta\big)(Y) = X(\eta(Y)) - \eta\big(\bar{\nabla}_X Y\big). \tag{B.2.19}$$

Further, $\bar{\nabla}$ is torsion-free and metric-compatible with γ:

- Torsion: For any $X, Y \in \Gamma(T\Sigma)$, we have

$$\begin{aligned} \bar{T}(X, Y) &= \bar{\nabla}_X Y - \bar{\nabla}_Y X - \underbrace{[X, Y]}_{\in \Gamma(T\Sigma)} = P(\nabla_X Y - \nabla_Y X - [X, Y]) \\ &= PT(X, Y) = 0 \end{aligned} \tag{B.2.20}$$

- Metricity: For any $X, Y, Z \in \Gamma(T\Sigma)$,

$$\begin{aligned} X(\gamma(Y, Z)) &= X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z) \\ &= g(P\nabla_X Y, Z) + g(Y, P\nabla_X Z) = \gamma\big(\bar{\nabla}_X Y, Z\big) + \gamma\big(Y, \bar{\nabla}_X Z\big). \end{aligned} \tag{B.2.21}$$

B.2.3.2 Intrinsic Curvature

Since $\bar{\nabla}$ defines a connection on $T\Sigma$, and by that on each of the leaves of the foliation, it comes with an associated Riemannian curvature tensor, defined by

$$\bar{R}(X, Y)Z = \bar{\nabla}_X \bar{\nabla}_Y Z - \bar{\nabla}_Y \bar{\nabla}_X Z - \bar{\nabla}_{[X,Y]} Z \tag{B.2.22}$$

In components, this reads

$$\big(\bar{R}(X, Y)Z\big)^\mu = \bar{R}^\mu{}_{\nu\rho\sigma} Z^\nu X^\rho Y^\sigma. \tag{B.2.23}$$

We further have an associated Ricci tensor and scalar,

$$\bar{R}_{\mu\nu} = \bar{R}^{\lambda}{}_{\mu\lambda\nu}, \quad \bar{R} = g^{\mu\nu}\bar{R}_{\mu\nu} = \gamma^{\mu\nu}\bar{R}_{\mu\nu}, \tag{B.2.24}$$

where the last equality holds because $\bar{R}^{\mu}{}_{\nu\rho\sigma}$ vanishes when contracted with the foliation normal n on any index.

Since $\bar{\nabla}$ is the Levi-Civita connection with respect to γ, $\bar{R}_{\mu\nu\rho\sigma}$ enjoys the same symmetries as the ambient Riemann tensor. Further, the Ricci identity

$$\big(\bar{\nabla}_\mu\bar{\nabla}_\nu - \bar{\nabla}_\nu\bar{\nabla}_\mu\big)V^\lambda = \bar{R}^{\lambda}{}_{\rho\mu\nu}V^\rho \tag{B.2.25}$$

holds.

B.2.4 Gauss-Codazzi-Ricci Equations

Theorem B.1 *(Gauss Equation)* With the definitions given in the above, the *Gauss equation*

$$(PR)^{\lambda}{}_{\rho\mu\nu} = \bar{R}^{\lambda}{}_{\rho\mu\nu} + K^{\lambda}{}_{\mu}K_{\rho\nu} - K^{\lambda}{}_{\nu}K_{\rho\mu} \tag{B.2.26}$$

holds. Contracting over $\lambda\mu$ leads to (after renaming indices)

$$(PR)_{\mu\nu} + R_{\lambda\mu\rho\nu}n^\lambda n^\rho = \bar{R}_{\mu\nu} + KK_{\mu\nu} - K^{\lambda}{}_{\mu}K_{\nu\lambda}. \tag{B.2.27}$$

Tracing with respect to $\gamma^{\mu\nu}$ (or equivalently, $g^{\mu\nu}$) leads to

$$R + 2R_{\mu\nu}n^\mu n^\nu = \bar{R} + K^2 - K_{\mu\nu}K^{\mu\nu}. \tag{B.2.28}$$

Proof (Hint) To show this, first compute $\bar{\nabla}_\mu\bar{\nabla}_\nu X^\sigma$ for $X \in \Gamma(T\Sigma)$. Use $K_{\mu\nu} = -P^{\lambda}{}_{\mu}\nabla_\lambda n_\nu$ to trade derivatives of n for the extrinsic curvature. □

Theorem B.2 *(Codazzi Equation)* Under the same assumptions, the *Codazzi equation*

$$P\big(R_{\rho\lambda\mu\nu}n^\rho\big) = \bar{\nabla}_\mu K_{\nu\lambda} - \bar{\nabla}_\nu K_{\mu\lambda} \tag{B.2.29}$$

holds. Tracing over $\lambda\mu$ with respect to $\gamma^{\lambda\mu}$ (or equivalently, $g^{\lambda\mu}$) leads to

$$P^{\mu}{}_{\lambda}n^\nu R_{\mu\nu} = \bar{\nabla}_\lambda K - \bar{\nabla}_\mu K^{\mu}{}_{\lambda} \tag{B.2.30}$$

Lemma B.3 With the setup from the above, we have:

(i) The following Lie derivatives of the projector/induced metric:

$$\begin{aligned} \mathcal{L}_n\gamma^{\mu\nu} &= 2K^{\mu\nu} + n^\mu a^\nu + a^\mu n^\nu, \\ \mathcal{L}_n P^{\mu}{}_{\nu} &= n^\mu a_\nu, \\ \mathcal{L}_n\gamma_{\mu\nu} &= -2K_{\mu\nu} \end{aligned} \tag{B.2.31}$$

(ii) If $T \in \Gamma\big(T^{(0,s)}\Sigma\big)$, that is, if T is a foliation-tangent covariant tensor, then so is $\mathcal{L}_n T$. That is,

$$\mathcal{L}_n T = P(\mathcal{L}_n T), \quad T \in \Gamma\big(T^{(0,s)}\Sigma\big). \tag{B.2.32}$$

(iii) The acceleration covector a_μ and the lapse α satisfy the identity

$$a_\mu = P^{\lambda}{}_{\mu}\nabla_\lambda \log\alpha = \bar{\nabla}_\mu \log\alpha. \tag{B.2.33}$$

Theorem B.4 *(Ricci Equation)* Again with the same setup, it holds the *Ricci equation*

$$P^{\alpha}{}_{\mu}n^\rho P^{\beta}{}_{\nu}n^\sigma R_{\alpha\rho\beta\sigma} = \mathcal{L}_n K_{\mu\nu} + K_{\mu\lambda}K^{\lambda}{}_{\nu} + \frac{1}{\alpha}\bar{\nabla}_\mu\bar{\nabla}_\nu\alpha \tag{B.2.34}$$

Note that on the left-hand side, the projections of the α and β indices are superfluous, since the $n \otimes n^\flat$-part of P vanishes when contracted due to the symmetries of the Riemann tensor. That is, we may write the Ricci equation more compactly as

$$R_{\mu\rho\nu\sigma} n^\rho n^\sigma = \mathcal{L}_n K_{\mu\nu} + K_{\mu\lambda} K^\lambda{}_\nu + \frac{1}{\alpha} \bar{\nabla}_\mu \bar{\nabla}_\nu \alpha. \tag{B.2.35}$$

Since both sides yield zero when contracted with n, the traces with respect to $g^{\mu\nu}$ and $\gamma^{\mu\nu}$ are identical and read

$$R_{\mu\nu} n^\mu n^\nu = \mathcal{L}_n K - K_{\mu\nu} K^{\mu\nu} + \frac{1}{\alpha} \bar{\nabla}^\mu \bar{\nabla}_\mu \alpha. \tag{B.2.36}$$

With the triad of the Gauss-Codazzi-Ricci equations, we can now isolate the final projection of the Ricci tensor, the purely spatial one, and find an expression for the full Ricci scalar in terms of the intrinsic and extrinsic quantities.

Theorem B.5 The above results combine into

$$\begin{aligned} (PR)_{\mu\nu} &= -\mathcal{L}_n K_{\mu\nu} + K K_{\mu\nu} - 2 K_{\mu\lambda} K^\lambda{}_\nu + \bar{R}_{\mu\nu} - \frac{1}{\alpha} \bar{\nabla}_\mu \bar{\nabla}_\nu \alpha, \\ R &= -2 \mathcal{L}_n K + K^2 + K_{\mu\nu} K^{\mu\nu} + \bar{R} - \frac{2}{\alpha} \bar{\nabla}^\mu \bar{\nabla}_\mu \alpha \end{aligned} \tag{B.2.37}$$

B.2.5 The Einstein Equations in the $3+1$-Formalism

We work with the Einstein equations in the convention where

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}. \tag{B.2.38}$$

Taking the trace of this equation with respect to $g^{\mu\nu}$, we obtain

$$\frac{1}{2} R = 2\Lambda - \frac{1}{2} 8\pi T, \tag{B.2.39}$$

with $T = g^{\mu\nu} T_{\mu\nu}$ the trace of the energy-momentum tensor. Inserting this into the Einstein equations yields their trace-reversed counterpart,

$$R_{\mu\nu} = 8\pi \Big(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \Big) + \Lambda g_{\mu\nu}. \tag{B.2.40}$$

Note that a second trace-reversal brings us back to the original form, implying that the Einstein equations and their trace-reverse are equivalent.

We define the energy density ρ, the energy current j^μ, and the stress tensor $S_{\mu\nu}$ as measured by Eulerian observers traveling along the integral lines of n by

$$\begin{aligned} \rho &= n^\mu n^\nu T_{\mu\nu}, \\ j_\mu &= -P^\alpha{}_\mu n^\nu T_{\alpha\nu} \\ S_{\mu\nu} &= P^\alpha{}_\mu P^\beta{}_\nu T_{\alpha\beta} \end{aligned} \tag{B.2.41}$$

These are simply the time-time, time-space and space-space projections of the energy-momentum tensor, and allow us to decompose it as

$$T_{\mu\nu} = \rho n_\mu n_\nu + j_\mu n_\nu + j_\nu n_\mu + S_{\mu\nu}. \tag{B.2.42}$$

Its trace T is given by

$$T = S - \rho \tag{B.2.43}$$

where $S = g^{\mu\nu}S_{\mu\nu} = \gamma^{\mu\nu}S_{\mu\nu}$.

Together with the results from the previous section, these definitions now allow us to cast the Einstein equations into a first-order system for a second-order in time evolution of the spatial metric $\gamma_{\mu\nu}$, with the extrinsic curvature $K_{\mu\nu}$ serving as an auxiliary variable:

Theorem B.6 The Einstein equations written in terms of $\alpha, n, \gamma, K, \rho, j$ and S read

$$\begin{aligned}
\mathcal{H} &:= 2\mathcal{E}_{\mu\nu}n^\mu n^\nu = \bar{R} + K^2 - K_{\mu\nu}K^{\mu\nu} - 2\Lambda - 16\pi\rho = 0,\\
\mathcal{M}_\mu &:= P^\lambda{}_\mu n^\nu \mathcal{E}_{\lambda\nu} = \bar{\nabla}_\mu K - \bar{\nabla}_\nu K^\nu{}_\mu + 8\pi j_\mu = 0,\\
\mathcal{L}_n\gamma_{\mu\nu} &= -2K_{\mu\nu},\\
\mathcal{L}_n K_{\mu\nu} &= KK_{\mu\nu} - 2K_{\mu\lambda}K^\lambda{}_\nu + \bar{R}_{\mu\nu} - \frac{1}{\alpha}\bar{\nabla}_\mu\bar{\nabla}_\nu\alpha\\
&\quad -\Lambda\gamma_{\mu\nu} - 8\pi\left(\left(S_{\mu\nu} - \frac{1}{2}S\gamma_{\mu\nu}\right) + \frac{1}{2}\rho\gamma_{\mu\nu}\right).
\end{aligned} \tag{B.2.44}$$

The first two equations are derived by taking the normal-normal and normal-tangential projections of the original Einstein equations $\mathcal{E} = G_{\mu\nu} + \Lambda g_{\mu\nu} - 8\pi T_{\mu\nu} = 0$ and subsequently identifying the definitions of ρ and j as well as applying the Gauss and Codazzi equations, respectively. These two equations involve no time derivatives, and are hence a set of four constraints.

The remaining two equations are first order in time (by the presence of the $\mathcal{L}_n$ normal derivatives), and hence dynamical. The first of the two is simply a consequence of the definition of the extrinsic curvature. The latter is the result of projecting the trace-reversed Einstein equations onto $T\Sigma$ by contracting with P on both indices, and subsequently applying the first equation in (B.2.37) to re-express the projected Ricci tensor $(PR)_{\mu\nu}$.

B.2.6 Adapted Coordinates

B.2.6.1 The Metric and Bases

We can amend the time function $t : \mathcal{M} \to \mathbb{R}$ (for which $dt \neq 0$) which defines the foliation Σ_t to a coordinate system *adapted to* Σ_t by introducing three additional functions $x^i : \mathcal{M} \to \mathbb{R}$ such that $dx^i \neq 0$ and $x^i|_{\Sigma_t}$ label points uniquely on any leaf Σ_t. This defines a basis $\{\partial_t, \partial_i\}$ of $T\mathcal{M}$ and an associated dual basis $\{dt, dx^i\}$ of $T^*\mathcal{M}$.

A natural question is to ask how the two time directions that we now have are related—we have the (coordinate-independent) timelike normal vector n, and additionally, the coordinate time direction ∂_t. To this end, we introduce the shift vector β, defined by

$$\beta := \partial_t - \alpha n \quad\Leftrightarrow\quad \partial_t = \alpha n + \beta \quad\Leftrightarrow\quad n = \frac{1}{\alpha}(\partial_t - \beta). \tag{B.2.45}$$

This vector is tangent to the foliation, since

$$dt(\beta) = dt(\partial_t) \underbrace{- \alpha\, dt}_{=n^\flat}(n) = 1 - g(n,n) = 0. \tag{B.2.46}$$

Hence, it quantifies the failure of ∂_t to be normal to the foliation; that is, how much the coordinate lines of t *shift* with respect to the geometric normal direction specified by n.

With the lapse α, the shift β, and the induced metric γ, we now have all the objects we need for the 3+1-decomposition of the metric tensor g. Its components in the basis $\{\partial_t, \partial_i\}$ read

$$
\begin{aligned}
g_{ij} &= g(\partial_i, \partial_j) = \gamma(\partial_i, \partial_j) =: \gamma_{ij} \\
g_{ti} &= g(\partial_t, \partial_i) = \gamma_{ij}\beta^j =: \beta_i, \\
g_{tt} &= g(\partial_t, \partial_t) = -\alpha^2 + \beta_i\beta^i.
\end{aligned}
\tag{B.2.47}
$$

In matrix form, this reads

$$
\big[g_{\mu\nu}\big]_{\mu,\nu\in\{t,i\}} = \begin{pmatrix} -\alpha^2 + \beta_k\beta^k & \beta_j \\ \beta_i & \gamma_{ij} \end{pmatrix},
\tag{B.2.48}
$$

and tensorially, we may write

$$
\begin{aligned}
g &= g_{\mu\nu}\, dx^\mu \otimes dx^\nu \\
&= -\alpha^2\, dt \otimes dt + \gamma_{ij}(\, dx^i + \beta^i\, dt)(\, dx^j + \beta^j\, dt).
\end{aligned}
\tag{B.2.49}
$$

The corresponding inverse metric has the components

$$
g^{tt} = -\frac{1}{\alpha^2}, \quad g^{ti} = \frac{\beta^i}{\alpha^2}, \quad g^{ij} = \gamma^{ij} - \frac{\beta^i\beta^j}{\alpha^2},
\tag{B.2.50}
$$

which takes the matrix form

$$
[g^{\mu\nu}]_{\mu,\nu\in\{t,i\}} = \begin{pmatrix} -1/\alpha^2 & \beta^j/\alpha^2 \\ \beta^i/\alpha^2 & \gamma^{ij} - \beta^i\beta^j/\alpha^2 \end{pmatrix}.
\tag{B.2.51}
$$

Here, γ^{ij} denotes the matrix inverse of γ_{ij}, characterised uniquely by the condition

$$
\gamma^{ij}\gamma_{jk} = \delta^i{}_k.
\tag{B.2.52}
$$

Tensorially, the inverse metric can be expressed compactly as

$$
g^{-1} = -n \otimes n + \gamma^{ij}\partial_i \otimes \partial_j.
\tag{B.2.53}
$$

As we have seen in the previous sections, we often care about normal projections of tensors, not their coordinate time components. Because of this, another useful basis is $\{n, \partial_i\}$, alongside its dual $\{-n^\flat, dx^i\} = \{\alpha\, dt, dx^i\}$, with indices $\perp, i$. Generically, this is *not* a coordinate basis. However, it does give the metric a nice block-diagonal structure:

$$
g_{\perp\perp} = g(n,n) = -1, \quad g_{\perp i} = g(n, \partial_i) = 0, \quad g_{ij} = g(\partial_i, \partial_j) = \gamma_{ij},
\tag{B.2.54}
$$

and

$$
g^{\perp\perp} = -1, \quad g^{\perp i} = 0, \quad g^{ij} = \gamma^{ij}.
\tag{B.2.55}
$$

For example, in terms of this basis, the Hamiltonian and momentum constraints are nothing but the Einstein equations

$$
\mathcal{H} = G_{\perp\perp} - 8\pi T_{\perp\perp} = 0, \quad \mathcal{M}_i = G_{\perp i} - 8\pi T_{\perp i}.
\tag{B.2.56}
$$

In this basis, foliation-tangent tensors, i.e. objects $T \in \Gamma\big(T^{(r,s)}\Sigma\big)$, only have components where all indices are spatial. That is, a component vanishes if any of its indices is $\perp$.

Back in the coordinate basis $\{\partial_t, \partial_i\}$, the same is true for upstairs indices; for example, for a vector $X \in \Gamma(T\Sigma)$, we have

$$
X^t = dt(X) = -\frac{1}{\alpha} n^\flat(X) = 0,
\tag{B.2.57}
$$

since $X \perp n$. For downstairs indices, this is generally not true, as, for instance,

$$X_t = g_{t\mu}X^\mu = g_{ti}X^i = \beta_i X^i. \tag{B.2.58}$$

However, the X_t-component is not independent of the other three, $X_i = \gamma_{ij}X^j$. Because of this, it is irrelevant, and hence, the full information about any foliation-tangent tensor $T \in \Gamma\big(T^{(r,s)}\Sigma\big)$ (such as $\gamma_{\mu\nu}$, $K_{\mu\nu}$, $\bar{R}_{\mu\nu\rho\sigma}$, $S_{\mu\nu}$ etc.) is stored in its purely spatial components. We also dont need the downstairs-, t components for contractions; it will always be contracted with an upstairs-t component, which is zero. Concretely, this implies that for example,

$$K_{\mu\nu}K^{\mu\nu} = K_{ij}K^{ij} \quad \text{and} \quad \bar{R}_{ij} = \bar{R}^{\mu}{}_{i\mu j} = \bar{R}^{k}{}_{ikj}. \tag{B.2.59}$$

B.2.6.2 Extrinsic and Intrinsic Curvature

Lemma B.7 For any vector field N that is normal to $T\Sigma$ (not necessarily the unit-normal n), and a contravariant tangential tensor $T \in \Gamma\big(T^{(0,s)}\Sigma\big)$, we have

$$\mathcal{L}_{fN}T = f\mathcal{L}_N T. \tag{B.2.60}$$

Note that this is not necessarily true if T is not foliation-tangential or has upstairs indices.

This lemma might seem kind of useless, but it allows us to write down an explicit expression for the extrinsic curvature in an adapted coordinate basis:

$$K_{\mu\nu} = -\frac{1}{2}\mathcal{L}_n\gamma_{\mu\nu} = -\frac{1}{2}\mathcal{L}_{\frac{1}{\alpha}(\partial_t-\beta)}\gamma_{\mu\nu} = -\frac{1}{2\alpha}\big(\partial_t\gamma_{\mu\nu} - \mathcal{L}_\beta\gamma_{\mu\nu}\big) \tag{B.2.61}$$

Since the extrinsic curvature itself is also a foliation-tangent tensor, an analogous result holds. Writing directly in the adapted basis (and dropping any t-components), we obtain

$$\begin{aligned} K_{ij} &= -\frac{1}{2\alpha}\big(\partial_t\gamma_{ij} - \mathcal{L}_\beta\gamma_{ij}\big), \\ \mathcal{L}_n K_{ij} &= \frac{1}{\alpha}\big(\partial_t K_{ij} - \mathcal{L}_\beta K_{ij}\big) \end{aligned} \tag{B.2.62}$$

Writing out the Lie derivatives along β, we obtain the following relationships:

$$\begin{aligned} \partial_t\gamma_{ij} &= \beta^k\partial_k\gamma_{ij} + 2\gamma_{k(i}\partial_{j)}\beta^k - 2\alpha K_{ij}, \\ \partial_t K_{ij} &= \beta^k\partial_k K_{ij} + 2K_{k(i}\partial_{j)}\beta^k + \alpha\mathcal{L}_n K_{ij}. \end{aligned} \tag{B.2.63}$$

These will be useful to recast the 3+1 Einstein equations we derived in the previous section into adapted coordinates, in a form where they clearly describe the first-order time evolution of (γ_{ij}, K_{ij}).

In the 3+1 Einstein equations, beyond the extrinsic curvature, the intrinsic connection $\bar{\nabla}$ as well as its associated curavture $\bar{R}$ appears. We therefore also need to write these in terms of the adapted coordinate basis. Since $\bar{\nabla}$ is the Levi-Civita connection on any of the leaves with respect to the induced metric γ, its components in adapted coordinates read

$$\bar{\Gamma}^k{}_{ij} = \frac{1}{2}\gamma^{k\ell}\big(\partial_i\gamma_{j\ell} + \partial_j\gamma_{i\ell} - \partial_\ell\gamma_{ij}\big) \tag{B.2.64}$$

Correspondingly, the associated Riemann curvature has the components

$$\bar{R}^k{}_{\ell ij} = \partial_i\bar{\Gamma}^k{}_{\ell j} - \partial_j\bar{\Gamma}^k{}_{\ell i} + \bar{\Gamma}^k{}_{ni}\bar{\Gamma}^n{}_{\ell j} - \bar{\Gamma}^k{}_{nj}\bar{\Gamma}^n{}_{\ell i} \tag{B.2.65}$$

B.2.6.3 The Einstein Equations in $3+1$-Adapted Coordinates

The equations (B.2.65) directly imply that the Einstein equations in adapted coordinates reads

$$\mathcal{H} = \bar{R} + K^2 - K_{ij}K^{ij} - 2\Lambda - 16\pi\rho = 0,$$

$$\mathcal{M}_i = \bar{\nabla}_i K - \bar{\nabla}_j K^j_{\ i} + 8\pi j_i = 0,$$

$$\begin{aligned}\partial_t \gamma_{ij} &= \beta^k \partial_k \gamma_{ij} + 2\gamma_{k(i}\partial_{j)}\beta^k - 2\alpha K_{ij}, \\ \partial_t K_{ij} &= \beta^k \partial_k K_{ij} + 2K_{k(i}\partial_{j)}\beta^k - \bar{\nabla}_i\bar{\nabla}_j\alpha + \alpha\Big(KK_{ij} - 2K_{ik}K^k_{\ j} + \bar{R}_{ij}\Big) \\ &\quad -\alpha\Lambda\gamma_{ij} - 8\pi\alpha\Big(S_{ij} - \frac{1}{2}(S-\rho)\gamma_{ij}\Big)\end{aligned} \tag{B.2.66}$$

The conservation equation $\nabla_\mu T^{\mu\nu} = 0$ further implies equations of motion for the energy density ρ and j_i. To derive them, we must first relate ρ and j_i to components of the energy-momentum tensor in adapted coordinates. To this end, we should recall that in adapted coordinates,

$$\begin{aligned} n^t &= \frac{1}{\alpha}, \quad & n^i &= -\frac{\beta^i}{\alpha}, \\ n_t &= -\alpha, \quad & n_i &= 0. \end{aligned} \tag{B.2.67}$$

Consequently, the projector $P^\mu_{\ \nu}$ has the components

$$\begin{aligned} P^t_{\ t} &= 0, \quad & P^t_{\ i} &= 0, \\ P^i_{\ t} &= \beta^i, \quad & P^i_{\ j} &= \delta^i_{\ j}, \end{aligned} \tag{B.2.68}$$

which may be summarised as $P^t_{\ \mu} = 0$ and $P^i_{\ \mu} = \delta^i_{\ \mu} + \beta^i\delta^t_{\ \mu}$ With this projector and the components of n, we can directly evaluate

$$\begin{aligned} \rho &= T^{\mu\nu}n_\mu n_\nu = \alpha^2 T^{tt}, \\ j^i &= -P^i_{\ \mu}T^{\mu\nu}n_\nu = \alpha(\beta^i T^{tt} + T^{it}), \\ S^{ij} &= P^i_{\ \mu}P^j_{\ \nu}T^{\mu\nu} = T^{ij} + 2\beta^{(i}T^{j)t} + \beta^i\beta^j T^{tt}. \end{aligned} \tag{B.2.69}$$

Here, we recall that $j^t = 0$, $S^{t\mu} = 0$, and that hence, we may raise and lower indices with γ_{ij}. Note that j_t and $S_{t\mu}$ are not necessarily zero, but by the reasoning given earlier, these components are not independent of the purely spatial ones and hence carry no additional information.

The above puts ρ, j^i and S^{ij} in terms of the components of $T^{\mu\nu}$. To evaluate projections of $\nabla_\mu T^{\mu\nu} = 0$, however, we need to invert this relationship, and express $T^{\mu\nu}$ in terms of ρ, j^i and S^{ij}. Carrying out this inversion leads to

$$T^{tt} = \frac{1}{\alpha^2}\rho, \qquad T^{it} = \frac{1}{\alpha}j^i - \frac{\beta^i}{\alpha^2}\rho, \qquad T^{ij} = S^{ij} - \frac{2}{\alpha}\beta^{(i}j^{j)} + \frac{\beta^i\beta^j}{\alpha^2}\rho. \tag{B.2.70}$$

Note that this is simply saying that

$$T = \rho n\otimes n + j\otimes n + n\otimes j + S. \tag{B.2.71}$$

The normal projection $\big(\nabla_\mu T^{\mu\nu}\big)n_\nu = 0$ then implies the equation of motion for ρ, reading

$$\partial_t\rho = \beta^k\partial_k\rho - 2j^k\partial_k\alpha + \alpha\big(K\rho + K_{ij}S^{ij} - \bar{\nabla}_i j^i\big). \tag{B.2.72}$$

This is derived by first integrating by parts, $\big(\nabla_\mu T^{\mu\nu}\big)n_\nu = \nabla_\mu(T^{\mu\nu}n_\nu) - T^{\mu\nu}\nabla_\mu n_\nu$, and then applying identities from the above to replace derivatives of n with the extrinsic curvature. The standard formula $\nabla_\mu X^\mu = \frac{1}{\sqrt{g}}\partial_\mu(\sqrt{g}X^\mu)$, together with $\sqrt{g} = \alpha\sqrt{\gamma}$, then allows the deduction of the result.

A similar equation of motion can be derived for $\partial_t j_i$ by projecting $\nabla_\mu T^{\mu\nu} = 0$ onto $T\Sigma$ using P, but I will not do this here.

B.3 Metric Derivative and Curvature Identities

B.3.1 Jacobi Formula

For the variation δ of a matrix M, we have the *Jacobi formula*

$$\delta \det M = \det M \cdot \operatorname{tr}(M^{-1}\delta M). \tag{B.3.1}$$

Using $g = \det\big[g_{\mu\nu}\big]$, this implies that

$$\delta g = g g^{\mu\nu}\delta g_{\mu\nu} = -g g_{\mu\nu}\delta g^{\mu\nu}. \tag{B.3.2}$$

For a derivative variation, $\delta = \partial_\lambda$, this can be expressed in terms of Christoffel symbols as

$$\partial_\lambda g = g g^{\mu\nu}\partial_\lambda g_{\mu\nu} = 2g\Gamma^{\mu}{}_{\mu\lambda}. \tag{B.3.3}$$

This expression further implies expressions for the divergence of $X \in \Gamma(T\mathcal{M})$ and the wave operator on $f \in C^\infty(\mathcal{M})$, reading

$$\begin{gathered}\operatorname{div} X = \nabla_\mu X^\mu = \frac{1}{\sqrt{g}}\partial_\mu(\sqrt{g}X^\mu) \\ \text{and} \\ \Box_g f = g^{\mu\nu}\nabla_\mu\nabla_\nu f = \frac{1}{\sqrt{g}}\partial_\mu(\sqrt{g}g^{\mu\nu}\partial_\nu f),\end{gathered} \tag{B.3.4}$$

respectively.

B.3.2 Connection and Curvature of Conformal Metrics

In the following, let $g_{\mu\nu}$ and $\tilde{g}_{\mu\nu}$ be two metrics on a d-dimensional manifold $\mathcal{M}$, related by

$$\tilde{g}_{\mu\nu} = e^{4\phi} g_{\mu\nu}. \tag{B.3.5}$$

Trivially, their inverses and determinants are then related by

$$\tilde{g}^{\mu\nu} = e^{-4\phi} g^{\mu\nu} \quad \text{and} \quad \tilde{g} = e^{4d\phi} g. \tag{B.3.6}$$

Denoting their Christoffel symbols by Γ and $\tilde{\Gamma}$, respectively, we have the identity

$$\tilde{\Gamma}^{\lambda}{}_{\mu\nu} = \Gamma^{\lambda}{}_{\mu\nu} + 2\Big(\delta^{\lambda}{}_{\mu}\partial_\nu\phi + \delta^{\lambda}{}_{\nu}\partial_\mu\phi - g_{\mu\nu}g^{\lambda\rho}\partial_\rho\phi\Big) \tag{B.3.7}$$

or equivalently,

$$\Gamma^{\lambda}{}_{\mu\nu} = \tilde{\Gamma}^{\lambda}{}_{\mu\nu} - 2\Big(\delta^{\lambda}{}_{\mu}\partial_\nu\phi + \delta^{\lambda}{}_{\nu}\partial_\mu\phi - g_{\mu\nu}g^{\lambda\rho}\partial_\rho\phi\Big). \tag{B.3.8}$$

We note that the difference between the two connection coefficients,

$$C^{\lambda}{}_{\mu\nu} := \tilde{\Gamma}^{\lambda}{}_{\mu\nu} - \Gamma^{\lambda}{}_{\mu\nu} = 2\Big(\delta^{\lambda}{}_{\mu}\tilde{\nabla}_\nu\phi + \delta^{\lambda}{}_{\nu}\tilde{\nabla}_\mu\phi - \tilde{g}_{\mu\nu}\tilde{g}^{\lambda\rho}\tilde{\nabla}_\rho\phi\Big) \tag{B.3.9}$$

is a tensor, and independent of whether one writes it in terms of $g_{\mu\nu}$ or $\tilde{g}_{\mu\nu}$ in the last term. This means that the Levi-Civita connection actions on an arbitrary tensor $T \in \Gamma\big(T^{(r,s)}\mathcal{M}\big)$ are related by

$$\tilde{\nabla}_\lambda T^{\mu\ldots}{}_{\nu\ldots} = \nabla_\lambda T^{\mu\ldots}{}_{\nu\ldots} + C^{\mu}{}_{\rho\lambda}T^{\rho\ldots}{}_{\nu\ldots} + \ldots - C^{\rho}{}_{\nu\lambda}T^{\mu\ldots}{}_{\rho\ldots} - \ldots \tag{B.3.10}$$

Concretely, using the action on vectors, one can derive that

$$R^{\lambda}{}_{\rho\mu\nu} = \tilde{R}^{\lambda}{}_{\rho\mu\nu} - \tilde{\nabla}_{\mu} C^{\lambda}{}_{\rho\nu} + \tilde{\nabla}_{\nu} C^{\lambda}{}_{\rho\mu} + C^{\lambda}{}_{\sigma\mu} C^{\sigma}{}_{\rho\nu} - C^{\lambda}{}_{\sigma\nu} C^{\sigma}{}_{\rho\mu}. \tag{B.3.11}$$

By contracting over $\lambda\mu$, we find a relationship between the Ricci tensors reading

$$R_{\mu\nu} = \tilde{R}_{\mu\nu} + 2\tilde{g}_{\mu\nu}\tilde{g}^{\rho\sigma}\tilde{\nabla}_{\rho}\tilde{\nabla}_{\sigma}\phi + 2(d-2)\big(\tilde{\nabla}_{\mu}\tilde{\nabla}_{\nu}\phi + 2\tilde{\nabla}_{\mu}\phi\tilde{\nabla}_{\nu}\phi - 2\tilde{g}_{\mu\nu}\tilde{g}^{\rho\sigma}\tilde{\nabla}_{\rho}\phi\tilde{\nabla}_{\sigma}\phi\big) \tag{B.3.12}$$

For the Ricci scalars—taking note that $\tilde{R} = \tilde{g}^{\mu\nu}\tilde{R}_{\mu\nu}$ and $R = g^{\mu\nu}R_{\mu\nu}$ are traced with respect to their respective metric—we obtain

$$R = e^{4\phi}\big(\tilde{R} + 4(d-1)\tilde{g}^{\mu\nu}\tilde{\nabla}_{\mu}\tilde{\nabla}_{\nu}\phi - 4(d-2)(d-1)\tilde{g}^{\mu\nu}\tilde{\nabla}_{\mu}\phi\tilde{\nabla}_{\nu}\phi\big). \tag{B.3.13}$$

B.3.3 Bochner Formula and the Ricci Tensor

For two scalar fields φ and ψ, the *Bochner formula*

$$\begin{aligned}\frac{1}{2}\Box_g\big(\nabla_{\mu}\varphi\nabla^{\mu}\psi\big) = \frac{1}{2}\big(\nabla_{\mu}\Box_g\varphi\big)\nabla^{\mu}\psi + \frac{1}{2}\nabla_{\mu}\varphi(\nabla^{\mu}\Box_g\psi)\\ +\nabla_{\mu}\nabla_{\nu}\varphi\nabla^{\mu}\nabla^{\nu}\psi + R^{\mu\nu}\nabla_{\mu}\varphi\nabla_{b}\psi.\end{aligned} \tag{B.3.14}$$

Using the coordinate functions $\varphi = x^{\mu}$, $\psi = x^{\nu}$, it follows that

$$R^{\mu\nu} = \frac{1}{2}g^{\lambda\rho}\partial_{\lambda}\partial_{\rho}g^{\mu\nu} - \frac{1}{2}\Gamma^{\lambda}\partial_{\lambda}g^{\mu\nu} + \partial^{(\mu}\Gamma^{\nu)} - \Gamma^{\mu}{}_{\lambda\rho}\Gamma^{\nu\lambda\rho}, \tag{B.3.15}$$

where $\Gamma^{\mu} = g^{\lambda\rho}\Gamma^{\mu}{}_{\lambda\rho}$. Explicitly lowering the indices leads to

$$\begin{aligned}R_{\mu\nu} = -\frac{1}{2}g^{\lambda\rho}\partial_{\lambda}\partial_{\rho}g_{\mu\nu} + g_{\lambda(\mu}\partial_{\nu)}\Gamma^{\lambda}\\ +\frac{1}{2}\Gamma^{\lambda}\partial_{\lambda}g_{\mu\nu} + g^{\lambda\rho}g^{\sigma\tau}\partial_{\lambda}g_{\mu\sigma}\partial_{\rho}g_{\nu\tau} - \Gamma_{\mu\rho\sigma}\Gamma_{\nu}{}^{\rho\sigma}.\end{aligned} \tag{B.3.16}$$

This expression will motivate why in BSSNOK, one promotes a version of Γ^{μ} to be a dynamical variable; in this case, the only second-order part in $R_{\mu\nu}$ is the well-behaved (in that case, elliptic) expression $g^{\lambda\rho}\partial_{\lambda}\partial_{\rho}g_{\mu\nu}$.

For numerical implementations, we should address one issue the above has, though. This issue is that the expression we obtained for the Ricci tensor depends on both ∂g as well as Γ. This increases register pressure in GPU-based implementations, which we would like to avoid; hence, we should make use of the identity

$$\partial_{\lambda}g_{\mu\nu} = \Gamma_{\mu\nu\lambda} + \Gamma_{\nu\mu\lambda} \tag{B.3.17}$$

to re-express everything in terms of Christoffel symbols. Doing so yields

$$\begin{aligned}R_{\mu\nu} = -\frac{1}{2}g^{\lambda\rho}\partial_{\lambda}\partial_{\rho}g_{\mu\nu} + g_{\lambda(\mu}\partial_{\nu)}\Gamma^{\lambda}\\ +\Gamma^{\lambda}\Gamma_{(\mu\nu)\lambda} + \Gamma_{\lambda\rho\mu}\Gamma^{\lambda\rho}{}_{\nu} + 2\Gamma_{\lambda\rho(\mu}\Gamma_{\nu)}{}^{\lambda\rho},\end{aligned} \tag{B.3.18}$$

which is now written entirely in terms of g, $\partial^2 g$, and the two flavours of Γ.

B.4 BSSNOK

In this section, we take what we have derived so far and build atop it the BSSNOK formalism. This involves

- defining the BSSNOK variables in terms of the ADM variables γ_{ij} and K_{ij}, scaling out the conformal factor to isolate the conformal metric $\tilde{\gamma}_{ij}$ and trace-free extrinsic curvature $\tilde{A}_{ij}$;
- deriving their first time derivative expressions from the known expressions for $\partial_t \gamma_{ij}$ and $\partial_t K_{ij}$ via the ADM evolution equations, as well as re-expressing the Hamiltonian and momentum constraints in terms of the new conformal variables;
- and finally, relating the physical induced connections and curvatures to their conformal counterparts using conformal transformations and the auxiliary connection quantities.

In doing so, we will use many of the identities from the previous section, and establish a number of additional ones to handle the conformal rescaling. Unfortunately, these algebraic definitions are rather lengthy and messy, but they ultimately lead to a strongly hyperbolic formulation of general relativity well-suited to be implemented in high-performance computing (HPC) code.

B.4.1 BSSNOK Reparametrisation and Evolution Equations

Definition B.8 *(BSSN Variables)* The dynamical variables in the BSSN formulation of GR are

(i) the *conformal factor* ϕ, defined as

$$\phi = -\frac{1}{12}\log\gamma, \tag{B.4.1}$$

(ii) the *conformal metric* $\tilde{\gamma}_{ij}$, given by

$$\tilde{\gamma}_{ij} = e^{4\phi}\gamma_{ij}, \tag{B.4.2}$$

(iii) the trace of the extrinsic curvature K, defined by

$$K = \gamma^{ij}K_{ij}, \tag{B.4.3}$$

(iv) the traceless conformal extrinsic curvature $\tilde{A}_{ij}$, expressed as

$$\tilde{A}_{ij} = e^{4\phi}\left(K_{ij} - \frac{1}{3}K\gamma_{ij}\right) \tag{B.4.4}$$

(v) and the contracted conformal connection $\tilde{\Gamma}^i$, defined as

$$\tilde{\Gamma}^i = \tilde{\gamma}^{k\ell}\tilde{\Gamma}^i{}_{k\ell}, \quad \text{with} \quad \tilde{\Gamma}^i{}_{k\ell} = \frac{1}{2}\tilde{\gamma}^{ij}\big(\partial_k\tilde{\gamma}_{\ell j} + \partial_\ell\tilde{\gamma}_{kj} - \partial_j\tilde{\gamma}_{k\ell}\big). \tag{B.4.5}$$

Corollary B.9 These definitions have a number of important immediate corollaries:

(i) The determinant of the conformal metric $\tilde{\gamma}_{ij}$ is 1, that is,

$$\tilde{\gamma} = e^{3\cdot 4\phi}\gamma = e^{-\log\gamma}\gamma = 1. \tag{B.4.6}$$

This has a few important consequences; firstly, the trace of its Christoffel symbols vanishes,

$$\tilde{\Gamma}^i{}_{ki} = \frac{1}{2}\tilde{\gamma}^{ij}\partial_k\tilde{\gamma}_{ij} = \frac{1}{\sqrt{\tilde{\gamma}}}\partial_k\sqrt{\tilde{\gamma}} = \partial_k 1 = 0. \tag{B.4.7}$$

Moreover, its contracted Christoffel symbols $\tilde{\Gamma}^i$ may be written as

$$\tilde{\Gamma}^i = -\partial_k\tilde{\gamma}^{ki} \tag{B.4.8}$$

(ii) $\tilde{A}_{ij}$ is traceless with respect to both $\tilde{\gamma}^{ij}$ and γ^{ij}, that is,

$$\tilde{\gamma}^{ij}\tilde{A}_{ij} = \gamma^{ij}\tilde{A}_{ij} = 0. \tag{B.4.9}$$

(iii) The definitions of $\phi, \tilde{\gamma}_{ij}, K$ and $\tilde{A}_{ij}$ can be inverted for γ_{ij} and K_{ij}, with the inversions reading

$$\gamma_{ij} = e^{-4\phi}\tilde{\gamma}_{ij} \quad \text{and} \quad K_{ij} = e^{-4\phi}\left(\tilde{A}_{ij} + \frac{1}{3}K\tilde{\gamma}_{ij}\right) \tag{B.4.10}$$

(iv) From the point of view of the dynamical evolution equations we are about to derive for $\tilde{\gamma}_{ij}, \tilde{A}_{ij}$ and $\tilde{\Gamma}^i$, these are two symmetric 3×3-matrix- and one 3-vector-valued functions. The above tells us that in addition to these evolution equations, we get additional constraints; algebraically, we must always have

$$\tilde{\gamma} = 1, \quad \tilde{\gamma}^{ij}\tilde{A}_{ij} = 0, \tag{B.4.11}$$

and, in addition, a differential constraint emerges from

$$\mathcal{G}^i := \tilde{\Gamma}^i - \tilde{\gamma}^{k\ell}\tilde{\Gamma}^i{}_{k\ell} = 0. \tag{B.4.12}$$

Although mathematically speaking, the evolution equations will preserve these, numerically, there is a slight drift away from them. The algebraic constraints are very easy to enforce numerically—during discrete evolution we can simply regularly remap

$$\tilde{\gamma}_{ij} \mapsto \tilde{\gamma}^{-1/3}\tilde{\gamma}_{ij}, \quad \tilde{A}_{ij} \mapsto \tilde{A}_{ij} - \frac{1}{3}\tilde{\gamma}_{ij}\tilde{A}^k{}_k. \tag{B.4.13}$$

To preserve the differential constraint $\mathcal{G}^i = 0$ even in numerical simulations, we will artificially add it to the right-hand side of the evolution equation for $\tilde{\Gamma}^i$, as

$$\partial_t\tilde{\Gamma}^i = -\sigma\mathcal{G}^i + ... \tag{B.4.14}$$

for some positive constant $\sigma > 0$. This creates exponential damping/decay of $\mathcal{G}^i$ towards 0, so that numerical error does not diverge.

(v) The trace of the square of K_{ij} reads

$$K_{ij}K^{ij} = \tilde{A}_{ij}\tilde{A}^{ij} + \frac{1}{3}K^2. \tag{B.4.15}$$

Here, indices of quantities with a tilde are raised and lowered using $\tilde{\gamma}$, whereas for quantities without tilde, γ is used. We will continue to use this convention in the following to omit writing hundreds of instances of the induced and conformal metrics.

Proposition B.10 *(BSSNOK Evolution Equations)* Below are the evolution equations for the BSSNOK variables $\phi, \tilde{\gamma}_{ij}, K, \tilde{A}_{ij}$ and $\tilde{\Gamma}^i$.

$$\mathcal{H} = \bar{R} + \frac{2}{3}K^2 - \tilde{A}_{ij}\tilde{A}^{ij} - 2\Lambda - 16\pi\rho = 0,$$

$$\mathcal{M}_i = \frac{2}{3}\partial_i K - \tilde{\nabla}_k \tilde{A}^k{}_i + 6\tilde{A}^k{}_i \partial_k \phi + 8\pi j_i = 0,$$

$$\partial_t \phi = \beta^k \partial_k \phi - \frac{1}{6}\partial_k \beta^k + \frac{1}{6}\alpha K,$$

$$\partial_t \tilde{\gamma}_{ij} = \beta^k \partial_k \tilde{\gamma}_{ij} + 2\tilde{\gamma}_{k(i}\partial_{j)}\beta^k - \frac{2}{3}\tilde{\gamma}_{ij}\partial_k \beta^k - 2\alpha\tilde{A}_{ij},$$

$$\partial_t K = \beta^k \partial_k K - \gamma^{ij}\bar{\nabla}_i \bar{\nabla}_j \alpha + \alpha\Big(\tilde{A}_{ij}\tilde{A}^{ij} + \frac{1}{3}K^2 - \Lambda + 4\pi(S+\rho)\Big),$$

$$\begin{aligned}\partial_t \tilde{A}_{ij} = {}& \beta^k \partial_k \tilde{A}_{ij} + 2\tilde{A}_{k(i}\partial_{j)}\beta^k - \frac{2}{3}\tilde{A}_{ij}\partial_k \beta^k + e^{4\phi}\left[\alpha\bar{R}_{ij} - \bar{\nabla}_i\bar{\nabla}_j\alpha - 8\pi\alpha S_{ij}\right]^{\mathrm{TF}} \\ & + \alpha\Big(K\tilde{A}_{ij} - 2\tilde{A}_{ik}\tilde{A}^k{}_j\Big)\end{aligned}$$

$$\begin{aligned}\partial_t \tilde{\Gamma}^i = {}& \beta^k \partial_k \tilde{\Gamma}^i - \tilde{\Gamma}^k \partial_k \beta^i + \frac{2}{3}\tilde{\Gamma}^i \partial_m \beta^m + \tilde{\gamma}^{j\ell}\partial_j\partial_\ell \beta^i + \frac{1}{3}\tilde{\gamma}^{ij}\partial_j\big(\partial_k \beta^k\big) \\ & - 2\tilde{A}^{ij}\partial_j \alpha + 2\alpha\tilde{\Gamma}^i{}_{jk}\tilde{A}^{jk} - \frac{4}{3}\alpha\tilde{\gamma}^{ij}\partial_j K - 12\alpha\tilde{A}^{ij}\partial_j\phi - 16\pi\alpha\tilde{\gamma}^{ik}j_k - \sigma\mathcal{G}^i \qquad \text{(B.4.16)}\end{aligned}$$

In the above, the ^{TF} exponent refers to the trace-free part of the bracketed expression, i.e.

$$X^{\mathrm{TF}}_{ij} = X_{ij} - \frac{1}{3}\gamma_{ij}\gamma^{k\ell}X_{k\ell} = X_{ij} - \frac{1}{3}\tilde{\gamma}_{ij}\tilde{\gamma}^{k\ell}X_{k\ell}. \qquad \text{(B.4.17)}$$

Noteworthily, it does not matter whether trace removal is carried out using γ_{ij} or $\tilde{\gamma}_{ij}$.

Remark B.11

(i) In the derivation of the right-hand side for $\partial_t K$, $\alpha\ \mathcal{H} = 0$ was subtracted. This does not affect the validity of the equality on-shell, but changes the principal part of the evolution system. Nonetheless, sometimes it is useful to have the same equation without this subtraction. For such situations, we provide it below:

$$\partial_t K = \beta^k \partial_k K - \gamma^{ij}\bar{\nabla}_i\bar{\nabla}_j\alpha + \alpha\big(K^2 + \bar{R}\big) - 3\alpha\Lambda + 4\pi\alpha(S - 3\rho) \qquad \text{(B.4.18)}$$

One such situation is in the derivation of the right-hand side for $\partial_t \tilde{A}_{ij}$ above, where it turns out to be more convenient to use the unsubtracted version. The reason to subtract the Hamiltonian constraint from this equation is to eliminate the intrinsic Ricci scalar $\bar{R}$. In the standard ADM formulation, $\bar{R}$ introduces second spatial derivatives of the metric into $\partial_t K$, which contributes to weak hyperbolicity and numerical instabilities. Subtracting $\alpha\ \mathcal{H}$ removes all second derivatives of the metric from the right-hand side of $\partial_t K$, which helps cast the BSSNOK evolution system into a strongly hyperbolic form.

(ii) This version of the equations is only preliminary as it contains expressions involving $\bar{R}_{ij}$, $\bar{R}$ and $\bar{\nabla}$, which are associated with the non-conformal quantity γ_{ij} which is not available directly to the evolution code. Concretely, the offending terms in the above equations which still contain objects tied to γ_{ij} are $\bar{R}$ in the Hamiltonian constraint, the Laplacian $\bar{\Delta}_\gamma \alpha = \gamma^{ij}\bar{\nabla}_i\bar{\nabla}_j\alpha$ and the intrinsic Ricci tensor $\bar{R}_{ij}$. We will need to re-express these in terms of conformal quantities—this requires the results presented below.

(iii) Since the evolution equation for $\tilde{\Gamma}^i$ contains an artificially introduced damping term $-\sigma\mathcal{G}^i$ with $\sigma > 0$, we present the full derivation of the equation below. We begin by expanding

$$
\begin{aligned}
\partial_t\tilde{\Gamma}^i &= -\partial_t\partial_j\tilde{\gamma}^{ij} = -\partial_j\partial_t\tilde{\gamma}^{ij} = \partial_j\big(\tilde{\gamma}^{ik}\tilde{\gamma}^{j\ell}\partial_t\tilde{\gamma}_{k\ell}\big) \\
&= \partial_j\Big(\tilde{\gamma}^{ik}\tilde{\gamma}^{j\ell}\Big(\beta^m\partial_m\tilde{\gamma}_{k\ell} + \tilde{\gamma}_{mk}\partial_\ell\beta^m + \tilde{\gamma}_{m\ell}\partial_k\beta^m - \frac{2}{3}\tilde{\gamma}_{k\ell}\partial_m\beta^m - 2\alpha\tilde{A}_{k\ell}\Big)\Big) \\
&= \partial_j\Big(-\beta^m\partial_m\tilde{\gamma}^{ij} + \tilde{\gamma}^{j\ell}\partial_\ell\beta^i + \tilde{\gamma}^{i\ell}\partial_\ell\beta^j - \frac{2}{3}\tilde{\gamma}^{ij}\partial_m\beta^m - 2\alpha\tilde{A}^{ij}\Big) \\
&= \cancel{-(\partial_j\beta^m)\partial_m\tilde{\gamma}^{ij}} - \beta^m\partial_m\underbrace{\partial_j\tilde{\gamma}^{ij}}_{=-\tilde{\Gamma}^i} + \underbrace{(\partial_j\tilde{\gamma}^{j\ell})}_{=-\tilde{\Gamma}^\ell}\partial_\ell\beta^i + \tilde{\gamma}^{j\ell}\partial_j\partial_\ell\beta^i + \cancel{(\partial_j\tilde{\gamma}^{i\ell})\partial_\ell\beta^j} + \underline{\tilde{\gamma}^{i\ell}\partial_j\partial_\ell\beta^j} \\
&\quad -\frac{2}{3}\underbrace{(\partial_j\tilde{\gamma}^{ij})}_{=-\tilde{\Gamma}^i}\partial_m\beta^m \underline{-\frac{2}{3}\tilde{\gamma}^{ij}\partial_j\partial_m\beta^m} - 2\tilde{A}^{ij}\partial_j\alpha - 2\alpha\partial_j\tilde{A}^{ij} \\
&= \beta^k\partial_k\tilde{\Gamma}^i - \tilde{\Gamma}^k\partial_k\beta^i + \frac{2}{3}\tilde{\Gamma}^i\partial_m\beta^m + \tilde{\gamma}^{j\ell}\partial_j\partial_\ell\beta^i + \frac{1}{3}\tilde{\gamma}^{ij}\partial_j\big(\partial_k\beta^k\big) \\
&\quad -2\tilde{A}^{ij}\partial_j\alpha - 2\alpha\partial_j\tilde{A}^{ij}
\end{aligned}
\tag{B.4.19}
$$

The last term is still in an unfortunate form, which we can improve by using the momentum constraint $\mathcal{M}^i = 0$. Concretely, it may be used in re-arranged form to replace $\tilde{\nabla}_j\tilde{A}^{ij}$ as

$$
\begin{aligned}
\partial_j\tilde{A}^{ij} &= \tilde{\nabla}_j\tilde{A}^{ij} - \tilde{\Gamma}^i{}_{jk}\tilde{A}^{jk} - \underbrace{\tilde{\Gamma}^j{}_{kj}}_{=0}\tilde{A}^{ki} \\
&= -\tilde{\Gamma}^i{}_{jk}\tilde{A}^{jk} + \frac{2}{3}\tilde{\gamma}^{ij}\partial_j K + 6\tilde{A}^{ij}\partial_j\phi + 8\pi j^i
\end{aligned}
\tag{B.4.20}
$$

Consequently, the evolution equation for $\tilde{\Gamma}^i$, with the artificial damping term $-\sigma\mathcal{G}^i$ introduced, reads

$$
\begin{aligned}
\partial_t\tilde{\Gamma}^i = \beta^k\partial_k\tilde{\Gamma}^i - \tilde{\Gamma}^k\partial_k\beta^i + \frac{2}{3}\tilde{\Gamma}^i\partial_m\beta^m + \tilde{\gamma}^{j\ell}\partial_j\partial_\ell\beta^i + \frac{1}{3}\tilde{\gamma}^{ij}\partial_j\big(\partial_k\beta^k\big) \\
-2\tilde{A}^{ij}\partial_j\alpha + 2\alpha\tilde{\Gamma}^i{}_{jk}\tilde{A}^{jk} - \frac{4}{3}\alpha\tilde{\gamma}^{ij}\partial_j K - 12\alpha\tilde{A}^{ij}\partial_j\phi - 16\pi\alpha j^i - \sigma\mathcal{G}^i
\end{aligned}
\tag{B.4.21}
$$

(iv) The keen-eyed reader might have noticed that for any of the BSSN variables, its time derivative contains similarly structured spatial derivative terms of that variable involving the shift β on the right hand side. These combinations are reminiscient of Lie derivatives—and in fact, they are—with a catch. The catch comes from the terms involving the divergence $\partial_i\beta^i$, which in a tensorial Lie derivative do not appear. However, we should recall that ϕ is a function of the *determinant* of the tensor γ_{ij}, and that in the definitions of $\tilde{\gamma}_{ij}$ as well as $\tilde{A}_{ij}$ include factors of

$$
e^{4\phi} = \gamma^{-1/3}. \tag{B.4.22}
$$

From this, we conclude that ϕ is not a scalar but a scalar *density* of weight $-\frac{1}{6}$, and that $\tilde{\gamma}_{ij}$ and $\tilde{A}_{ij}$ are $(0,2)$-tensor *densities* of weight $-\frac{2}{3}$. We recall that for a tensor density $T^{\mu\ldots}{}_{\nu\ldots}$ of weight w, its Lie derivative along some vector field X is given by

$$
\begin{aligned}
\mathcal{L}_X T^{\mu\ldots}{}_{\nu\ldots} = X^\lambda\partial_\lambda T^{\mu\ldots}{}_{\nu\ldots} - (\partial_\lambda X^\mu)T^{\lambda\ldots}{}_{\nu\ldots} - \ldots \\
+(\partial_\nu X^\lambda)T^{\mu\ldots}{}_{\lambda\ldots} + \ldots + w(\partial_\lambda X^\lambda)T^{\mu\ldots}{}_{\nu\ldots}
\end{aligned}
\tag{B.4.23}
$$

A similar expression can be obtained for connections as well. Taking $X = \beta, T = \phi, \tilde{\gamma}_{ij}, K, \tilde{A}_{ij}$ and $\tilde{\Gamma}^i$, we identify that the BSSNOK evolution equations in (B.4.16) may be written in the (slightly) more compact form

$$
\begin{aligned}
\partial_t \phi &= \mathcal{L}_\beta \phi + \frac{1}{6}\alpha K, \\
\partial_t \tilde{\gamma}_{ij} &= \mathcal{L}_\beta \tilde{\gamma}_{ij} - 2\alpha \tilde{A}_{ij}, \\
\partial_t K &= \mathcal{L}_\beta K - \gamma^{ij}\bar{\nabla}_i \bar{\nabla}_j \alpha + \alpha\Big(\tilde{A}_{ij}\tilde{A}^{ij} + \frac{1}{3}K^2 - \Lambda + 4\pi(S+\rho)\Big), \\
\partial_t \tilde{A}_{ij} &= \mathcal{L}_\beta \tilde{A}_{ij} + e^{4\phi}\left[\alpha \bar{R}_{ij} - \bar{\nabla}_i \bar{\nabla}_j \alpha - 8\pi\alpha S_{ij}\right]^{\mathrm{TF}} + \alpha\Big(K\tilde{A}_{ij} - 2\tilde{A}_{ik}\tilde{A}^k{}_j\Big), \\
\partial_t \tilde{\Gamma}^i &= \mathcal{L}_\beta \tilde{\Gamma}^i - 2\tilde{A}^{ij}\partial_j \alpha + 2\alpha \tilde{\Gamma}^i{}_{jk}\tilde{A}^{jk} - \frac{4}{3}\alpha\tilde{\gamma}^{ij}\partial_j K - 12\alpha\tilde{A}^{ij}\partial_j \phi - 16\pi\alpha\tilde{\gamma}^{ik}j_k - \sigma\mathcal{G}^i.
\end{aligned}
\tag{B.4.24}
$$

Note that the trace of the extrinsic curvature, K, is the only true scalar, so $\mathcal{L}_\beta K = \beta^i \partial_i K$. Moreover, it should be noted that $\mathcal{L}_\beta \tilde{\Gamma}^i$ is *not* to be interpreted as the Lie derivative of a vector field nor vector density, but rather expanded as

$$
\mathcal{L}_\beta \tilde{\Gamma}^i = \mathcal{L}_\beta\big(\tilde{\Gamma}^i{}_{jk}\tilde{\gamma}^{jk}\big) = \big(\mathcal{L}_\beta \tilde{\Gamma}^i{}_{jk}\big)\tilde{\gamma}^{jk} + \tilde{\Gamma}^i{}_{jk}\big(\mathcal{L}_\beta \tilde{\gamma}^{jk}\big). \tag{B.4.25}
$$

The Lie derivative of the connection coefficients is then evaluated using the analogon of equation (B.4.23) for connections, and gives rise to the terms involving second derivatives of β in the last of equations (B.4.16).

Although this compactification of the BSSN evolution equations does not introduce any benefits to its numerical implementation—we still have to compute all the individual terms the Lie derivatives consist of—writing them down in this form enables a more straightforward interpretation. For any of the $T = \phi, \tilde{\gamma}_{ij}, K, \tilde{A}_{ij}, \tilde{\Gamma}^i$, we have the main Lie-advection piece

$$
\partial_t T = \mathcal{L}_\beta T. \tag{B.4.26}
$$

In absence of any other terms, this implies that the BSSN variables are Lie-transported along the normal vector field $\alpha n = \partial_t - \beta$ and thus remain invariant under its generated diffeomorphisms. However, there are other terms; these source the Lie transport by introducing curvature, matter and gauge contributions, such as $e^{4\phi}\alpha\bar{R}_{ij}$, $8\pi e^{4\phi}\alpha S^{\mathrm{TF}}_{ij}$ and $\gamma^{ij}\bar{\nabla}_i\bar{\nabla}_j\alpha$, respectively. Put differently, this means that the failure of the BSSN variables to be Lie-transported across the leaves of the foliation is caused by global curvature, the presence of matter, and the choice of the foliation and coordinates on it.

B.4.2 From Bar to Tilde

As alluded to before, the BSSN equations (B.4.16) are preliminary in the sense that they still contain expressions associated with the induced metric γ_{ij}, which is not part of the BSSN variables. Although technically, one can always reconstruct γ_{ij} from ϕ and $\tilde{\gamma}_{ij}$ as $\gamma_{ij} = e^{-4\phi}\tilde{\gamma}_{ij}$, this is computationally inefficient and induces more memory usage and traffic. In this section, we address the offending terms and reexpress them in terms of the BSSN variables $\phi, \tilde{\gamma}_{ij}, K, \tilde{A}_{ij}$ and $\tilde{\Gamma}^i$. Concretely, the terms that appear which we need to reformulate are

$$
\bar{R}_{ij}, \quad \bar{R}, \quad \bar{\nabla}_i\bar{\nabla}_j\alpha \quad \text{and} \quad \Delta_\gamma \alpha = \gamma^{ij}\bar{\nabla}_i\bar{\nabla}_j\alpha. \tag{B.4.27}
$$

We begin with the last of the four, as it is the simplest. Using the standard formula for the Laplacian, we get

$$
\begin{aligned}
\Delta_\gamma \alpha &= \frac{1}{\sqrt{\gamma}}\partial_i\big(\sqrt{\gamma}\gamma^{ij}\partial_j\alpha\big) = \frac{1}{\sqrt{e^{-12\phi}}}\partial_i\Big(\sqrt{e^{-12\phi}}e^{4\phi}\tilde{\gamma}^{ij}\partial_j\alpha\Big) \\
&= e^{6\phi}\partial_i\big(e^{-2\phi}\tilde{\gamma}^{ij}\partial_j\alpha\big) = e^{4\phi}\big(\partial_i\big(\tilde{\gamma}^{ij}\partial_j\alpha\big) - 2\tilde{\gamma}^{ij}\partial_i\phi\partial_j\alpha\big) \\
&= e^{4\phi}\big(\Delta_{\tilde{\gamma}}\alpha - 2\tilde{\gamma}^{ij}\partial_i\phi\partial_j\alpha\big).
\end{aligned}
\tag{B.4.28}
$$

In the last step, we made use of the fact that $\tilde{\gamma} = 1$.

Next up, we consider the Hessian of α. For this, we recall the relationship (B.3.10), which allows us to infer

$$\begin{aligned}\bar{\nabla}_i\bar{\nabla}_j\alpha &= \bar{\nabla}_i\partial_j\alpha = \tilde{\nabla}_i\partial_j\alpha + C^k{}_{ji}\partial_k\alpha \\ &= \tilde{\nabla}_i\tilde{\nabla}_j\alpha + 2\big(\delta^k{}_i\partial_j\phi + \delta^k{}_j\partial_i\phi - \tilde{\gamma}_{ij}\tilde{\gamma}^{k\ell}\partial_\ell\phi\big)\partial_k\alpha \\ &= \tilde{\nabla}_i\tilde{\nabla}_j\alpha + 4\partial_{(i}\phi\partial_{j)}\alpha - 2\tilde{\gamma}_{ij}\tilde{\gamma}^{k\ell}\partial_k\phi\partial_\ell\alpha.\end{aligned} \tag{B.4.29}$$

Note that since $\tilde{\gamma}^{ij}\tilde{\gamma}_{ij} = 3$, this is consistent with equation (B.4.28).

Now turning to $\bar{R}_{ij}$, we can simply employ our result (B.3.12) for the conformal Ricci tensor, which for $d = 3$ implies

$$\bar{R}_{ij} = \tilde{R}_{ij} + 2\big(\tilde{\nabla}_i\tilde{\nabla}_j\phi + \tilde{\gamma}_{ij}\tilde{\gamma}^{k\ell}\tilde{\nabla}_k\tilde{\nabla}_\ell\phi\big) + 4\big(\partial_i\phi\partial_j\phi - \tilde{\gamma}_{ij}\tilde{\gamma}^{k\ell}\partial_k\phi\partial_\ell\phi\big). \tag{B.4.30}$$

Here, according to equation (B.3.18), the conformal Ricci tensor $\tilde{R}_{ij}$ is given by

$$\begin{aligned}\tilde{R}_{ij} = &-\frac{1}{2}\tilde{\gamma}^{k\ell}\partial_k\partial_\ell\tilde{\gamma}_{ij} + \tilde{\gamma}_{k(i}\partial_{j)}\tilde{\Gamma}^k \\ &+ \tilde{\Gamma}^k\tilde{\Gamma}_{(ij)k} + \tilde{\Gamma}_{k\ell i}\tilde{\Gamma}^{k\ell}{}_j + 2\tilde{\Gamma}_{k\ell(i}\tilde{\Gamma}_{j)}{}^{k\ell}.\end{aligned} \tag{B.4.31}$$

Here it becomes apparent why introducing $\tilde{\Gamma}^i$ as an additional dynamical variable is a good idea: since it only appears with first derivatives, the principal part of the conformal Ricci tensor is the elliptic operator

$$\tilde{\gamma}^{k\ell}\partial_k\partial_\ell\tilde{\gamma}_{ij} \tag{B.4.32}$$

which is much more well-behaved numerically than the second-order terms introduced by $\partial_i\tilde{\Gamma}^j$ would be. The introduction of $\tilde{\Gamma}^i$ turns exactly these terms into mere first-order contributions, so that they do not mess up the principal symbol.

Lastly, for the Ricci scalar, we may use equation (B.3.13), which for $d = 3$ reads

$$R = e^{4\phi}\big(\tilde{R} + 8\tilde{\gamma}^{ij}\big(\tilde{\nabla}_i\tilde{\nabla}_j\phi - \partial_i\phi\partial_j\phi\big)\big). \tag{B.4.33}$$

For quicker reference, we provide all these results again below:

$$\begin{aligned}\bar{\nabla}_i\bar{\nabla}_j\alpha &= \tilde{\nabla}_i\tilde{\nabla}_j\alpha + 4\partial_{(i}\phi\partial_{j)}\alpha - 2\tilde{\gamma}_{ij}\tilde{\gamma}^{k\ell}\partial_k\phi\partial_\ell\alpha, \\ \gamma^{ij}\bar{\nabla}_i\bar{\nabla}_j\alpha &= e^{4\phi}\big(\Delta_{\tilde{\gamma}}\alpha - 2\tilde{\gamma}^{ij}\partial_i\phi\partial_j\alpha\big), \\ \bar{R}_{ij} &= \tilde{R}_{ij} + 2\big(\tilde{\nabla}_i\tilde{\nabla}_j\phi + \tilde{\gamma}_{ij}\tilde{\gamma}^{k\ell}\tilde{\nabla}_k\tilde{\nabla}_\ell\phi\big) + 4\big(\partial_i\phi\partial_j\phi - \tilde{\gamma}_{ij}\tilde{\gamma}^{k\ell}\partial_k\phi\partial_\ell\phi\big) \\ &\quad \text{with} \quad \tilde{R}_{ij} = -\frac{1}{2}\tilde{\gamma}^{k\ell}\partial_k\partial_\ell\tilde{\gamma}_{ij} + \tilde{\gamma}_{k(i}\partial_{j)}\tilde{\Gamma}^k \\ &\qquad\qquad + \tilde{\Gamma}^k\tilde{\Gamma}_{(ij)k} + \tilde{\Gamma}_{k\ell i}\tilde{\Gamma}^{k\ell}{}_j + 2\tilde{\Gamma}_{k\ell(i}\tilde{\Gamma}_{j)}{}^{k\ell}, \\ \bar{R} &= e^{4\phi}\big(\tilde{R} + 8\tilde{\gamma}^{ij}\big(\tilde{\nabla}_i\tilde{\nabla}_j\phi - \partial_i\phi\partial_j\phi\big)\big).\end{aligned} \tag{B.4.34}$$

B.4.3 Gauge Dynamics

B.4.3.1 Lapse: Bona-Massó and $1 + \log$ Slicing

One objective in choosing a gauge is to prevent the spatial foliation Σ_t from encountering a spacetime singularity as the coordinate time t advances. Numerical evolution cannot reliably

represent the divergent curvature associated with a physical singularity, so it is desirable to choose a slicing that avoids reaching it. The Penrose singularity theorems establish generic geodesic incompleteness under suitable assumptions, meaning that generally, we have singularities to avoid.

A common strategy is to drive the lapse function α towards zero in regions approaching a singularity. Since the proper-time separation between neighbouring slices along the normal congruence is $d\tau = \alpha\, dt$, a collapsing lapse suppresses the accumulation of proper time as coordinate time advances. This mechanism, known as *singularity avoidance*, can allow the slices to approach a limiting hypersurface without reaching the physical singularity. Whether this occurs depends on the spacetime and the chosen slicing condition.

Our first goal is to establish the general Bona-Massó evolution equation for α, motivated by harmonic coordinates, which are commonly used in establishing hyperbolicity and well-posedness properties of the Einstein equations. The harmonic coordinate condition for the time coordinate, $\Box_g t = 0$, can be written in coordinates adapted to the foliation as

$$\begin{aligned} 0 \overset{!}{=} \Box_g t = \nabla_\mu(g^{\mu\nu}\partial_\nu t) = \nabla_\mu(g^{\mu t}) = \nabla_\mu\left(-\frac{1}{\alpha}n^\mu\right) \\ = -\frac{1}{\alpha}\nabla_\mu n^\mu + \frac{1}{\alpha^2}n^\mu\nabla_\mu\alpha \end{aligned} \tag{B.4.35}$$

Making use of $\nabla_\mu n^\mu = -K$ and

$$n^\mu\nabla_\mu\alpha = \frac{1}{\alpha}(\partial_t - \beta^i\partial_i)\alpha, \tag{B.4.36}$$

we arrive at

$$\partial_t\alpha = \beta^i\partial_i\alpha - \alpha^2 K. \tag{B.4.37}$$

Let us examine what this harmonic slicing condition implies, and how we might modify it to improve singularity avoidance—harmonic slicing does not, in general, guarantee that the foliation avoids a singularity.

Recalling that

$$K_{ij} = -\frac{1}{2\alpha}\big(\partial_t\gamma_{ij} - \bar{\nabla}_i\beta_j - \bar{\nabla}_j\beta_i\big), \tag{B.4.38}$$

we obtain, upon taking the trace,

$$\begin{aligned} K &= -\frac{1}{2\alpha}\big(\gamma^{ij}\partial_t\gamma_{ij} - 2\bar{\nabla}_i\beta^i\big) \\ &= -\frac{1}{\alpha}\partial_t\log\sqrt{\gamma} + \frac{1}{\alpha}\bar{\nabla}_i\beta^i. \end{aligned} \tag{B.4.39}$$

Substituting this expression into equation (B.4.37) yields

$$\partial_t\alpha = \beta^i\partial_i\alpha + \alpha\partial_t\log\sqrt{\gamma} - \alpha\bar{\nabla}_i\beta^i. \tag{B.4.40}$$

To establish a simple scaling relation between the lapse and the spatial volume density, we now make the simplifying assumption that $\beta^i = 0$. The equation becomes

$$\partial_t\alpha = \alpha\partial_t\log\sqrt{\gamma} \quad\Leftrightarrow\quad \partial_t\log\alpha = \partial_t\log\sqrt{\gamma}. \tag{B.4.41}$$

Integrating in time gives

$$\alpha(t,x) = \alpha_0(x)\frac{\sqrt{\gamma(t,x)}}{\sqrt{\gamma_0(x)}}. \tag{B.4.42}$$

Thus, under harmonic slicing with vanishing shift, the lapse is proportional to the spatial volume density at each spatial coordinate. If $\sqrt{\gamma}$ approaches zero, the lapse does so at the same rate. This relation motivates the search for a slicing condition that suppresses the lapse more strongly as the spatial volume density decreases.

We are therefore led to generalise equation (B.4.37) by allowing the response of the lapse to the trace of the extrinsic curvature to depend on the lapse itself. Concretely, we employ the *Bona-Massó slicing condition*,

$$\partial_t \alpha = \beta^i \partial_i \alpha - \alpha^2 f(\alpha) K, \tag{B.4.43}$$

where $f(\alpha)$ is a freely specificable gauge function. Different choices of f produce different lapse dynamics and can substantially alter the singularity-avoidance properties of the resulting foliation.

A particularly useful choice is $f(\alpha) = 2/\alpha$, which gives the standard *1+log slicing condition*

$$\partial_t \alpha = \beta^i \partial_i \alpha - 2\alpha K. \tag{B.4.44}$$

To examine its scaling behaviour, we again assume vanishing shift. Substituting the relation between K and the spatial volume density gives

$$\partial_t \alpha = -2\alpha K = 2\partial_t \log \sqrt{\gamma} = \partial_t \log \gamma. \tag{B.4.45}$$

Integrating in time, we find

$$\alpha(t, x) = \alpha_0(x) + \log\left(\frac{\gamma(t, x)}{\gamma_0(x)}\right). \tag{B.4.46}$$

This is the origin of the name $1 + \log$ slicing—in the commonly used normalisation $\alpha_0 = 1$ and $\gamma_0 = 1$, the relation takes the form $\alpha = 1 + \log \gamma$.

The integration relation shows that the lapse decreases logarithmically as the spatial volume density decreases. In particular, the formal condition $\alpha = 0$ corresponds to

$$\frac{\gamma}{\gamma_0} = e^{-\alpha_0}. \tag{B.4.47}$$

For an initially unit lapse, the lapse therefore collapses when the spatial volume density has decreased by a factor e^{-1}, rather than requiring it to approach zero first. This illustrates how $1 +$ log slicing can drive the lapse towards zero more strongly than harmonic slicing, producing the desirable singularity-avoidance behaviour required in black hole evolutions.

At first sight, the integrated relation appears to permit the lapse to become negative if γ decreases sufficiently far. However, this interpretation treats the relation as though γ could evolve independently of the lapse—in the coupled Einstein evolution system, the lapse itself influences the evolution of the spatial metric. Intuitively speaking, the evolution of γ halts before its value can decrease into the range leading to negative α. Another way of seeing why α remains positive if it is positive initially is that $\alpha = 0$ leads to a vanishing right-hand side of the lapse evolution equation (B.4.44)—it is a steady state.

As a final remark, let us consider the behaviour of the BSSNOK equations in a region where the lapse vanishes identically. In this case, the lapse-dependent terms and their spatial derivatives vanish within that region, and the evolution equations reduce to

$$
\begin{aligned}
\partial_t \phi &= \mathcal{L}_\beta \phi, \\
\partial_t \tilde{\gamma}_{ij} &= \mathcal{L}_\beta \tilde{\gamma}_{ij}, \\
\partial_t K &= \mathcal{L}_\beta K, \\
\partial_t \tilde{A}_{ij} &= \mathcal{L}_\beta \tilde{A}_{ij}, \\
\partial_t \tilde{\Gamma}^i &= \mathcal{L}_\beta \tilde{\Gamma}^i - \sigma \mathcal{G}^i.
\end{aligned}
\tag{B.4.48}
$$

Consequently, a frozen lapse causes the BSSNOK equations to reduce to a pure Lie-advective form—up to the algebraic constraint correction term in the last equation. In hindsight, this is expected: even though the evolution comes to a standstill with respect to t along the integral lines of n, the spatial position of these integral lines may still vary as t progresses—this is encoded by the shift. Thus, even if evolution is frozen ($\alpha \to 0$), we still need to advect the field configuration across our simulation domain to account for coordinate motion. Note: At the boundary of a region where $\alpha = 0$, its first and second derivatives may be non-zero. This introduces further terms into the equations for K, $\tilde{A}_{ij}$ and $\partial_t \tilde{\Gamma}^i$.

B.4.3.2 Shift: Γ-Driver

Although a singularity-avoiding lapse condition can prevent the spatial foliation from encountering a spacetime singularity, we must further ensure that the spatial coordinates remain well-behaved throughout the evolution. Undesirable coordinate motion can distort the numerical grid, which makes the evolution equations difficult to resolve accurately and potentially destabilises the simulation.

As we have seen before, the shift vector $\beta = \beta^i \partial_i$ controls the motion of spatial coordinates relative to the normal observers. This coordinate motion is not inherently undesirable: a coordinate system that rotates over time can be useful, for example, to co-rotate with a merging binary system. However, excessive or poorly controlled coordinate motion can lead to significant distortion of the grid. Similarly expansion and contraction of the coordinate grid can redistribute physical resolution, which may be beneficial or detrimental depending on the region and the structures being resolved, although this is of course no substitute for adaptive mesh refinement.

The goal of a suitable shift condition is therefore not to eliminate coordinate motion, but to control it so that the resulting coordinates remain useful for numerical evolution. To this end, we first need a way to quantify the rate at which the grid is distorted; otherwise, it would be difficult to write down a formal condition for controlling said distortion.

For this purpose, we distinguish changes in the local volume scale from changes in the shape of the spatial metric. Recall that in BSSNOK, the physical and conformal spatial metrics are related by

$$
\gamma_{ij} = \gamma^{1/3} \tilde{\gamma}_{ij}. \tag{B.4.49}
$$

This decomposition separates the metric into a determinant part, $\gamma^{1/3}$, and a shape part, the unit-determinant conformal metric $\tilde{\gamma}_{ij}$. The former encodes the local spatial volume scale relative to the coordinates, whereas the latter describes the angular relations and relative lengths of the spatial directions ∂_i independently of this volume scale.

To quantify distortion—that is, changes in shape—we therefore examine the time derivative $\partial_t \tilde{\gamma}_{ij}$. Since $\tilde{\gamma}_{ij}$ is a tensor density of weight $-2/3$, its time derivative has the same weight (under time-independent, spatial coordinate transformations). Multiplying by $\gamma^{1/3}$, which has weight $+2/3$, returns us to a genuine spatial tensor (under spatial coordinate transformations). This motivates the definition of the *distortion tensor* as

$$
\Sigma_{ij} = \frac{1}{2} \gamma^{1/3} \partial_t \tilde{\gamma}_{ij} \tag{B.4.50}
$$

Here, the factor of 1/2 is a conventional normalisation. The distortion tensor thus provides a measure of the coordinate-time rate of change of the shape of the spatial metric, separated from changes in its local volume density.

To get a better interpretation of the dynamical meaning of Σ_{ij} under BSSNOK evolution, let us use the evolution equations (B.4.16) to re-express it in terms of other variables. Concretely, replacing $\partial_t \tilde{\gamma}_{ij}$ leads to

$$\Sigma_{ij} = \frac{1}{2}\gamma^{1/3}\mathcal{L}_\beta \tilde{\gamma}_{ij} - \alpha A_{ij} \tag{B.4.51}$$

Here, A_{ij} is the trace-free extrinsic curvature before rescaling into $\tilde{A}_{ij}$,

$$A_{ij} = K_{ij} - \frac{1}{3}K\gamma_{ij}. \tag{B.4.52}$$

These are physically sourced distortions: we cannot control them directly, as they are governed by the evolution. The first term, however, depends on β—a quantity we control. This is how we will exert influence over the distortion. In due course, we will examine this term further. Before that, however, let us consider what we have found so far. Dynamically, the distortion tensor consists of two contributions—one we control by means of the shift, and the other prescribed by the physical evolution of the system. We can think of this as a kind of "game": the evolution wants to distort the grid by producing a nonzero A_{ij}—it is then our job to use the β-dependent term to cancel it out as much as possible in order for the distortion to remain minimal. If, however, we were to choose β poorly, the first term acts as a further source of grid distortions.

To analyse the β-dependent term further, we expand it using the Leibniz rule:

$$\begin{aligned}
\frac{1}{2}\gamma^{1/3}\mathcal{L}_\beta\tilde{\gamma}_{ij} &= \frac{1}{2}\mathcal{L}_\beta\big(\gamma^{1/3}\tilde{\gamma}_{ij}\big) - \frac{1}{2}\tilde{\gamma}_{ij}\mathcal{L}_\beta\big(\gamma^{1/3}\big) \\
&= \frac{1}{2}\mathcal{L}_\beta\gamma_{ij} - \frac{1}{2}\tilde{\gamma}_{ij}\left(\beta^k\partial_k\gamma^{1/3} + \frac{2}{3}\gamma^{1/3}\partial_k\beta^k\right) \\
&= \frac{1}{2}\big(\bar{\nabla}_i\beta_j + \bar{\nabla}_j\beta_i\big) - \frac{1}{2}\tilde{\gamma}_{ij}\gamma^{1/3}\left(\frac{1}{3}\underbrace{\gamma^{-1}\beta^k\partial_k\gamma}_{\substack{=\beta^k\partial_k\log\gamma \\ =2\beta^k\partial_k\log\sqrt{\gamma}}} + \frac{2}{3}\partial_k\beta^k\right) \\
&= \bar{\nabla}_{(i}\beta_{j)} - \frac{1}{3}\gamma_{ij}\bar{\nabla}_k\beta^k.
\end{aligned} \tag{B.4.53}$$

The β-dependent contribution is hence nothing but the trace-free symmetric part of the gradient field $\bar{\nabla}_i\beta_j$—that is, its shear. Hence, the β-dependent part of Σ_{ij} reacts to shear of t-coordinate lines, caused by coordinate motion in the direction of β. We insert back into the dynamical expression (B.4.51) for the distortion tensor to obtain

$$\Sigma_{ij} = \bar{\nabla}_{(i}\beta_{j)} - \frac{1}{3}\gamma_{ij}\bar{\nabla}_k\beta^k - \alpha A_{ij}, \tag{B.4.54}$$

which makes the split of the distortion tensor into a shear part—which we control—and a physical part—which is fixed by evolution—manifest. Notably, as intuition already suggested earlier, neither the rotation nor the expansion of $\bar{\nabla}_i\beta_j$ appears directly in this expression. One should keep in mind, however, that rotation and expansion affect the evolution of A_{ij}, and can hence indirectly contribute to distortion.

Now that we know that we are working with, we can start to design our gauge condition for β accordingly. Since we want to minimise Σ_{ij}, a naive guess might be to just set it to zero and at each time step solve the elliptic system

$$\bar{\nabla}_{(i}\beta_{j)} - \frac{1}{3}\gamma_{ij}\bar{\nabla}_k\beta^k = \alpha A_{ij} \tag{B.4.55}$$

for β. Unfortunately, however, solving an elliptic system at each iteration step is computationally intensive and hence not really viable. In addition, and more subtly, doing so is generally impossible anyways: the equation above is symmetric traceless, meaning that we have a total of five independent equations. Since we only have three degrees of freedom (the components of β) to satisfy them with, for an arbitrary source A_{ij}, it is generically overdetermined and in most cases will not admit a solution.

We are hence forced to do the next-best thing—attempt to minimise some spatial norm of Σ_{ij} on each slice. Picking as a functional to minimise the L^2-norm

$$Q\big[\Sigma_{ij}, t\big] = \int_{\Sigma_t} \Sigma_{ij}\Sigma^{ij}\, d\mathrm{vol}_\gamma, \tag{B.4.56}$$

we can use the condition of stationarity to find its minima. Of course, we do not vary Q with respect to Σ_{ij} itself—this would just recover the condition that $\Sigma_{ij} = 0$ which we have found to be of no merit. Instead, we vary the quantity we actually control: the shift β. When sending $\beta \to \beta + \delta\beta$, the distortion Σ_{ij} varies as

$$\Sigma_{ij} \to \Sigma_{ij} + \delta\Sigma_{ij}, \quad \delta\Sigma_{ij} = \bar{\nabla}_{(i}\delta\beta_{j)} - \frac{1}{3}\gamma_{ij}\bar{\nabla}_k\delta\beta^k. \tag{B.4.57}$$

Consequently, the variation of Q with respect to β is

$$\delta Q = 2\int_{\Sigma_t} \Sigma^{ij}\delta\Sigma_{ij}\, d\mathrm{vol}_\gamma = 2\int_{\Sigma_t} \Sigma^{ij}\Big(\bar{\nabla}_{(i}\delta\beta_{j)} - \frac{1}{3}\gamma_{ij}\bar{\nabla}_k\delta\beta^k\Big) d\mathrm{vol}_\gamma. \tag{B.4.58}$$

Since Σ_{ij} is traceless, the second term will not contribute, so that

$$\delta Q = 2\int_{\Sigma_t} \Sigma^{ij}\bar{\nabla}_i\delta\beta_j\, d\mathrm{vol}_\gamma = -2\int_{\Sigma_t} \big(\bar{\nabla}_i\Sigma^{ij}\big)\delta\beta_j\, d\mathrm{vol}_\gamma. \tag{B.4.59}$$

Note that this implicitly assumed the distortion and variation to vanish quickly enough as $r \to \infty$ for integration by parts not to introduce boundary terms. Demanding $\delta Q = 0$, we obtain the criticality condition

$$\bar{\nabla}_i\Sigma^{ij} = 0. \tag{B.4.60}$$

Though this is still an elliptic system, which a priori, we need to solve at each timestep, we now have the correct number of equations; three equations for three degrees of freedom.

B.5 Initial Data

In this section, we discuss different approaches used to decompose and solve the constraint equations

$$\mathcal{H} = \bar{R} + K^2 - K_{ij}K^{ij} - 2\Lambda - 16\pi\rho = 0,$$

$$\mathcal{M}_i = \bar{\nabla}_i K - \bar{\nabla}_k K^k{}_i + 8\pi j_i = 0. \tag{B.5.1}$$

These are 4 equations for the total of 12 functions in γ_{ij} and K_{ij}, meaning that there is a significant amount of freedom in initial data. To separate the freely specifiable data from data fixed by these equations, we perform different kinds of decompositions which have been developed historically to approach the problem of computing physically meaningful initial data.

B.5.1 York-Lichnerowicz Conformal Transverse-Traceless (CTT) Decomposition

The York-Lichnerowicz conformal traceless split employs the variables

$$\begin{aligned} \gamma_{ij} &= \psi^4 \hat{\gamma}_{ij} \quad \Leftrightarrow \quad \gamma^{ij} = \psi^{-4}\hat{\gamma}^{ij}, \\ K_{ij} &= A_{ij} + \frac{1}{3}\gamma_{ij}K, \\ A_{ij} &= \psi^{-2}\hat{A}_{ij} \quad \Leftrightarrow \quad A^{ij} = \psi^{-10}\hat{A}^{ij}. \end{aligned} \tag{B.5.2}$$

Concretely, this means that $\hat{\gamma}_{ij}$ is a conformal metric,

$$\hat{\gamma}_{ij} = \psi^{-4}\gamma_{ij}, \tag{B.5.3}$$

for which we do *not* necessarily require $\det\hat{\gamma} = 1$, A_{ij} is the traceless extrinsic curvature

$$A_{ij} = K_{ij} - \frac{1}{3}\gamma_{ij}K, \tag{B.5.4}$$

and $\hat{A}_{ij} = \psi^2 A_{ij}$ its conformal rescaling.

We further denote by $\hat{\nabla}$ the Levi-Civita connection associated with $\hat{\gamma}_{ij}$, whose coefficients—according to equation (B.3.8) with $\psi = e^{-\phi} \Leftrightarrow \phi = -\log\psi$—are related to those of $\bar{\nabla}$ by

$$\bar{\Gamma}^k{}_{ij} = \hat{\Gamma}^k{}_{ij} + \underbrace{\frac{2}{\psi}\big(\delta^k{}_i\partial_j\psi + \delta^k{}_j\partial_i\psi - \hat{\gamma}_{ij}\hat{\gamma}^{k\ell}\partial_\ell\psi\big)}_{=:C^k{}_{ij}}. \tag{B.5.5}$$

Besides the connection $\bar{\nabla}$ needing to be re-expressed in terms of the variables $\big(\psi, \hat{\gamma}_{ij}, K, \hat{A}_{ij}\big)$, we also need to relate the Ricci scalar appearing in the Hamiltonian constraint to quantities associated to $\hat{\gamma}_{ij}$ and ψ. In $d = 3$, equation (B.3.13) tells us that

$$\bar{R} = e^{4\phi}\big(\hat{R} + 8\hat{\gamma}^{ij}\hat{\nabla}_i\hat{\nabla}_j\phi - 8\hat{\gamma}^{ij}\hat{\nabla}_i\phi\hat{\nabla}_j\phi\big). \tag{B.5.6}$$

Again using $\phi = -\log\psi$, we expand

$$\begin{aligned} \hat{\gamma}^{ij}\hat{\nabla}_i\hat{\nabla}_j\phi &= -\hat{\gamma}^{ij}\hat{\nabla}_i\hat{\nabla}_j\log\psi = -\hat{\gamma}^{ij}\hat{\nabla}_i\big(\psi^{-1}\hat{\nabla}_j\psi\big) \\ &= -\psi^{-1}\hat{\gamma}^{ij}\hat{\nabla}_i\hat{\nabla}_j\psi + \psi^{-2}\hat{\gamma}^{ij}\hat{\nabla}_i\psi\hat{\nabla}_j\psi \\ \hat{\gamma}^{ij}\hat{\nabla}_i\phi\hat{\nabla}_j\phi &= \psi^{-2}\hat{\gamma}^{ij}\hat{\nabla}_i\psi\hat{\nabla}_j\psi, \end{aligned} \tag{B.5.7}$$

whence the Ricci scalar becomes

$$\bar{R} = \psi^{-4}\hat{R} - 8\psi^{-5}\hat{\gamma}^{ij}\hat{\nabla}_i\hat{\nabla}_j\psi. \tag{B.5.8}$$

With this in hand, we can write the Hamiltonian constraint as

$$\widehat{\mathcal{H}} = 8\hat{\gamma}^{ij}\hat{\nabla}_i\hat{\nabla}_j\psi - \hat{R}\psi - \frac{2}{3}K^2\psi^5 + \hat{A}_{ij}\hat{A}^{ij}\psi^{-7} + (2\Lambda + 16\pi\rho)\psi^5 = 0. \tag{B.5.9}$$

For the momentum constraint, we need to do some more work. We rewrite it slightly to become

$$\mathcal{M}^j = \bar{\nabla}_i(\gamma^{ij}K - K^{ij}) + 8\pi j^j = 0, \tag{B.5.10}$$

and rewrite the first term separately as follows:

$$\bar{\nabla}_i(\gamma^{ij}K - K^{ij}) = \bar{\nabla}_i\Big(\gamma^{ij}K - A^{ij} - \frac{1}{3}\gamma^{ij}K\Big) = \frac{2}{3}\gamma^{ij}\bar{\nabla}_i K - \bar{\nabla}_i A^{ij}. \tag{B.5.11}$$

For the divergence term, we get

$$\begin{aligned}\bar{\nabla}_i A^{ij} &= \bar{\nabla}_i\left(\psi^{-10}\hat{A}^{ij}\right) = \hat{\nabla}_i\left(\psi^{-10}\hat{A}^{ij}\right) + \psi^{-10}C^i{}_{ki}\hat{A}^{kj} + \psi^{-10}C^j{}_{ki}\hat{A}^{ik} \\ &= \psi^{-10}\hat{\nabla}_i\hat{A}^{ij} - 10\psi^{-11}\hat{A}^{ij}\partial_i\psi + 6\psi^{-11}\hat{A}^{kj}\partial_k\psi + 4\psi^{-11}\hat{A}^{jk}\partial_k\psi \\ &= \psi^{-10}\hat{\nabla}_i\hat{A}^{ij}.\end{aligned} \tag{B.5.12}$$

Hence, inserting back into the momentum constraint, we get

$$-\psi^{10}\mathcal{M}^i = \hat{\nabla}_k\hat{A}^{ki} - \frac{2}{3}\psi^6\hat{\gamma}^{ik}\hat{\nabla}_k K - 8\pi\psi^{10}j^i = 0. \tag{B.5.13}$$

Although we could leave it at this, we can make the following important observation which will allow us to split the degrees of freedom even further: the traceless conformal extrinsic curvature $\hat{A}^{ij}$ appears in the momentum constraint only through its divergence $\hat{\nabla}_j\hat{A}^{ij}$. This means that any divergence-free components remain completely unconstrained and can be chosen arbitrarily. According to the following lemma, the remaining longitudinal component of $\hat{A}^{ij}$ can be written in terms of a vector potential:

Lemma B.12 Let $\hat{A}^{ij}$ be a symmetric traceless tensor. Then, there exists a symmetric, traceless and transverse tensor Q^{ij}, i.e.

$$\hat{\nabla}_j Q^{ij} = 0, \quad Q^i{}_i = 0, \tag{B.5.14}$$

and a vector field X^i such that

$$\hat{A}^{ij} = Q^{ij} + (\mathbb{L}X)^{ij} = Q^{ij} + \hat{\nabla}^i X^j + \hat{\nabla}^j X^i - \frac{2}{3}\hat{\gamma}^{ij}\hat{\nabla}_k X^k. \tag{B.5.15}$$

The operator $\mathbb{L}$ is called the *Killing operator*, and $(\mathbb{L}X)^i{}_i = 0$ by construction.

We insert this into the divergence term that appears in the momentum constraint to obtain

$$\begin{aligned}\hat{\nabla}_k\hat{A}^{ki} &= \hat{\nabla}_k\hat{\nabla}^k X^i + \hat{\nabla}_k\hat{\nabla}^i X^k - \frac{2}{3}\hat{\nabla}^i\hat{\nabla}_k X^k \\ &= \hat{\nabla}_k\hat{\nabla}^k X^i + \frac{1}{3}\hat{\nabla}^i\hat{\nabla}_k X^k + \hat{R}^i{}_k X^k.\end{aligned} \tag{B.5.16}$$

Thus, the Hamiltonian and momentum constraints in this decomposition read

$$\widehat{\mathcal{H}} = 8\hat{\gamma}^{ij}\hat{\nabla}_i\hat{\nabla}_j\psi - \hat{R}\psi - \frac{2}{3}K^2\psi^5 + \hat{A}_{ij}\hat{A}^{ij}\psi^{-7} + (2\Lambda + 16\pi\rho)\psi^5 = 0,$$

$$\widehat{\mathcal{M}}^i = \hat{\nabla}_k\hat{\nabla}^k X^i + \frac{1}{3}\hat{\nabla}^i\hat{\nabla}_k X^k + \hat{R}^i{}_k X^k - \frac{2}{3}\psi^6\hat{\gamma}^{ik}\partial_k K - 8\pi\psi^{10}j^i = 0. \tag{B.5.17}$$

The implications of this decomposition are very convenient; the data is now clearly split into eight freely specifiable functions—5 from the background conformal metric $\hat{\gamma}_{ij}$, one from the mean curvature K, and two from the symmetric transverse-traceless extrinsic curvature component Q^{ij}. The remaining four variables—the scalar ψ and the three components of X^i—are then fixed by the constraint equations above. Further, the momentum constraint is linear in the vector potential X^i, which will enable us to find analytical solutions in certain cases.

B.5.1.1 Brill-Lindquist Data

The most convenient way to solve an equation is always to specialise to the most simplified version of the problem. Here, this is achieved by assuming vacuum ($\rho = j^i = 0$), no cosmological constant ($\Lambda = 0$), and time-symmetry ($K_{ij} = 0$, so $K = Q^{ij} = X^i = 0$), as well as a flat conformal

background, $\hat{\gamma}_{ij} = \delta_{ij}$. These assumptions collapse the constraint equations (B.5.17) down to just a single scalar equation,

$$\Delta\psi = 0, \tag{B.5.18}$$

where $\Delta = \delta^{ij}\partial_i\partial_j$ is the flat-space Laplacian. If we require solutions to be asymptotically flat and smooth, we must have $\psi \to 1$ as $r \to \infty$ and $\|\psi\|_\infty < \infty$, which by Liouville's theorem forces $\psi \equiv 1$ everywhere. However, if we allow for isolated singular points, then solutions of the form

$$\psi(\mathbf{x}) = 1 + \frac{C}{r} \tag{B.5.19}$$

with $\mathbf{x} = (x^i)$ and $r = \sqrt{\delta_{ij}x^i x^j}$ are allowed as well. Given that in isotropic coordinates, the Schwarzschild metric reads

$$g_{\text{SS}} = -\left(\frac{1 - \frac{M}{2r}}{1 + \frac{M}{2r}}\right)^2 dt^2 + \left(1 + \frac{M}{2r}\right)^4 \delta_{ij}\, dx^i\, dx^j, \tag{B.5.20}$$

we identify

$$\psi(\mathbf{x}) = 1 + \frac{M}{2r}, \tag{B.5.21}$$

whence $C = M/2$ is related to the mass of the black hole. By linearity of equation (B.5.18), we can superimpose multiple such solutions to obtain multi-black hole initial data,

$$\psi(\mathbf{x}) = 1 + \sum_{i=1}^{n} \frac{M_{(a)}}{2|\mathbf{x} - \mathbf{x}_{(a)}|} \tag{B.5.22}$$

where $M_{(a)}$ is the mass and $\mathbf{x}_{(a)}$ the initial position of the i-th black hole. This form of initial data is known as *Brill-Lindquist data*.

Remark B.13 There are a couple of observations to be made here.

(i) As can be seen in the Schwarzschild example, the singularities of ψ do not actually correspond to physical singularities inside the black hole, but rather represent the coordinate singularities at spatial infinity of the additional asymptotically flat ends that are introduced by the presence of black holes.

(ii) The time-symmetry assumption $K_{ij} = 0$ implies that the mean curvature vanishes, $K = 0$. This is the condition for a submanifold to be maximal—in essence, it ensures that the initial data slice has a maximal 3-volume within the four-dimensional spacetime.

(iii) The masses $M_{(a)}$ are the *bare* masses of the black holes, not the physical ADM masses of their associated asymptotically flat ends. Near the puncture $\mathbf{x}_{(a)}$—that is, near spatial infinity of the flat end, where $\mathbf{x} \to \mathbf{x}_{(a)}$—the conformal factor ψ behaves as

$$\psi(\mathbf{x}) \sim 1 + \frac{M_{(a)}}{2|\mathbf{x} - \mathbf{x}_{(a)}|} + \left(\sum_{j \neq i} \frac{M_{(j)}}{2|\mathbf{x} - \mathbf{x}_{(j)}|}\right). \tag{B.5.23}$$

Although the leading "$1 + M_{(a)}/2r$" ensures that $M_{(a)}$ appears as a term in the ADM mass of the i-th end, the remaining terms in parentheses provide an additional, regular contribution to the surface integral that is best interpreted as a binding energy. Explicitly, it can be computed that

$$M_{\mathrm{ADM},(i)} = M_{(a)} \left(1 + \sum_{j \neq i} \frac{M_{(j)}}{2|\mathbf{x}_{(a)} - \mathbf{x}_{(j)}|} \right) \tag{B.5.24}$$

I did not check this explicitly yet. Nonetheless, the total ADM mass of the "main end" at $r \to \infty$ is

$$M_{\mathrm{ADM},\infty} = \sum_{i=1}^{n} M_{(a)}, \tag{B.5.25}$$

as is easily verified by the fact that asymptotically, $|\mathbf{x} - \mathbf{x}_{(a)}| \sim r$ and hence,

$$\psi(\mathbf{x}) \sim 1 + \frac{\sum_{i=1}^{n} M_{(a)}}{2r} \quad \text{as} \quad r \to \infty. \tag{B.5.26}$$

(iv) Although Brill-Lindquist can accommodate multi-black hole initial data, it does so in a very limited way. The black holes are initially stationary (although of course, they do not remain stationary during evolution), and have no spin. In astrophysical scenarios, this is typically rather uninteresting; we usually want to simulate black hole mergers where the black holes have both linear and angular momentum. For this reason, in the next section, we will loosen our simplifying assumptions, with the goal of enabling us to place black holes with arbitrary mass, momentum and spin into our simulation domain.

B.5.1.2 Bowen-York Initial Data and the Puncture Method

To improve on the lack of initial linear and angular momentum of Brill-Lindquist initial data, in this section, we make our assumptions more general. Concretely, we drop time symmetry $K_{ij} = 0$, and instead only demand maximal slicing ($K = 0$) and that the transverse part vanishes, $Q^{ij} = 0$. To arrive at Bowen-York initial data, we keep the flat conformal background assumption $\hat{\gamma}_{ij} = \delta_{ij}$ in place, and continue to work in vacuum with a vanishing cosmological constant.

These assumptions turn the constraint equations into

$$\begin{aligned} \widehat{\mathcal{H}} &= 8\Delta\psi + \hat{A}_{ij}\hat{A}^{ij}\psi^{-7} = 0, \\ \widehat{\mathcal{M}}^i &= \Delta X^i + \frac{1}{3}\partial^i\partial_k X^k = 0, \end{aligned} \tag{B.5.27}$$

where $\Delta = \delta^{ij}\partial_i\partial_j$ is the flat space Laplacian, indices are raised with $\hat{\gamma}^{ij} = \delta^{ij}$, and $\hat{A}^{ij}$ is computed as

$$\hat{A}^{ij} = (\mathbb{L}X)^{ij} = \partial^i X^j + \partial^j X^i - \frac{2}{3}\delta^{ij}\partial_k X^k. \tag{B.5.28}$$

We can see that under these assumptions, only X^i appears in the momentum constraints, decoupling it fully from ψ. Further, the momentum constraint is now a linear equation for X^i—which we can actually solve analytically—and which allows for solutions to be superimposed. Once X^i is determined, we can evaluate $\hat{A}^{ij}$, leaving us to solve the nonlinear Hamiltonian constraint for ψ. Although this has to be approached numerically, we will consider an ansatz that splits off the analytical Brill-Lindquist terms, making it so that we only have to solve for a regular function—instead of the full ψ which typically contains singular points.

Let us begin by discussing the two most important types of solutions to the momentum constraint equation, starting with angular momentum.

Proposition B.14 *(Bowen-York Data: Angular Momentum)* The momentum constraint equation (B.5.27) is solved by

$$X^i_{\text{ang}} = \varepsilon^{ijk}\frac{x_j}{r^3}J_k, \tag{B.5.29}$$

which has the associated traceless conformal extrinsic curvature tensor

$$\hat{A}^{ij}_{\text{ang}} = -\frac{6}{r^5}x^{(i}\varepsilon^{j)k\ell}x_k J_\ell. \tag{B.5.30}$$

One can verify that for asymptotically flat initial data, where $\psi \to 1$ for $r \to \infty$, $J^i_{\text{ADM}} = J^i$ for the AF end produced by the puncture.

Proposition B.15 *(Bowen-York Data: Linear Momentum)* The momentum constraint equation (B.5.27) is also solved by

$$X^i_{\text{lin}} = -\frac{1}{4r}\left(7P^i + \frac{x^i x_k}{r^2}P^k\right) \tag{B.5.31}$$

for a constant vector P^i. The associated traceless conformal extrinsic curvature tensor is

$$\hat{A}^{ij}_{\text{lin}} = \frac{3}{2r^3}\left[2P^{(i}x^{j)} + \left(\frac{x^i x^j}{r^2} - \delta^{ij}\right)x_m P^m\right]. \tag{B.5.32}$$

For asymptotically flat initial data, where $\psi \to 1$ for $r \to \infty$, $P^i_{\text{ADM}} = P^i$ for the AF end produced by the puncture, so that X^i_{lin} has the interpretation of the linear momentum of a black hole.

We now have everything we need to close the Bowen-York initial data story. Given positions $\mathbf{x}_{(a)}$, masses $M_{(a)}$, linear momenta $\mathbf{P}_{(a)}$ and angular momenta $\mathbf{J}_{(a)}$, we set up the total $\hat{A}^{ij}$ as

$$\hat{A}^{ij}(\mathbf{x}) = \sum_{a=1}^{n}\hat{A}^{ij}_{\text{lin}}\big(\mathbf{x}-\mathbf{x}_{(a)};\mathbf{P}_{(a)}\big) + \sum_{a=1}^{n}\hat{A}^{ij}_{\text{ang}}\big(\mathbf{x}-\mathbf{x}_{(a)};\mathbf{J}_{(a)}\big). \tag{B.5.33}$$

It now remains to solve the Hamiltonian constraint for ψ. To assign the masses to the punctures, it is sufficient—mathematically speaking—to impose the boundary conditions that

$$\psi(\mathbf{x}) \sim \frac{M_{(a)}}{2|\mathbf{x}-\mathbf{x}_{(a)}|} \quad \text{for} \quad \mathbf{x} \to \mathbf{x}_{(a)}, \quad \psi(\mathbf{x}) \to 1 \quad \text{for} \quad r \to \infty \tag{B.5.34}$$

However, this is hard to enforce numerically, and will cause the function ψ which is being solved for to have singularities that are difficult to resolve on a grid. Given that we already know the Brill-Lindquist solution, we rather build on that; we take the ansatz

$$\psi(\mathbf{x}) = u(\mathbf{x}) + \underbrace{1 + \sum_{a=1}^{n}\frac{M_{(a)}}{2|\mathbf{x}-\mathbf{x}_{(a)}|}}_{=:\psi_0(\mathbf{x})}, \tag{B.5.35}$$

and solve for the (now regular) function $u(\mathbf{x})$ instead. Since away from the punctures at $\mathbf{x}_{(a)}$, the Laplacian acting on the $\sim 1/r$ terms vanishes, the equation for u reads

$$\Delta u + \frac{1}{8}\hat{A}_{ij}\hat{A}^{ij}(u+\psi_0)^{-7} = 0. \tag{B.5.36}$$

We have thus now reduced the problem of solving the rather complicated constraint equations to an almost fully analytical solution, with as final step solving a non-linear scalar equation for a regular function. This final reduction step is often termed the *puncture method*, due to Brandt and Brügmann.

The solutions that result are black holes in specifiable positions, with tunable mass as well as linear and angular momentum. However, the initial data produced this way is not perfect, in the sense that it contains more than just black holes. This is evident from the following line of reasoning: It can be shown that the Kerr spacetime—i.e. the spacetime containing a singular rotating black hole—does not allow for any conformally flat spatial hypersurfaces. Our initial data, however, is able to produce a hypersurface that contains a black hole with spin, starting out from the assumption that it is a conformally flat slice ($\hat{\gamma}_{ij} = \delta_{ij}$). This means that, besides a Kerr black hole, such a slice must contain additional gravitational radiation, as otherwise we would contradict the theorem that Kerr allows no conformally flat spatial slices. For slow spins, this radiation is typically of no concern, as it propagates away quickly and hence only contaminates the first part of the gravitational wave signal.

B.5.2 (Extended) Conformal Thin Sandwich ((X)CTS)

B.6 An Initial Overview of Z4

B.6.1 Time Evolution of Violated Constraints in Standard GR

It is rather simple to show that mathematically speaking, if the constraints are satisfied initially, they are satisfied for the entirety of the evolution—this is a direct consequence of $\nabla^\mu G_{\mu\nu} = 0$. Unfortunately, we cannot perfectly satisfy $\mathcal{H} = \mathcal{M}_i = 0$ when computing initial data numerically, and machine as well as discretisation error introduce additional deviations, so that at best, $\mathcal{H} \approx \mathcal{M}_i \approx 0$. Motivated by this, let us see how nonzero values of $\mathcal{H}$ and $\mathcal{M}_i$ evolve under evolution.

Since we are only interested in a mathematical result for now, and not a numerically stable implementation, we can make our lives easier by starting from the Einstein equations written as $\mathcal{E}_{\mu\nu} = G_{\mu\nu} + \Lambda g_{\mu\nu} - 8\pi T_{\mu\nu} = 0$, without needing to decompose or invoke adapted coordinates just yet. We recall that the constraints are defined as

$$\mathcal{H} = 2\mathcal{E}_{\mu\nu} n^\mu n^\nu, \quad \mathcal{M}_\mu = P^\lambda{}_\mu n^\nu \mathcal{E}_{\lambda\nu}. \tag{B.6.1}$$

By the contracted Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$, the metric compatibility of the connection, $\nabla_\lambda g_{\mu\nu} = 0$, and conservation of energy-momentum $\nabla^\mu T_{\mu\nu} = 0$, we have

$$\nabla^\mu \mathcal{E}_{\mu\nu} = 0. \tag{B.6.2}$$

Besides this identity, we will also need to assume that the evolution equations in (B.2.44) hold. We recall that these are derived from the tangential-tangential projection of the *trace-reversed* Einstein equations, that is, from

$$P^\lambda{}_\mu P^\rho{}_\nu \left(\mathcal{E}_{\lambda\rho} - \frac{1}{2} g_{\lambda\rho} \mathcal{E} \right) = 0. \tag{B.6.3}$$

Since in the derivations that follow, the projection $P^\lambda{}_\mu P^\rho{}_\nu \mathcal{E}_{\lambda\rho}$ will appear, let us first examine what this evaluates to assuming that the above holds. Taking the trace with respect to $g^{\mu\nu}$ gives us

$$0 = P^{\lambda\rho}\mathcal{E}_{\lambda\rho} - \frac{3}{2}\mathcal{E} = \underbrace{g^{\lambda\rho}\mathcal{E}_{\lambda\rho}}_{=\mathcal{E}} + \underbrace{n^\lambda n^\rho \mathcal{E}_{\lambda\rho}}_{=\frac{1}{2}\mathcal{H}} - \frac{3}{2}\mathcal{E} = \frac{1}{2}\mathcal{H} - \frac{1}{2}\mathcal{E}. \tag{B.6.4}$$

Thus, $\mathcal{E} = \mathcal{H}$, and by equation (B.6.3),

$$P^\lambda{}_\mu P^\rho{}_\nu \mathcal{E}_{\lambda\rho} = \frac{1}{2}\gamma_{\mu\nu}\mathcal{H}. \tag{B.6.5}$$

To derive an evolution equation for $\mathcal{H}$, we project $\nabla^\mu \mathcal{E}_{\mu\nu}$ onto n^ν, to obtain

$$
\begin{aligned}
0 = n^\nu \nabla^\mu \mathcal{E}_{\mu\nu} &= \nabla^\mu \big(\mathcal{E}_{\mu\nu} n^\nu\big) - \mathcal{E}_{\mu\nu} \nabla^\mu n^\nu \\
&= \nabla^\mu \Big(-n_\mu \underbrace{n^\lambda \mathcal{E}_{\lambda\nu} n^\nu}_{=\frac{1}{2}\mathcal{H}} + \underbrace{P_\mu{}^\lambda \mathcal{E}_{\lambda\nu} n^\nu}_{=\mathcal{M}_\mu} \Big) + \mathcal{E}_{\mu\nu}(K^{\mu\nu} + n^\mu a^\nu) \\
&= -\frac{1}{2}\nabla_\mu (n^\mu \mathcal{H}) + \nabla^\mu \mathcal{M}_\mu + \underbrace{\mathcal{E}_{\mu\nu} P^\mu{}_\lambda P^\nu{}_\rho}_{=\frac{1}{2}\gamma_{\lambda\rho}\mathcal{H}} K^{\nu\rho} + \mathcal{M}_\mu a^\mu \\
&= -\frac{1}{2} n^\mu \nabla_\mu \mathcal{H} - \frac{1}{2}\underbrace{(\nabla_\mu n^\mu)}_{=-K}\mathcal{H} + \nabla^\nu \big(P^\mu{}_\nu \mathcal{M}_\mu\big) + \frac{1}{2} K\mathcal{H} + \mathcal{M}_\mu a^\mu \\
&= -\frac{1}{2} n^\mu \nabla_\mu \mathcal{H} + K\mathcal{H} + \underbrace{P^\mu{}_\nu \nabla^\nu \mathcal{M}_\mu}_{=\bar{\nabla}^\mu \mathcal{M}_\mu} + \underbrace{\mathcal{M}_\mu \nabla^\nu (n^\mu n_\nu)}_{=\mathcal{M}_\mu a^\mu} + \mathcal{M}_\mu a^\mu \\
&= -\frac{1}{2} n^\mu \nabla_\mu \mathcal{H} + K\mathcal{H} + \bar{\nabla}^\mu \mathcal{M}_\mu + 2\mathcal{M}_\mu a^\mu
\end{aligned}
\tag{B.6.6}
$$

Solving for $n^\mu \nabla_\mu \mathcal{H}$, we find

$$
n^\mu \nabla_\mu \mathcal{H} = 2K\mathcal{H} + 2\bar{\nabla}^\mu \mathcal{M}_\mu + 4\mathcal{M}_\mu a^\mu. \tag{B.6.7}
$$

Switching to adapted coordinates and making use of

$$
n = \frac{1}{\alpha}(\partial_t - \beta^i \partial_i), \quad a_\mu = \frac{1}{\alpha}\bar{\nabla}_\mu \alpha \tag{B.6.8}
$$

we finally arrive at

$$
\partial_t \mathcal{H} = \beta^i \partial_i \mathcal{H} + 2\alpha K\mathcal{H} + 2\alpha \bar{\nabla}^i \mathcal{M}_i + 4\mathcal{M}_i \bar{\nabla}^i \alpha. \tag{B.6.9}
$$

We can proceed similarly for the momentum constraint, this time projecting onto the foliation-tangent space:

$$
\begin{aligned}
0 = P^\lambda{}_\nu (\nabla^\mu \mathcal{E}_{\mu\lambda}) &= \nabla^\mu \big(P^\lambda{}_\nu \mathcal{E}_{\mu\lambda}\big) - \mathcal{E}_{\mu\lambda} \nabla^\mu P^\lambda{}_\nu \\
&= \nabla^\mu \Big(-n_\mu \underbrace{n^\rho P^\lambda{}_\nu \mathcal{E}_{\rho\lambda}}_{=\mathcal{M}_\nu} + \underbrace{P^\rho{}_\mu P^\lambda{}_\nu \mathcal{E}_{\rho\lambda}}_{=\frac{1}{2}\gamma_{\mu\nu}\mathcal{H}} \Big) - \mathcal{E}_{\mu\lambda} \nabla^\mu (n^\lambda n_\nu) \\
&= -\nabla^\mu \big(n_\mu \mathcal{M}_\nu\big) + \frac{1}{2}\nabla^\mu \big(\gamma_{\mu\nu}\mathcal{H}\big) + \mathcal{E}_{\mu\lambda}\big(K^{\mu\lambda} + n^\mu a^\lambda\big) n_\nu + \mathcal{E}_{\mu\lambda} n^\lambda \big(K^\mu{}_\nu + n^\mu a_\nu\big) \\
&= -n^\mu \nabla_\mu \mathcal{M}_\nu - \mathcal{M}_\nu \underbrace{\nabla^\mu n_\mu}_{=-K} + \frac{1}{2}\bar{\nabla}_\nu \mathcal{H} + \frac{1}{2}\mathcal{H}\nabla^\mu \big(n_\mu n_\nu\big) + \frac{1}{2}\mathcal{H} K n_\nu + \mathcal{M}_\mu a^\mu n_\nu \\
&\quad + K^\mu{}_\nu \mathcal{M}_\mu + \frac{1}{2}\mathcal{H} a_\nu \\
&= -n^\mu \nabla_\mu \mathcal{M}_\nu + K\mathcal{M}_\nu + \frac{1}{2}\bar{\nabla}_\nu \mathcal{H} + \frac{1}{2}\mathcal{H} a_\nu + \mathcal{M}_\mu a^\mu n_\nu + K^\mu{}_\nu \mathcal{M}_\mu + \frac{1}{2}\mathcal{H} a_\nu
\end{aligned}
\tag{B.6.10}
$$

We again want to find an expression in adapted coordinates, and then solve for $\partial_t \mathcal{M}_i$. Since $n_i = 0$, we can immediately drop the term proportional to n_ν to find

$$
n^\mu \nabla_\mu \mathcal{M}_i = K\mathcal{M}_i + K^j{}_i \mathcal{M}_j + \frac{1}{2}\bar{\nabla}_i \mathcal{H} + \frac{1}{\alpha}\mathcal{H}\bar{\nabla}_i \alpha \tag{B.6.11}
$$

Since $\mathcal{M}_\mu$ is a covector, we cannot simply replace the normal derivative on the left-hand side with $\frac{1}{\alpha}(\partial_t - \beta^i \partial_i)$ as before. Instead, let us expand the covariant derivative explicitly. For any spatial covector X_μ with $n^\mu X_\mu = 0$,

$$\mathcal{L}_n X_\nu = n^\mu \nabla_\mu X_\nu + X_\mu \nabla_\nu n^\mu \tag{B.6.12}$$

implying that

$$\begin{aligned} n^\mu \nabla_\mu X_\nu &= \mathcal{L}_n X_\nu - X_\mu \nabla_\nu n^\mu \\ &= \mathcal{L}_n X_\nu + X_\mu \big(K^\mu{}_\nu + n_\nu a^\mu\big) \\ &= \mathcal{L}_n X_\nu + K^\mu{}_\nu X_\mu + n_\nu X_\mu a^\mu. \end{aligned} \tag{B.6.13}$$

Further using equation (B.2.60), we can rewrite the spatial components of this expression as

$$n^\nu \nabla_\nu X_i = \frac{1}{\alpha}\partial_t X_i - \frac{1}{\alpha}\mathcal{L}_\beta X_i + K^j{}_i X_j. \tag{B.6.14}$$

Inserting this result back into the original equation and expanding the Lie derivative along β, we get

$$\partial_t \mathcal{M}_i = \beta^j \partial_j \mathcal{M}_i + \mathcal{M}_j \partial_i \beta^j + \alpha K \mathcal{M}_i + \frac{1}{2}\alpha \bar{\nabla}_i \mathcal{H} + \mathcal{H}\bar{\nabla}_i \alpha \tag{B.6.15}$$

For quicker reference, let us summarise the results for both $\partial_t \mathcal{H}$ and $\partial_t \mathcal{M}_i$ below, making compact again the Lie derivatives along β:

$$\begin{aligned} \partial_t \mathcal{H} &= \mathcal{L}_\beta \mathcal{H} + 2\alpha K \mathcal{H} + 2\alpha \bar{\nabla}^i \mathcal{M}_i + 4\mathcal{M}_i \bar{\nabla}^i \alpha, \\ \partial_t \mathcal{M}_i &= \mathcal{L}_\beta \mathcal{M}_i + \alpha K \mathcal{M}_i + \frac{1}{2}\alpha \bar{\nabla}_i \mathcal{H} + \mathcal{H}\bar{\nabla}_i \alpha. \end{aligned} \tag{B.6.16}$$

Remarks:

(i) The simplest check we can carry out to verify whether this result is plausible is to see whether evolution preserves initially satisfied constraints. This is indeed the case, since all terms on the right-hand sides of both $\partial_t \mathcal{H}$ and $\partial_t \mathcal{M}_i$ are proportional to either $\mathcal{H}$, $\mathcal{M}_i$, or spatial derivatives thereof. Hence, if $\mathcal{H} = \mathcal{M}_i = 0$ initially, then the right-hand sides vanish initially. Since $\mathcal{H} = \mathcal{M}_i$ is itself a solution of the constraint-propagation equations, uniqueness of the corresponding initial-value problem implies that initially staisfied constraints remain satisfied throughout the evolution.

(ii) Although the evolution equations (B.6.16) look rather involved, all terms can be given a physical interpretation:

- The terms $\mathcal{L}_\beta \mathcal{H}$ and $\mathcal{L}_\beta \mathcal{M}_i$ describe advection by the shift, separating coordinate motion from evolution along the normal direction. The terms proportional to αK then modify the evolution according to the expansion or contraction of the normal congruence, decreasing the magnitude of the constraints when the congruence expands and increasing it when the congruence contracts.
- The terms $2\alpha\bar{\nabla}^i \mathcal{M}_i$ and $\frac{1}{2}\alpha\bar{\nabla}_i \mathcal{H}$ couple spatial variations of the momentum and Hamiltonian constraints, respectively. The remaining terms involving $\bar{\nabla}_i \alpha$ provide additional lower-order coupling when the lapse varies spatially.

B.6.2 Propagation of Constraints in GR/ADM

The time evolution of $\mathcal{H}$ and $\mathcal{M}_i$ can be used to derive second-order wave-like equations for $\mathcal{H}$ and $\mathcal{M}_i$, which reveal the propagation behaviour (or lack thereof) of constraint violations across the spacetime. In this section, we carry out such a derivation in a simplified scenario. Specifically, we make the simplifying assumption of unit lapse and vanishing shift, i.e.

$$\alpha = 1, \quad \beta^i = 0. \tag{B.6.17}$$

This choice is locally fully general: around any spacelike hypersurface, one can construct Gaussian normal coordinates for which the above holds, at least within a sufficiently small neighbourhood of the hypersurface.

In this gauge—and replacing ∂_t with an overdot—the constraint evolution (B.6.16) breaks down to

$$\begin{aligned}\dot{\mathcal{H}} &= 2K\mathcal{H} + 2\bar{\nabla}^i\mathcal{M}_i,\\ \dot{\mathcal{M}}_i &= K\mathcal{M}_i + \frac{1}{2}\bar{\nabla}_i\mathcal{H}.\end{aligned} \tag{B.6.18}$$

We begin by taking a second time derivative of $\mathcal{H}$. This leads to

$$\begin{aligned}\ddot{\mathcal{H}} &= 2\dot{K}\mathcal{H} + 2K\dot{\mathcal{H}} + 2\partial_t\big(\gamma^{ij}\big(\partial_i\mathcal{M}_j - \bar{\Gamma}^k{}_{ij}\mathcal{M}_k\big)\big)\\ &= 2\dot{K}\mathcal{H} + 2K\dot{\mathcal{H}} + 2\dot{\gamma}^{ij}\bar{\nabla}_i\mathcal{M}_j + 2\bar{\nabla}^i\dot{\mathcal{M}}_i - 2\gamma^{ij}\dot{\bar{\Gamma}}^k{}_{ij}\mathcal{M}_k\end{aligned} \tag{B.6.19}$$

For $\dot{\mathcal{H}}$ and $\dot{\mathcal{M}}_i$, we can simply insert the simplified evolution equations from above. The terms involving $\dot{\gamma}^{ij}$ and $\dot{\bar{\Gamma}}^k{}_{ij}$ require slightly more attention. Starting with $\dot{\gamma}^{ij}$ and keeping in mind that $\alpha = 1$, $\beta^i = 0$, we use the ADM equation (B.2.66) for $\partial_t\gamma_{ij}$ to rewrite

$$\dot{\gamma}^{ij} = -\gamma^{ik}\gamma^{j\ell}\underbrace{\partial_t\gamma_{k\ell}}_{=-2K_{ij}} = 2K^{ij}. \tag{B.6.20}$$

For the time derivative of the connection coefficients, we observe that it is the (infinitesimal) difference of connections, and hence a tensor. In normal coordinates, it follows that

$$\dot{\bar{\Gamma}}^k{}_{ij} = \frac{1}{2}\gamma^{k\ell}\big(\bar{\nabla}_i\partial_t\gamma_{j\ell} + \bar{\nabla}_j\partial_t\gamma_{i\ell} - \bar{\nabla}_\ell\partial_t\gamma_{ij}\big). \tag{B.6.21}$$

Again using $\partial_t\gamma_{ij} = -2K_{ij}$, we arrive at

$$\begin{aligned}\gamma^{ij}\dot{\bar{\Gamma}}^k{}_{ij} &= -\gamma^{ij}\gamma^{k\ell}\big(\bar{\nabla}_iK_{j\ell} + \bar{\nabla}_jK_{i\ell} - \bar{\nabla}_\ell K_{ij}\big)\\ &= \bar{\nabla}^kK - 2\bar{\nabla}_iK^{ik}\end{aligned} \tag{B.6.22}$$

Inserting all this back into $\ddot{\mathcal{H}}$, we arrive at

$$\begin{aligned}\ddot{\mathcal{H}} &= 2\dot{K}\mathcal{H} + 2K\big(2K\mathcal{H} + 2\bar{\nabla}^i\mathcal{M}_i\big) + 4K^{ij}\bar{\nabla}_i\mathcal{M}_j + 2\bar{\nabla}^i\Big(K\mathcal{M}_i + \frac{1}{2}\bar{\nabla}_i\mathcal{H}\Big)\\ &\quad -2\big(\bar{\nabla}^kK - 2\bar{\nabla}_iK^{ik}\big)\mathcal{M}_k\\ &= \big(2\dot{K} + 4K^2\big)\mathcal{H} + 4(\gamma^{ij}K + K^{ij})\bar{\nabla}_i\mathcal{M}_j + \cancel{2(\bar{\nabla}^iK)\mathcal{M}_i} + 2K\bar{\nabla}^i\mathcal{M}_i + \bar{\nabla}^i\bar{\nabla}_i\mathcal{H}\\ &\quad \cancel{-2(\bar{\nabla}^kK)\mathcal{M}} + 4\big(\bar{\nabla}_iK^{ik}\big)\mathcal{M}_k\\ &= \bar{\nabla}^i\bar{\nabla}_i\mathcal{H} + \big(2\dot{K} + 4K^2\big)\mathcal{H} + 6K\bar{\nabla}^i\mathcal{M}_i + 4\bar{\nabla}_i\big(K^{ij}\mathcal{M}_j\big)\end{aligned} \tag{B.6.23}$$

We can rearrange this into a wave equation for $\mathcal{H}$,

$$\ddot{\mathcal{H}} - \bar{\nabla}^i\bar{\nabla}_i\mathcal{H} = \big(2\dot{K} + 4K^2\big)\mathcal{H} + 6K\bar{\nabla}^i\mathcal{M}_i + 4\bar{\nabla}_i\big(K^{ij}\mathcal{M}_j\big). \tag{B.6.24}$$

We now carry out the same procedure for $\ddot{\mathcal{M}}_i$, where we make use of the fact that $\bar{\nabla}_i\mathcal{H} = \partial_i\mathcal{H}$, since it is a scalar:

$$\begin{aligned}\ddot{\mathcal{M}}_i &= \dot{K}\mathcal{M}_i + K\dot{\mathcal{M}}_i + \frac{1}{2}\bar{\nabla}_i\dot{\mathcal{H}} \\ &= \dot{K}\mathcal{M}_i + K\Big(K\mathcal{M}_i + \frac{1}{2}\bar{\nabla}_i\mathcal{H}\Big) + \frac{1}{2}\bar{\nabla}_i\big(2K\mathcal{H} + 2\bar{\nabla}^k\mathcal{M}_k\big) \\ &= \big(\dot{K} + K^2\big)\mathcal{M}_i + \frac{1}{2}K\bar{\nabla}_i\mathcal{H} + \bar{\nabla}_i(K\mathcal{H}) + \bar{\nabla}_i\bar{\nabla}^k\mathcal{M}_k\end{aligned} \tag{B.6.25}$$

To isolate the principal symbol, we move the second-order terms to the left. This leads us to—where for a complete summary, we repeat the equation for $\mathcal{H}$ as well—the propagation equations

$$\begin{aligned}\ddot{\mathcal{H}} - \bar{\nabla}^i\bar{\nabla}_i\mathcal{H} &= \big(2\dot{K} + 4K^2\big)\mathcal{H} + 6K\bar{\nabla}^i\mathcal{M}_i + 4\bar{\nabla}_i\big(K^{ij}\mathcal{M}_j\big), \\ \ddot{\mathcal{M}}_i - \bar{\nabla}_i\bar{\nabla}^k\mathcal{M}_k &= \big(\dot{K} + K^2\big)\mathcal{M}_i + \frac{3}{2}K\bar{\nabla}_i\mathcal{H} + \big(\bar{\nabla}_iK\big)\mathcal{H}.\end{aligned} \tag{B.6.26}$$

The equation for $\mathcal{H}$ is a wave equation; the Hamiltonian constraint hence propagates hyperbolically. The lower-order terms on the right-hand side introduce coupling to $\mathcal{M}_i$, as well as an effective mass term dependent upon the extrinsic curvature.

The equation for the momentum constraint is similar, but has a fundamental difference. Instead of the spatial derivatives forming a Laplacian (i.e., the divergence of the gradient), instead, they constitute the gradient of the divergence. This implies that the momentum constraint does not propagate like a standard wave—let us examine its behaviour more precisely. We consider a spatial 3-covector $\mathbf{X} = (X_i)$ on flat spacetime, satisfying the equation

$$\ddot{X}_i - \partial_i\partial^k X_k = 0. \tag{B.6.27}$$

We employ the mode ansatz

$$\mathbf{X}(t, \mathbf{x}) = \mathbf{C}e^{i(\omega t - \mathbf{k}\cdot\mathbf{x})} \tag{B.6.28}$$

where $\mathbf{k}$ is the wave vector and $\mathbf{C}$ the polarisation vector. Inserting this ansatz, we find

$$-\omega^2 C_i + k_i k^k C_k = 0 \quad\Leftrightarrow\quad \omega^2\mathbf{C} = (\mathbf{k}\cdot\mathbf{C})\mathbf{k}. \tag{B.6.29}$$

There are two distinct ways to satisfy this equation to obtain a non-trivial solution $\mathbf{X}$:

(i) Either, the polarisation is longitudinal, $\mathbf{C} = C\mathbf{k}$ for some constant C, and $\mathbf{k}^2 = \omega^2$,

(ii) or, the polarisation is transverse, meaning that $\mathbf{k}\cdot\mathbf{C} = 0$ and hence necessarily, $\omega^2 = 0$.

This means that the question of whether the constraints $\mathcal{H}, \mathcal{M}_i$ in GR propagate has an answer that is more subtle than a pure yes or no. We have just found that while the Hamiltonian constraint $\mathcal{H}$, as well as the longitudinal modes of $\mathcal{M}_i$—the contributions for which $\bar{\nabla}^i\mathcal{M}_i = 0$—propagate hyperbolically, the transverse modes of $\mathcal{M}_i$ stay fixed in place. Hence, in numerical simulations, such transverse momentum constraint violations do not get propagated to the boundary, but instead remain where they are produced, and potentially grow to scales that make the simulation become unphysical and crash.

B.6.3 The Auxiliary Field Z_μ

To improve the stability of numerical simulations of GR, we would ideally like to modify the propagation system (B.6.26) such that all constraint-violation modes have hyperbolic propagation. However, the constraint propagation equations cannot be altered independently: they follow from the choice of evolution equations. Thus to change the constraint propagation, we must instead modify the evolution system for the dynamical variables and allow the constraint propagation to change as a consequence. In principle, such modifications could be chosen to produce a more

desirable constraint subsystem, but constructing them indirectly is difficult, particularly because we would like the dynamics on the constraint manifold to remain exactly those of GR.

In essence, since the constraints are defined as

$$\mathcal{H} = 2n^\mu n^\nu \mathcal{E}_{\mu\nu}, \qquad \mathcal{M}_\mu = P^\lambda{}_\mu n^\nu \mathcal{E}_{\lambda\nu}, \tag{B.6.30}$$

they are nothing but a measure of how well certain projections of the full Einstein equations $\mathcal{E}_{\mu\nu} = 0$ are satisfied. Suppose we introduce new degrees of freedom to the Einstein equations, which likewise measure the deviation from $\mathcal{E}_{\mu\nu} = 0$, but whose equations of motion we can control directly. Such degrees of freedom would necessarily be related to the constraints, and could therefore provide a more direct means of controlling their propagation behaviour.

Since we want to control four constraints, a reasonable guess is to introduce four degrees of freedom—arranged into a covector we call Z_μ. The motivation to pick a covector is that $\mathcal{H}$ and $\mathcal{M}_\mu$ are really just further projections of the partial projection $\mathcal{E}_{\mu\nu}n^\nu$, which is a covector.

Next, we should think about how to modify the Einstein equations, $\mathcal{E}_{\mu\nu} = 0$, to incorporate additional terms involving Z_μ. To obtain the simplest possible equations of motion, we should only add terms linear in Z_μ—this has the additional effect that $Z_\mu = 0$ recovers the original Einstein equations. Further, whatever we add should be symmetric in $\mu\nu$, and should contain derivatives of Z_μ so that it is a dynamical field, rather than merely algebraically fixed. These requirements are satisfied by the terms $\nabla_{(\mu}Z_{\nu)}$ and $g_{\mu\nu}\nabla^\lambda Z_\lambda$. Although there are more terms we could construct from the available tensors, such as $R_{\mu\nu}\nabla^\lambda Z_\lambda$, the incorporation of extra curvature couplings will only make the dynamics of Z_μ more involved, so we do not add any such terms. Using the two terms we have identified, we modify the Einstein equations into

$$\mathcal{E}_{\mu\nu} + \nabla_\mu Z_\nu + \nabla_\nu Z_\mu - g_{\mu\nu}\nabla^\lambda Z_\lambda = 0. \tag{B.6.31}$$

The specific choice of relative normalisation and sign between the symmetrised gradient and divergence terms will become clear shortly; let us first review the fundamental properties of this modification.

Remarks:

(i) This modification can be thought of as redefining the Ricci tensor,

$$R_{\mu\nu} \to \tilde{R}_{\mu\nu} + \nabla_\mu Z_\nu + \nabla_\nu Z_\mu, \tag{B.6.32}$$

which causes the associated Ricci scalar to turn into

$$R \to \tilde{R} = R + 2\nabla^\lambda Z_\lambda. \tag{B.6.33}$$

Hence, the Einstein tensor becomes

$$\begin{aligned} G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R \to \tilde{G}_{\mu\nu} &= \tilde{R}_{\mu\nu} - \frac{1}{2}g_{\mu\nu}\tilde{R} \\ &= G_{\mu\nu} + \nabla_\mu Z_\nu + \nabla_\nu Z_\mu - g_{\mu\nu}\nabla^\lambda Z_\nu. \end{aligned} \tag{B.6.34}$$

Since $\mathcal{E}_{\mu\nu} = G_{\mu\nu} + \Lambda g_{\mu\nu} - 8\pi T_{\mu\nu}$, this reproduces the modified Einstein equations above:

$$\mathcal{E}_{\mu\nu} = 0 \to \tilde{\mathcal{E}}_{\mu\nu} = \mathcal{E}_{\mu\nu} + \nabla_\mu Z_\nu + \nabla_\nu Z_\mu - g_{\mu\nu}\nabla^\lambda Z_\lambda = 0. \tag{B.6.35}$$

(ii) Next, let us examine how the constraints $\mathcal{H}$ and $\mathcal{M}_\mu$ are related to Z_μ. To do so, we simply take the corresponding projections of the modified Einstein tensor, where the $\mathcal{E}_{\mu\nu}$ term reproduces the constraints and the remaining terms give us their relationship to Z_μ. It makes sense to also introduce names for the normal and tangential projections of Z_μ, writing

$$\Theta = n^\mu Z_\mu, \quad Z^\perp_\mu = P^\nu{}_\mu Z_\nu. \tag{B.6.36}$$

The full vector is then recovered as

$$Z_\mu = -\Theta n_\mu + Z^\perp_\mu. \tag{B.6.37}$$

To obtain relationships between $\mathcal{H}, \mathcal{M}_\mu$ and the newly introduced variables Θ and $Z^\perp_\mu$, we take the corresponding projections of Einstein's equations, starting with the doubly normal one:

$$\begin{aligned}
0 &= 2n^\mu n^\nu \big(\mathcal{E}_{\mu\nu} + \nabla_\mu Z_\nu + \nabla_\nu Z_\mu - g_{\mu\nu}\nabla^\lambda Z_\lambda\big) \\
&= \mathcal{H} + 4n^\mu n^\nu \nabla_\mu Z_\nu + 2\nabla^\lambda Z_\lambda \\
&= \mathcal{H} + 4n^\mu \nabla_\mu \underbrace{(n^\nu Z_\nu)}_{=\Theta} - 4Z_\nu \underbrace{n^\mu \nabla_\mu n^\nu}_{=a^\nu \perp n^\mu} - 2\nabla^\mu\big(n_\mu \Theta\big) + 2\nabla^\mu Z^\perp_\mu \\
&= \mathcal{H} + 4n^\mu \nabla_\mu \Theta - 4a^\mu Z^\perp_\mu - 2\big(\nabla^\mu n_\mu\big)\Theta - 2n_\mu \nabla^\mu \Theta + 2(\gamma^{\mu\nu} - n^\mu n^\nu)\nabla_\mu Z^\perp_\nu \\
&= \mathcal{H} + 2n^\mu \nabla_\mu \Theta - 4a^\mu Z^\perp_\mu + 2K\Theta + 2\bar\nabla^\mu Z^\perp_\mu + 2a^\nu Z^\perp_\nu \\
&= \mathcal{H} + 2n^\mu \nabla_\mu \Theta + 2K\Theta + 2\bar\nabla^\mu Z^\perp_\mu - 2a^\mu Z^\perp_\mu.
\end{aligned} \tag{B.6.38}$$

Here, the raised index of $\bar\nabla$ is to be interpreted as being raised with the metric the connection is compatible with, i.e. with $\gamma^{\mu\nu}$. The above implies that

$$\mathcal{H} = -2n^\mu \nabla_\mu \Theta - 2K\Theta - 2\bar\nabla^\mu Z^\perp_\mu + 2a^\mu Z^\perp_\mu, \tag{B.6.39}$$

or in adapted coordinates,

$$\mathcal{H} = -\frac{2}{\alpha}\partial_t \Theta + \frac{2}{\alpha}\beta^i \partial_i \Theta - 2K\Theta - 2\bar\nabla^i Z_i + \frac{2}{\alpha}\gamma^{ij} Z_i \partial_j \alpha. \tag{B.6.40}$$

Note that since $n_\mu \sim \delta^t_\mu$, $Z^\perp_i = Z_i$. We can proceed similarly for $\mathcal{M}_\mu$:

$$\begin{aligned}
0 &= n^\mu P^\nu{}_\lambda \big(\mathcal{E}_{\mu\nu} + \nabla_\mu Z_\nu + \nabla_\nu Z_\mu - g_{\mu\nu}\nabla^\rho Z_\rho\big) \\
&= \mathcal{M}_\lambda + n^\mu P^\nu{}_\lambda \nabla_\mu Z_\nu + n^\mu P^\nu{}_\lambda \nabla_\nu Z_\mu - \underbrace{n^\mu \gamma_{\mu\lambda}}_{=0} \nabla^\rho Z_\rho \\
&= \mathcal{M}_\lambda + n^\mu \nabla_\mu \underbrace{\big(P^\nu{}_\lambda Z_\nu\big)}_{=Z^\perp_\lambda} - \big(n^\mu \nabla_\mu P^\nu{}_\lambda\big) Z_\nu + \underbrace{P^\nu{}_\lambda \nabla_\nu}_{=\bar\nabla_\lambda} \underbrace{\big(n^\mu Z_\mu\big)}_{\Theta} - Z_\mu P^\nu{}_\lambda \underbrace{\nabla_\nu n^\mu}_{=-K_\nu{}^\mu - n_\nu a^\mu} \\
&= \mathcal{M}_\lambda + n^\mu \nabla_\mu Z^\perp_\lambda - n^\mu \nabla_\mu (n^\nu n_\lambda) Z_\nu + \bar\nabla_\lambda \Theta + K^\mu{}_\lambda Z^\perp_\mu \\
&= \mathcal{M}_\lambda + n^\mu \nabla_\mu Z^\perp_\lambda + K^\mu{}_\lambda Z^\perp_\mu - n_\lambda a^\mu Z^\perp_\mu - a_\lambda \Theta + \bar\nabla_\lambda \Theta
\end{aligned} \tag{B.6.41}$$

This implies

$$\mathcal{M}_\lambda = -n^\mu \nabla_\mu Z^\perp_\lambda - K^\mu{}_\lambda Z^\perp_\mu + n_\lambda a^\mu Z^\perp_\mu - a_\lambda \Theta + \bar\nabla_\lambda \Theta \tag{B.6.42}$$

or in adapted coordinates, making use of equation (B.6.14),

$$\mathcal{M}_i = -\frac{1}{\alpha}\partial_t Z_i + \frac{1}{\alpha}\mathcal{L}_\beta Z_i - 2K^j{}_i Z_j - \frac{1}{\alpha}(\partial_i \alpha)\Theta + \partial_i \Theta. \tag{B.6.43}$$

In summary, we have the covariant identities

$$\begin{aligned}
\mathcal{H} &= -2n^\mu \nabla_\mu \Theta - 2K\Theta - 2\bar\nabla^\mu Z^\perp_\mu + 2a^\mu Z^\perp_\mu, \\
\mathcal{M}_\lambda &= -n^\mu \nabla_\mu Z^\perp_\lambda - K^\mu{}_\lambda Z^\perp_\mu + n_\lambda a^\mu Z^\perp_\mu - a_\lambda \Theta + \bar\nabla_\lambda \Theta,
\end{aligned} \tag{B.6.44}$$

and in adapted coordinates,

$$\begin{aligned}\mathcal{H} &= -\frac{2}{\alpha}\partial_t\Theta + \frac{2}{\alpha}\beta^i\partial_i\Theta - 2K\Theta - 2\bar{\nabla}^i Z_i + \frac{2}{\alpha}\gamma^{ij}Z_i\partial_j\alpha,\\ \mathcal{M}_i &= -\frac{1}{\alpha}\partial_t Z_i + \frac{1}{\alpha}\mathcal{L}_\beta Z_i - 2K^j{}_i Z_j - \frac{1}{\alpha}(\partial_i\alpha)\Theta + \partial_i\Theta.\end{aligned} \tag{B.6.45}$$

Hence, Θ and $Z_\mu^\perp$, or equivalently Z_μ, can be seen as a sort of "vector potential" for the constraints; we obtain the constraints by acting on Z_μ with a linear differential operator.

Note that the adapted coordinate expressions also give us the evolution equations for Θ and Z_i—we can rearrange them for their time derivatives and obtain

$$\begin{aligned}\partial_t\Theta &= \beta^i\partial_i\Theta - \alpha K\Theta - \alpha\bar{\nabla}^i Z_i + \gamma^{ij}Z_i\partial_j\alpha - \frac{1}{2}\alpha\mathcal{H},\\ \partial_t Z_i &= \mathcal{L}_\beta Z_i - 2\alpha K^j{}_i Z_j - (\partial_i\alpha)\Theta + \alpha\partial_i\Theta - \alpha\mathcal{M}_i.\end{aligned} \tag{B.6.46}$$

This allows us to arrive at an important conclusion: while (Θ, Z_i) form a mostly closed evolution subsystem, they are actively sourced by the constraint violations $\mathcal{H}$ and $\mathcal{M}_i$.

(iii) Although in the previous remark, we already derived evolution equations in adapted coordinates, their character is rather opaque. Although we could technically proceed as in the previous section—take a second time derivative and replace any occuring first time derivative with the above—this would be a severely tedious undertaking. It is much easier to compute second-order evolution equations for Z_μ directly from the covariant modified Einstein equations

$$\mathcal{E}_{\mu\nu} + \nabla_\mu Z_\nu + \nabla_\nu Z_\mu - g_{\mu\nu}\nabla^\lambda Z_\lambda = 0 \tag{B.6.47}$$

instead. In particular, this derivation will show us why the $-g_{\mu\nu}\nabla^\lambda Z_\lambda$ term is necessary; without it, the principal symbol would not be hyperbolic.

To proceed with the derivation, we recall that $\nabla^\mu\mathcal{E}_{\mu\nu} = 0$. Hence, if the modified Einstein equations hold, the remaining terms together must be divergence-free as well, i.e.

$$\begin{aligned}0 &= \nabla^\mu\big(\nabla_\mu Z_\nu + \nabla_\nu Z_\mu - g_{\mu\nu}\nabla^\lambda Z_\lambda\big)\\ &= \Box_g Z_\nu + \nabla^\mu\nabla_\nu Z_\mu - \nabla_\nu\nabla^\mu Z_\mu\\ &= \Box_g Z_\nu - R^\lambda{}_\mu{}^\mu{}_\nu Z_\lambda\\ &= \Box_g Z_\nu + R^\lambda{}_\nu Z_\lambda.\end{aligned} \tag{B.6.48}$$

As alluded to earlier, the $-g_{\mu\nu}\nabla^\lambda Z_\lambda$ term is needed—without it, we get additional second order terms because we cannot invoke the Ricci identity.

We conclude that Z_μ satisfies a wave equation containing an additional curvature contribution which allows the components of Z_μ to be "rotated into each other"—importantly, Z_μ propagates hyperbolically, at the speed of light. In particular, since the constraint violations $\mathcal{H}$ and $\mathcal{M}_i$ derive from the action of a linear operator on Z_μ, this means that in evolution governed by the modified Einstein equations (B.6.31), *all* constraint violations propagate as waves. Further—since the trace-reversed Einstein equations give an expression for $R_{\mu\nu}$ in terms of the energy-momentum tensor and first derivatives of Z_μ—we can see that both the presence of matter as well as gradients in Z_μ cause its components to mix and/or oscillate.

(iv) Since we have introduced four additional degrees of freedom, we should ask if all of them are physical *in the framework of the generalised theory*. Within GR, of course, they are all non-physical; viewing the modified Einstein equations as a new physical system, though, we

should check if Z_μ has gauge freedom. Clearly—due to their linearity in Z_μ—the modified Einstein equations remain invariant under transformations $Z_\mu \to Z_\mu + X_\mu$, provided that

$$\nabla_\mu X_\nu + \nabla_\nu X_\mu = g_{\mu\nu}\nabla^\lambda X_\lambda. \tag{B.6.49}$$

Though this might look like some special case of the conformal Killing equation, the reality is much simpler (in $d \neq 2$): taking the trace yields

$$(d-2)\nabla^\lambda X_\lambda = 0. \tag{B.6.50}$$

This means that the condition breaks down to the Killing equation

$$\nabla_\mu X_\nu + \nabla_\nu X_\mu = 0. \tag{B.6.51}$$

Hence, a "physical" configuration in the sense of the modified theory fixes Z_μ only up to a Killing field. The amount of physical degrees of freedom that are lost due to such transformations depends on the configuration of the spacetime. In general, the background will have no Killing vectors, and hence Z_μ has no gauge freedom. However, for e.g. stationary/static or even more symmetric backgrounds, dimension of the isometry algebra is nonzero and finite, so that the moduli space—quotient of the configuration space of Z_μ quotiented by the finite dimensional isometry algebra—has some of its physical configurations identified.

B.6.4 Adding Damping

Up until now, we have introduced the additional degrees of freedom Z_μ and modified the Einstein equations in such a way that

(i) Z_μ propagates as a wave;

(ii) and $\mathcal{H}$ and $\mathcal{M}_i$ emerge from the action of a linear first-order differential operator on Z_μ, in particular causing the implication $Z_\mu = 0 \Rightarrow \mathcal{H}, \mathcal{M}_i = 0$.

Although this is already a significant improvement over unmodified GR—constraints now propagate—we would ideally introduce an additional mechanism that drives Z_μ towards zero throughout its evolution. This would make small enough constraint violations decay over time, in consequence driving simulations towards the constraint manifold and hence improving stability. This is the goal of this section.

Before modifying the equations further to introduce damping, we should consider the implications for general covariance. The un-damped modified Einstein equations are fully covariant and independent of any choice of foliation. Introducing damping, however, inherently requires selecting a preferred timelike direction along which constraint violations decay—naturally provided by the unit normal $n = -\alpha\, dt^\sharp$ of a *chosen* spacelike foliation.

Consequently, manifest four-dimensional covariance is broken. From the point of view of fundamental physics, this would be a significant drawback; however, Z4 is not intended as a new physical theory, but rather as a numerical tool developed to stabilise evolutions and drive solutions back onto the constraint manifold. Because numerical relativity fundamentally relies on a 3+1 split and thus the choice of a foliation, breaking manifest covariance for the sake of constraint damping is entirely justified, as long as the dynamics on the constraint manifold reproduce those of GR.

To introduce damping along n into the wave equation satisfied by Z_μ, we must introduce terms of the form $n^\mu \nabla_\mu X$, where X is either Z_μ itself or one of its projections. Ideally, we would like to have control over the damping of both Θ and $Z_\mu^\perp$ individually, whence we should have multiple parameters we can control. For a damping term of the form $n^\mu \nabla_\mu X$ to emerge in the wave equation for Z_μ, we have to add additional terms to the modified Einstein equations. Specifically,

these terms must contain both Z_μ and n_μ linearly, which immediately leads us to the symmetric combinations

$$n_\mu Z_\nu + n_\nu Z_\mu \quad \text{and} \quad g_{\mu\nu} n^\mu Z_\mu. \tag{B.6.52}$$

These are the only symmetric tensors linear in both n_μ and Z_μ. We are thus motivated to introduce the tensor

$$D_{\mu\nu} = -\kappa_1 \left[n_\mu Z_\nu + n_\nu Z_\mu + \kappa_2 g_{\mu\nu} n^\lambda Z_\lambda \right] \tag{B.6.53}$$

to the modified Einstein equations. The specific way the tunable constants κ_1 and κ_2 are introduced will become clearer momentarily. We use $D_{\mu\nu}$ to modify the Einstein equations into

$$\mathcal{E}_{\mu\nu} + \nabla_\mu Z_\nu + \nabla_\nu Z_\mu - g_{\mu\nu} \nabla^\lambda Z_\lambda + D_{\mu\nu} = 0. \tag{B.6.54}$$

We could check if we do in fact produce a $n^\mu \nabla_\mu Z_\nu$ term by computing the divergence of the above. Clearly, though, taking the divergence $\nabla^\mu D_{\mu\nu}$ will yield a plethera of terms that we don't really want to deal with, so let's entertain another approach. Recall that in the previous section, we derived *first-order* equations of motion for Θ and $Z_\mu^\perp$ by projecting the modified Einstein equations onto the normal-normal and normal-tangential directions. By repeating this, we generate additional terms from the projections of $D_{\mu\nu}$; to arrive at damping, we need the first-order decay structure

$$n^\mu \nabla_\mu \Theta \sim -\Theta + ..., \tag{B.6.55}$$

and an analogous structure for $Z_\mu^\perp$.

Let us compute the relevant projections of $D_{\mu\nu}$. We start with the doubly normal projection,

$$\begin{aligned} 2 n^\mu n^\nu D_{\mu\nu} &= -\kappa_1 n^\mu n^\nu \Big[n_\mu Z_\nu + n_\nu Z_\mu + \kappa_2 g_{\mu\nu} \underbrace{n^\lambda Z_\lambda}_{=\Theta} \Big] \\ &= -2\kappa_1 [-\Theta - \Theta - \kappa_2 \Theta] \\ &= 2\kappa_1 (\kappa_2 + 2) \Theta. \end{aligned} \tag{B.6.56}$$

Hence, appending this to the doubly-normal-projected undamped Einstein equation (B.6.38), we get

$$\begin{aligned} 0 &= \mathcal{H} + 2 n^\mu \nabla_\mu \Theta + 2K\Theta + 2\bar{\nabla}^\mu Z_\mu^\perp - 2 a^\mu Z_\mu^\perp + 2\kappa_1 (\kappa_2 + 2)\Theta \\ \Leftrightarrow \quad n^\mu \nabla_\mu \Theta &= -\kappa_1 (\kappa_2 + 2)\Theta + K\Theta - \bar{\nabla}^\mu Z_\mu^\perp + a^\mu Z_\mu^\perp - \frac{1}{2}\mathcal{H} \end{aligned} \tag{B.6.57}$$

Switching back to adapted coordinates, we get

$$\partial_t \Theta = \beta^i \partial_i \Theta - \alpha \kappa_1 (\kappa_2 + 2)\Theta + \alpha K \Theta - \alpha \bar{\nabla}^i Z_i + \gamma^{ij} Z_i \partial_j \alpha - \frac{1}{2}\alpha \mathcal{H}. \tag{B.6.58}$$

The normal-tangential projection is even simpler:

$$\begin{aligned} n^\mu P^\nu{}_\lambda D_{\mu\nu} &= -\kappa_1 n^\mu P^\nu{}_\lambda \Big[n_\mu Z_\nu + n_\nu Z_\mu + \kappa_2 g_{\mu\nu} \underbrace{n^\lambda Z_\lambda}_{=\Theta} \Big] \\ &= -\kappa_1 \left[-Z_\lambda^\perp + 0 + 0 \right] \\ &= \kappa_1 Z_\lambda^\perp. \end{aligned} \tag{B.6.59}$$

Consequently, the normal-tangential-projected undamped Einstein equations (B.6.41) turn into

$$
\begin{aligned}
0 &= \mathcal{M}_\lambda + n^\mu \nabla_\mu Z^\perp_\lambda + K^\mu{}_\lambda Z^\perp_\mu - n_\lambda a^\mu Z^\perp_\mu - a_\lambda \Theta + \bar{\nabla}_\lambda \Theta \\
\Leftrightarrow \quad n^\mu \nabla_\mu Z^\perp_\lambda &= -\kappa_1 Z^\perp_\lambda - K^\mu{}_\lambda Z^\perp_\mu + n_\lambda a^\mu Z^\perp_\mu + a_\lambda \Theta - \bar{\nabla}_\lambda \Theta - \mathcal{M}_\lambda
\end{aligned}
\tag{B.6.60}
$$

In adapted coordinates, this rearranges to

$$
\partial_t Z_i = \mathcal{L}_\beta Z_i - \alpha\kappa_1 Z_i - 2\alpha K^j{}_i Z_j - (\partial_i \alpha)\Theta + \alpha \partial_i \Theta - \alpha \mathcal{M}_i \tag{B.6.61}
$$

In summary, the new equations of motion for Θ and Z_i read

$$
\begin{aligned}
\partial_t \Theta &= \beta^i \partial_i \Theta - \alpha\kappa_1(\kappa_2 + 2)\Theta + \alpha K \Theta - \alpha \bar{\nabla}^i Z_i + \gamma^{ij} Z_i \partial_j \alpha - \frac{1}{2}\alpha \mathcal{H} \\
\partial_t Z_i &= \mathcal{L}_\beta Z_i - \alpha\kappa_1 Z_i - 2\alpha K^j{}_i Z_j - (\partial_i \alpha)\Theta + \alpha \partial_i \Theta - \alpha \mathcal{M}_i
\end{aligned}
\tag{B.6.62}
$$

Both the covariant and adapted coordinate forms of these equations show that both Θ and Z_i are exponentially damped, with damping factors $\alpha\kappa_1(\kappa_2 + 2)$ and $\alpha\kappa_1$, respectively.

B.6.5 The Modified Z4-ADM Equations of Motion

So far, we have studied in-depth the normal-normal and normal-tangential projections of the modified Einstein equations

$$
\mathcal{E}_{\mu\nu} + Q_{\mu\nu} - D_{\mu\nu} = 0, \tag{B.6.63}
$$

with

$$
\begin{aligned}
\mathcal{E}_{\mu\nu} &= R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} - 8\pi T_{\mu\nu}, \\
Q_{\mu\nu} &= \nabla_\mu Z_\nu + \nabla_\nu Z_\mu - g_{\mu\nu} \nabla^\lambda Z_\lambda, \\
D_{\mu\nu} &= \kappa_1 \big(n_\mu Z_\nu + n_\nu Z_\mu + \kappa_2 g_{\mu\nu} n^\lambda Z_\lambda \big).
\end{aligned}
\tag{B.6.64}
$$

This led us to the equations of motion for Θ and $Z^\perp_\mu$, given in adapted coordinates by

$$
\begin{aligned}
\partial_t \Theta &= \beta^i \partial_i \Theta - \alpha\kappa_1(\kappa_2 + 2)\Theta + \alpha K \Theta - \alpha \bar{\nabla}^i Z_i + \gamma^{ij} Z_i \partial_j \alpha - \frac{1}{2}\alpha \mathcal{H} \\
\partial_t Z_i &= \mathcal{L}_\beta Z_i - \alpha\kappa_1 Z_i - 2\alpha K^j{}_i Z_j - (\partial_i \alpha)\Theta + \alpha \partial_i \Theta - \alpha \mathcal{M}_i.
\end{aligned}
\tag{B.6.65}
$$

Here, the constraints $\mathcal{H}$ and $\mathcal{M}_i$ act as sources, and are still given by

$$
\begin{aligned}
\mathcal{H} &= \bar{R} + K^2 - K_{ij} K^{ij} - 2\Lambda - 16\pi\rho, \\
\mathcal{M}_i &= \bar{\nabla}_i K - \bar{\nabla}_k K^k{}_i + 8\pi j_i.
\end{aligned}
\tag{B.6.66}
$$

Since these are now part of a dynamical system—that of Θ and Z_i—we do not require them to be zero explicitly. Naturally, it is understood that they should be close to zero so the simulation quickly converges towards pure GR dynamics through damping; we allow for numerical error, i.e. $\mathcal{H}, \mathcal{M}_i \approx 0$.

What we have not studied yet are the tangential-tangential projections of the (trace reversed) modified Einstein equations. While the modifications do not alter the fundamental relationship between the induced metric γ_{ij} and the extrinsic curvature K_{ij}—it remains given by

$$
\partial_t \gamma_{ij} = \beta^k \partial_k \gamma_{ij} + 2\gamma_{k(i} \partial_{j)} \beta^k - 2\alpha K_{ij} \tag{B.6.67}
$$

—the modifications we made to the Einstein equations will have an effect on the right-hand side of $\partial_t K_{ij}$. We recall from Section B.2.5 and its preceding sections that the derivation of the right-hand side for $\partial_t K_{ij}$ starts from the geometric identity

$$P^{\lambda}{}_{\mu} P^{\rho}{}_{\nu} R_{\lambda\rho} = -\mathcal{L}_n K_{\mu\nu} + K K_{\mu\nu} - 2K_{\mu\lambda} K^{\lambda}{}_{\nu} + \bar{R}_{\mu\nu} - \frac{1}{\alpha} \bar{\nabla}_{\mu} \bar{\nabla}_{\nu} \alpha. \tag{B.6.68}$$

The left-hand side was replaced using a projection of the trace-reversed Einstein equations. For Z4, we thus need the trace-reversed *modified* Einstein equations. We recall that the trace reversal of $\mathcal{E}_{\mu\nu}$ is

$$\hat{\mathcal{E}}_{\mu\nu} \coloneqq \mathcal{E}_{\mu\nu} - \frac{1}{2} g_{\mu\nu} \mathcal{E} = R_{\mu\nu} - \Lambda g_{\mu\nu} - 8\pi \Big(T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} (S - \rho) \Big). \tag{B.6.69}$$

For $Q_{\mu\nu}$, it is easily derived that

$$\hat{Q}_{\mu\nu} \coloneqq Q_{\mu\nu} - \frac{1}{2} g_{\mu\nu} Q = \nabla_{\mu} Z_{\nu} + \nabla_{\nu} Z_{\mu}, \tag{B.6.70}$$

as well as

$$\hat{D}_{\mu\nu} \coloneqq D_{\mu\nu} - \frac{1}{2} g_{\mu\nu} D = \kappa_1 \big(n_{\mu} Z_{\nu} + n_{\nu} Z_{\mu} - (\kappa_2 + 1) g_{\mu\nu} n^{\lambda} Z_{\lambda} \big) \tag{B.6.71}$$

The trace-reversed modified Einstein equations are thus simply

$$\hat{\mathcal{E}}_{\mu\nu} + \hat{Q}_{\mu\nu} - \hat{D}_{\mu\nu} = 0. \tag{B.6.72}$$

To take its tangential-tangential projection, we need the projections of the individual terms. For $\hat{\mathcal{E}}_{\mu\nu}$, this reads

$$\begin{aligned} P^{\lambda}{}_{\mu} P^{\rho}{}_{\nu} \hat{\mathcal{E}}_{\lambda\rho} &= P^{\lambda}{}_{\mu} P^{\rho}{}_{\nu} R_{\lambda\rho} - \Lambda \gamma_{\mu\nu} - 8\pi \Big(S_{\mu\nu} - \frac{1}{2} \gamma_{\mu\nu} (S - \rho) \Big) \\ &= -\mathcal{L}_n K_{\mu\nu} + K K_{\mu\nu} - 2K_{\mu\lambda} K^{\lambda}{}_{\nu} + \bar{R}_{\mu\nu} - \frac{1}{\alpha} \bar{\nabla}_{\mu} \bar{\nabla}_{\nu} \alpha \\ &\quad - \Lambda \gamma_{\mu\nu} - 8\pi \Big(S_{\mu\nu} - \frac{1}{2} \gamma_{\mu\nu} (S - \rho) \Big). \end{aligned} \tag{B.6.73}$$

The trace-reversed Z4 modifications project as

$$\begin{aligned} P^{\lambda}{}_{\mu} P^{\rho}{}_{\nu} \hat{Q}_{\lambda\rho} &= 2P^{\lambda}{}_{(\mu} P^{\rho}{}_{\nu)} \nabla_{\lambda} \big(-n_{\rho} \Theta + Z^{\perp}_{\rho} \big) \\ &= 2K_{\mu\nu} \Theta + 2\bar{\nabla}_{(\mu} Z^{\perp}_{\nu)} \end{aligned} \tag{B.6.74}$$

and

$$P^{\lambda}{}_{\mu} P^{\rho}{}_{\nu} \hat{D}_{\lambda\rho} = -\kappa_1 (\kappa_2 + 1) \gamma_{\mu\nu} \Theta. \tag{B.6.75}$$

Reassembling the trace-reversed modified Einstein equations and taking $\mathcal{L}_n K_{\mu\nu}$ to the other side, we hence get the covariant equation of motion

$$\begin{aligned} \mathcal{L}_n K_{\mu\nu} = K K_{\mu\nu} - 2K_{\mu\lambda} K^{\lambda}{}_{\nu} + \bar{R}_{\mu\nu} - \frac{1}{\alpha} \bar{\nabla}_{\mu} \bar{\nabla}_{\nu} \alpha - \Lambda \gamma_{\mu\nu} - 8\pi \Big(S_{\mu\nu} - \frac{1}{2} \gamma_{\mu\nu} (S - \rho) \Big) \\ + 2\bar{\nabla}_{(\mu} Z^{\perp}_{\nu)} + 2K_{\mu\nu} \Theta + \kappa_1 (\kappa_2 + 1) \gamma_{\mu\nu} \Theta. \end{aligned} \tag{B.6.76}$$

Translating this to adapted coordinates, we arrive at the Z4-ADM equations,

$$
\begin{aligned}
\mathcal{H} &= \bar{R} + K^2 - K_{ij}K^{ij} - 2\Lambda - 16\pi\rho, \\
\mathcal{M}_i &= \bar{\nabla}_i K - \bar{\nabla}_k K^k{}_i + 8\pi j_i, \\
\partial_t \Theta &= \beta^i \partial_i \Theta - \alpha\kappa_1(\kappa_2 + 2)\Theta + \alpha K\Theta - \alpha\bar{\nabla}^i Z_i + \gamma^{ij} Z_i \partial_j \alpha - \frac{1}{2}\alpha\mathcal{H} \\
\partial_t Z_i &= \mathcal{L}_\beta Z_i - \alpha\kappa_1 Z_i - 2\alpha K^j{}_i Z_j - (\partial_i \alpha)\Theta + \alpha\partial_i\Theta - \alpha\mathcal{M}_i, \\
\partial_t \gamma_{ij} &= \beta^k \partial_k \gamma_{ij} + 2\gamma_{k(i}\partial_{j)}\beta^k - 2\alpha K_{ij}, \\
\partial_t K_{ij} &= \beta^k \partial_k K_{ij} + 2K_{k(i}\partial_{j)}\beta^k \\
&\quad + \alpha\Big(K K_{ij} - 2K_{ik}K^k{}_j + \bar{R}_{ij}\Big) - \bar{\nabla}_i\bar{\nabla}_j\alpha \\
&\quad - \Lambda\alpha\gamma_{ij} - 8\pi\alpha\Big(S_{ij} - \frac{1}{2}\gamma_{ij}(S - \rho)\Big) \\
&\quad + 2\alpha\bar{\nabla}_{(i}Z_{j)} + 2\alpha K_{ij}\Theta + \kappa_1(\kappa_2 + 1)\alpha\gamma_{ij}\Theta
\end{aligned}
\tag{B.6.77}
$$

These are what we will from now on refer to as the *Z4-ADM equations of motion*. They reduce to the ADM equations of motion (B.2.66) in the case where $Z_\mu = 0$; if non-zero but small enough, the exponential damping introduced by the right-hand sides of $\partial_t\Theta$ and $\partial_t Z_i$ naturally drives Z_μ towards zero. This means that Z4 converges towards unmodified, physical GR dynamics even if the constraints are not satisfied perfectly initially.

B.6.6 Outlook: Z4c/CCZ4

Unfortunately, the Z4-ADM equations (B.6.77) do not constitute a strongly hyperbolic, numerically stable system of PDEs. While the propagation and damping of constraints significantly improve performance over standard ADM, long-term stable evolution still requires the "BSSN treatment". This involves reparametrising the system into conformal and traceless variables—the previously introduced $\big(\phi, \tilde{\gamma}_{ij}, K, \tilde{A}_{ij}\big)$—and introducing the auxiliary conformal connectino functions $\tilde{\Gamma}^i$ to ensure a well-posed principal symbol. Applying this transformation to Z4 yields formulations such as Z4c or CCZ4, which, when coupled with appropriate gauge conditions, form a robust, fully closed system for numerical relativity.

C Computer Algebra Systems and Code Generation

C.1 ASTs

C.1.1 General ASTs

C.1.2 Tensor ASTs

C.2 Code Generation